Chapter III
Riemann-Roch Theory

§ 1. Primes

Having set up the general theory of valued fields, we now return to algebraic number fields. We want to develop their basic theory from the valuation-theoretic point of view. This approach has two prominent advantages compared to the ideal-theoretic treatment given in the first chapter. The first one results from the possibility of passing to completions, thereby enacting the important "local-to-global principle". This will be done in chapter VI. The other advantage lies in the fact that the archimedean valuations bring into the picture the points at infinity, which were hitherto lacking, as the "primes at infinity". In this way a perfect analogy with the function fields is achieved. This is the task to which we now turn.

(1.1) Definition. *A* **prime** (*or* **place**) $\mathfrak{p}$ *of an algebraic number field* K *is a class of equivalent valuations of* K. *The nonarchimedean equivalence classes are called* **finite** *primes and the archimedean ones* **infinite** *primes.*

The infinite primes $\mathfrak{p}$ are obtained, according to chap. II, (8.1), from the embeddings $\tau : K \to \mathbb{C}$. There are two sorts of these: the **real primes**, which are given by embeddings $\tau : K \to \mathbb{R}$, and the **complex primes**, which are induced by the pairs of complex conjugate non-real embeddings $K \to \mathbb{C}$. $\mathfrak{p}$ is real or complex depending whether the completion $K_\mathfrak{p}$ is isomorphic to $\mathbb{R}$ or to $\mathbb{C}$. The infinite primes will be referred to by the formal notation $\mathfrak{p} \mid \infty$, the finite ones by $\mathfrak{p} \nmid \infty$.

In the case of a finite prime, the letter $\mathfrak{p}$ has a multiple meaning: it also stands for the prime ideal of the ring $\mathcal{O}$ of integers of K, or for the maximal ideal of the associated valuation ring, or even for the maximal ideal of the completion. However, this will nowhere create any risk of confusion. We write $\mathfrak{p}|p$ if p is the characteristic of the residue field $\kappa(\mathfrak{p})$ of the finite prime $\mathfrak{p}$. For an infinite prime we adopt the convention that the completion $K_\mathfrak{p}$ also serves as its own "residue field, " i.e., we put

$$\kappa(\mathfrak{p}) := K_\mathfrak{p}, \quad \text{when } \mathfrak{p}|\infty .$$

To each prime $\mathfrak{p}$ of K we now associate a canonical homomorphism

$$v_{\mathfrak{p}} : K^* \to \mathbb{R}$$

from the multiplicative group K^* of K. If $\mathfrak{p}$ is finite, then $v_{\mathfrak{p}}$ is the $\mathfrak{p}$-adic exponential valuation which is normalized by the condition $v_{\mathfrak{p}}(K^*) = \mathbb{Z}$. If $\mathfrak{p}$ is infinite, then $v_{\mathfrak{p}}$ is given by

$$v_{\mathfrak{p}}(a) = -\log |\tau a|,$$

where $\tau : K \to \mathbb{C}$ is an embedding which defines $\mathfrak{p}$.

For an arbitrary prime $\mathfrak{p}|p$ (p prime number or $p = \infty$) we put furthermore

$$f_{\mathfrak{p}} = \left[\kappa(\mathfrak{p}) : \kappa(p)\right],$$

so that $f_{\mathfrak{p}} = [K_{\mathfrak{p}} : \mathbb{R}]$ if $\mathfrak{p}|\infty$, and

$$\mathfrak{N}(\mathfrak{p}) = \begin{cases} p^{f_{\mathfrak{p}}}, & \text{if } \mathfrak{p} \nmid \infty, \\ e^{f_{\mathfrak{p}}}, & \text{if } \mathfrak{p} \mid \infty. \end{cases}$$

This convention suggests that we consider e as being an **infinite prime number**, and the extension $\mathbb{C}|\mathbb{R}$ as being *unramified* with inertia degree 2. We define the absolute value $|\ \ |_{\mathfrak{p}} : K \to \mathbb{R}$ by

$$|a|_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{-v_{\mathfrak{p}}(a)}$$

for $a \neq 0$ and $|0|_{\mathfrak{p}} = 0$. If $\mathfrak{p}$ is the infinite prime associated to the embedding $\tau : K \to \mathbb{C}$, then one finds

$$|a|_{\mathfrak{p}} = |\tau a|, \quad \text{resp.} \quad |a|_{\mathfrak{p}} = |\tau a|^2,$$

if $\mathfrak{p}$ is real, resp. complex.

If $L|K$ is a finite extension of K, then we denote the primes of L by $\mathfrak{P}$ and write $\mathfrak{P}|\mathfrak{p}$ to signify that the valuations in the class $\mathfrak{P}$, when restricted to K, give those of $\mathfrak{p}$. In the case of an infinite prime $\mathfrak{P}$, we define the **inertia degree**, resp. the **ramification index**, by

$$f_{\mathfrak{P}|\mathfrak{p}} = [L_{\mathfrak{P}} : K_{\mathfrak{p}}], \quad \text{resp. } e_{\mathfrak{P}|\mathfrak{p}} = 1.$$

For arbitrary primes $\mathfrak{P}|\mathfrak{p}$ we then have the

(1.2) Proposition. (i) $\sum_{\mathfrak{P}|\mathfrak{p}} e_{\mathfrak{P}|\mathfrak{p}} f_{\mathfrak{P}|\mathfrak{p}} = \sum_{\mathfrak{P}|\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = [L : K]$,

(ii) $\mathfrak{N}(\mathfrak{P}) = \mathfrak{N}(\mathfrak{p})^{f_{\mathfrak{P}|\mathfrak{p}}}$,

(iii) $v_{\mathfrak{P}}(a) = e_{\mathfrak{P}|\mathfrak{p}} v_{\mathfrak{p}}(a)$ *for* $a \in K^*$,

(iv) $v_{\mathfrak{p}}(N_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}(a)) = f_{\mathfrak{P}|\mathfrak{p}} v_{\mathfrak{P}}(a)$ *for* $a \in L^*$,

(v) $|a|_{\mathfrak{P}} = |N_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}(a)|_{\mathfrak{p}}$ *for* $a \in L$.

The normalized valuations $|\ |_{\mathfrak{p}}$ satisfy the following **product formula**, which demonstrates how important it is to include the infinite primes.

(1.3) Proposition. *Given any $a \in K^*$, one has $|a|_{\mathfrak{p}} = 1$ for almost all $\mathfrak{p}$, and*

$$\prod_{\mathfrak{p}} |a|_{\mathfrak{p}} = 1 .$$

Proof: We have $v_{\mathfrak{p}}(a) = 0$ and therefore $|a|_{\mathfrak{p}} = 1$ for all $\mathfrak{p} \nmid \infty$ which do not occur in the prime decomposition of the principal ideal (a) (see chap. I, § 11, p. 69). This therefore holds for almost all $\mathfrak{p}$. From (1.2) and formula (8.4) of chap. II,

$$N_{K|\mathbb{Q}}(a) = \prod_{\mathfrak{p}|p} N_{K_{\mathfrak{p}}|\mathbb{Q}_p}(a)$$

(which includes the case $p = \infty$, $\mathbb{Q}_p = \mathbb{R}$), we obtain the product formula for K from the product formula for $\mathbb{Q}$, which was proved already in chap. II, (2.1):

$$\prod_{\mathfrak{p}} |a|_{\mathfrak{p}} = \prod_{p} \prod_{\mathfrak{p}|p} |a|_{\mathfrak{p}} = \prod_{p} \prod_{\mathfrak{p}|p} \bigl| N_{K_{\mathfrak{p}}|\mathbb{Q}_p}(a) \bigr|_p = \prod_{p} \bigl| N_{K|\mathbb{Q}}(a) \bigr|_p = 1 . \quad \square$$

We denote by $J(\mathcal{O})$ the group of fractional ideals of K, by $P(\mathcal{O})$ the subgroup of fractional principal ideals, and by

$$Pic(\mathcal{O}) = J(\mathcal{O})/P(\mathcal{O})$$

the ideal class group Cl_K of K.

Let us now extend the notion of fractional ideal of K by taking into account also the infinite primes.

(1.4) Definition. *A* **replete ideal** *of K is an element of the group*

$$J(\bar{\mathcal{O}}) := J(\mathcal{O}) \times \prod_{\mathfrak{p}|\infty} \mathbb{R}_+^* ,$$

where $\mathbb{R}_+^*$ denotes the multiplicative group of positive real numbers.

In order to unify notation, we put, for any infinite prime $\mathfrak{p}$ and any real number $\nu \in \mathbb{R}$,

$$\mathfrak{p}^{\nu} := e^{\nu} \in \mathbb{R}_+^* .$$

Given a system of real numbers $\nu_{\mathfrak{p}}$, $\mathfrak{p}|\infty$, let $\prod_{\mathfrak{p}|\infty}\mathfrak{p}^{\nu_{\mathfrak{p}}}$ always denote the vector

$$\prod_{\mathfrak{p}|\infty}\mathfrak{p}^{\nu_{\mathfrak{p}}} = (\dots, e^{\nu_{\mathfrak{p}}}, \dots) \in \prod_{\mathfrak{p}|\infty}\mathbb{R}_+^*,$$

and *not* the product of the quantities $e^{\nu_{\mathfrak{p}}}$ in $\mathbb{R}$. Then every replete ideal $\mathfrak{a} \in J(\bar{o})$ admits the unique product representation

$$\mathfrak{a} = \prod_{\mathfrak{p}\nmid\infty}\mathfrak{p}^{\nu_{\mathfrak{p}}} \times \prod_{\mathfrak{p}|\infty}\mathfrak{p}^{\nu_{\mathfrak{p}}} =: \prod_{\mathfrak{p}}\mathfrak{p}^{\nu_{\mathfrak{p}}},$$

where $\nu_{\mathfrak{p}} \in \mathbb{Z}$ for $\mathfrak{p}\nmid\infty$, and $\nu_{\mathfrak{p}} \in \mathbb{R}$ for $\mathfrak{p}|\infty$. Put

$$\mathfrak{a}_f = \prod_{\mathfrak{p}\nmid\infty}\mathfrak{p}^{\nu_{\mathfrak{p}}} \quad \text{and} \quad \mathfrak{a}_\infty = \prod_{\mathfrak{p}|\infty}\mathfrak{p}^{\nu_{\mathfrak{p}}},$$

so that $\mathfrak{a} = \mathfrak{a}_f \times \mathfrak{a}_\infty$. $\mathfrak{a}_f$ is a fractional ideal of K, and $\mathfrak{a}_\infty$ is an element of $\prod_{\mathfrak{p}|\infty}\mathbb{R}_+^*$. At the same time, we view $\mathfrak{a}_f$, resp. $\mathfrak{a}_\infty$, as replete ideals

$$\mathfrak{a}_f \times \prod_{\mathfrak{p}|\infty} 1, \quad \text{resp. } (1) \times \mathfrak{a}_\infty.$$

Thus for all elements of $J(\bar{o})$ the decomposition

$$\mathfrak{a} = \mathfrak{a}_f \cdot \mathfrak{a}_\infty$$

applies. To $a \in K^*$ we associate the **replete principal ideal**

$$[a] = \prod_{\mathfrak{p}}\mathfrak{p}^{\nu_{\mathfrak{p}}(a)} = (a) \times \prod_{\mathfrak{p}|\infty}\mathfrak{p}^{\nu_{\mathfrak{p}}(a)}.$$

These replete ideals form a subgroup $P(\bar{o})$ of $J(\bar{o})$. The factor group

$$Pic(\bar{o}) = J(\bar{o})/P(\bar{o})$$

is called the **replete ideal class group**, or **replete Picard group**.

(1.5) Definition. *The* **absolute norm** *of a replete ideal* $\mathfrak{a} = \prod_{\mathfrak{p}}\mathfrak{p}^{\nu_{\mathfrak{p}}}$ *is defined to be the positive real number*

$$\mathfrak{N}(\mathfrak{a}) = \prod_{\mathfrak{p}}\mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}.$$

The absolute norm is multiplicative and induces a surjective homomorphism

$$\mathfrak{N} : J(\bar{o}) \to \mathbb{R}_+^*.$$

The absolute norm of a replete principal ideal $[a]$ is equal to 1 in view of the product formula (1.3),

$$\mathfrak{N}([a]) = \prod_{\mathfrak{p}}\mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}(a)} = \prod_{\mathfrak{p}}|a|_{\mathfrak{p}}^{-1} = 1.$$

We therefore obtain a surjective homomorphism

$$\mathfrak{N} : Pic(\bar{o}) \to \mathbb{R}_+^*$$

on the replete Picard group.

The relations between the replete ideals of a number field K and those of an extension field L are afforded by the two homomorphisms

$$J(\bar{\mathcal{O}}_K) \underset{N_{L|K}}{\overset{i_{L|K}}{\rightleftarrows}} J(\bar{\mathcal{O}}_L),$$

which are defined by

$$i_{L|K}\big(\prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}\big) = \prod_{\mathfrak{p}} \prod_{\mathfrak{P}|\mathfrak{p}} \mathfrak{P}^{e_{\mathfrak{P}|\mathfrak{p}}\nu_{\mathfrak{p}}},$$

$$N_{L|K}\big(\prod_{\mathfrak{P}} \mathfrak{P}^{\nu_{\mathfrak{P}}}\big) = \prod_{\mathfrak{p}} \prod_{\mathfrak{P}|\mathfrak{p}} \mathfrak{p}^{f_{\mathfrak{P}|\mathfrak{p}}\nu_{\mathfrak{P}}}.$$

Here the various product signs have to be read according to our convention. These homomorphisms satisfy the

(1.6) Proposition.

(i) *For a chain of fields* $K \subseteq L \subseteq M$, *one has* $N_{M|K} = N_{L|K} \circ N_{M|L}$ *and* $i_{M|K} = i_{M|L} \circ i_{L|K}$.

(ii) $N_{L|K}(i_{L|K}\mathfrak{a}) = \mathfrak{a}^{[L:K]}$ *for* $\mathfrak{a} \in J(\bar{\mathcal{O}}_K)$.

(iii) $\mathfrak{N}(N_{L|K}(\mathfrak{A})) = \mathfrak{N}(\mathfrak{A})$ *for* $\mathfrak{A} \in J(\bar{\mathcal{O}}_L)$.

(iv) *If* $L|K$ *is Galois with Galois group* G, *then for every prime ideal* $\mathfrak{P}$ *of* $\mathcal{O}_L$, *one has* $N_{L|K}(\mathfrak{P})\mathcal{O}_L = \prod_{\sigma\in G} \sigma\mathfrak{P}$.

(v) *For any replete principal ideal* $[a]$ *of* K, *resp.* L, *one has*

$$i_{L|K}([a]) = [a], \quad \text{resp. } N_{L|K}([a]) = [N_{L|K}(a)].$$

(vi) $N_{L|K}(\mathfrak{A}_{\mathfrak{f}}) = N_{L|K}(\mathfrak{A})_{\mathfrak{f}}$ *is the ideal of* K *generated by the norms* $N_{L|K}(a)$ *of all* $a \in \mathfrak{A}_{\mathfrak{f}}$.

Proof: (i) is based on the transitivity of inertia degree and ramification index. (ii) follows from (1.2) in view of the fundamental identity $\sum_{\mathfrak{P}|\mathfrak{p}} f_{\mathfrak{P}|\mathfrak{p}} e_{\mathfrak{P}|\mathfrak{p}} = [L:K]$. (iii) holds for any replete "prime ideal" $\mathfrak{P}$ of L, by (1.2):

$$\mathfrak{N}\big(N_{L|K}(\mathfrak{P})\big) = \mathfrak{N}(\mathfrak{p}^{f_{\mathfrak{P}|\mathfrak{p}}}) = \mathfrak{N}(\mathfrak{p})^{f_{\mathfrak{P}|\mathfrak{p}}} = \mathfrak{N}(\mathfrak{P}),$$

and therefore for all replete ideals of L.

(iv) The prime ideal $\mathfrak{p}$ lying below $\mathfrak{P}$ decomposes in the ring $\mathcal{O}$ of integers of L as $\mathfrak{p} = (\mathfrak{P}_1 \cdots \mathfrak{P}_r)^e$, with prime ideals $\mathfrak{P}_i = \sigma_i\mathfrak{P}$, $\sigma_i \in G/G_{\mathfrak{P}}$, which are conjugates of $\mathfrak{P}$ and thus have the same inertia degree f. Therefore

$$N_{L|K}(\mathfrak{P})\mathcal{O} = \mathfrak{p}^f\mathcal{O} = \prod_{i=1}^{r} \mathfrak{P}_i^{ef} = \prod_{i=1}^{r} \prod_{\tau\in G_{\mathfrak{P}}} \sigma_i\tau\mathfrak{P} = \prod_{\sigma\in G} \sigma\mathfrak{P}.$$

(v) For any element $a \in K^*$, (1.2) implies that $v_{\mathfrak{P}}(a) = e_{\mathfrak{P}|\mathfrak{p}} v_{\mathfrak{p}}(a)$. Hence

$$i_{L|K}([a]) = i_{L|K}\Big(\prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(a)}\Big) = \prod_{\mathfrak{p}} \prod_{\mathfrak{P}|\mathfrak{p}} \mathfrak{P}^{e_{\mathfrak{P}|\mathfrak{p}} v_{\mathfrak{p}}(a)} = \prod_{\mathfrak{P}} \mathfrak{P}^{v_{\mathfrak{P}}(a)} = [a].$$

If, on the other hand, $a \in L^*$, then (1.2) and chap. II, (8.4) imply that $v_{\mathfrak{p}}(N_{L|K}(a)) = \sum_{\mathfrak{P}|\mathfrak{p}} f_{\mathfrak{P}|\mathfrak{p}} v_{\mathfrak{P}}(a)$. Hence

$$N_{L|K}([a]) = N_{L|K}\Big(\prod_{\mathfrak{P}} \mathfrak{P}^{v_{\mathfrak{P}}(a)}\Big) = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(N_{L|K}(a))} = [N_{L|K}(a)].$$

(vi) Let $\mathfrak{a}_{\mathfrak{f}}$ be the ideal of K which is generated by all $N_{L|K}(a)$, with $a \in \mathfrak{A}_{\mathfrak{f}}$. If $\mathfrak{A}_{\mathfrak{f}}$ is a principal ideal (a), then $\mathfrak{a}_{\mathfrak{f}} = (N_{L|K}(a)) = N_{L|K}(\mathfrak{A}_{\mathfrak{f}})$, by (v). But the argument which yielded (v) applies equally well to the localizations $\mathcal{O}_{\mathfrak{p}}|o_{\mathfrak{p}}$ of the extension $\mathcal{O}|o$ of maximal orders of $L|K$. $\mathcal{O}_{\mathfrak{p}}$ has only a finite number of prime ideals, and is therefore a principal ideal domain (see chap. I, § 3, exercise 4). We thus get

$$(\mathfrak{a}_{\mathfrak{f}})_{\mathfrak{p}} = N_{L|K}\big((\mathfrak{A}_{\mathfrak{f}})_{\mathfrak{p}}\big) = N_{L|K}(\mathfrak{A}_{\mathfrak{f}})_{\mathfrak{p}}$$

for all prime ideals $\mathfrak{p}$ of o, and consequently $\mathfrak{a}_{\mathfrak{f}} = N_{L|K}(\mathfrak{A}_{\mathfrak{f}})$. □

Since the homomorphisms $i_{L|K}$ and $N_{L|K}$ map replete principal ideals to replete principal ideals, they induce homomorphisms of the replete Picard groups of K and L, and we obtain the

(1.7) Proposition. *For every finite extension $L|K$, the following two diagrams are commutative:*

$$\begin{tikzcd} Pic(\bar{o}_L) \arrow[r, "\mathfrak{N}"] \arrow[d, shift left, "N_{L|K}"] & \mathbb{R}_+^* \arrow[d, shift left, "\mathrm{id}"] \\ Pic(\bar{o}_K) \arrow[r, "\mathfrak{N}"] \arrow[u, shift left, "i_{L|K}"] & \mathbb{R}_+^*. \arrow[u, shift left, "{[L:K]}"] \end{tikzcd}$$

Let us now translate the notions we have introduced into the function-theoretic language of divisors. In chap. I, § 12, we defined the divisor group $Div(o)$ to consist of all formal sums

$$D = \sum_{\mathfrak{p}\nmid\infty} \nu_{\mathfrak{p}} \mathfrak{p},$$

where $\nu_{\mathfrak{p}} \in \mathbb{Z}$, and $\nu_{\mathfrak{p}} = 0$ for almost all $\mathfrak{p}$. Contained in this group is the group $\mathcal{P}(o)$ of principal divisors $\operatorname{div}(f) = \sum_{\mathfrak{p}\nmid\infty} v_{\mathfrak{p}}(f)\mathfrak{p}$, which allowed us to define the divisor class group

$$CH^1(o) = Div(o)/\mathcal{P}(o).$$

It follows from the main theorem of ideal theory, chap. I, (3.9), that this group is isomorphic to the ideal class group Cl_K, which is a finite group (see chap. I, (12.14)). We now extend these concepts as follows.

(1.8) Definition. *A* **replete divisor** (*or* **Arakelov divisor**) *of K is a formal sum*

$$D = \sum_{\mathfrak{p}} \nu_{\mathfrak{p}} \mathfrak{p},$$

where $\nu_{\mathfrak{p}} \in \mathbb{Z}$ for $\mathfrak{p} \nmid \infty$, $\nu_{\mathfrak{p}} \in \mathbb{R}$ for $\mathfrak{p} | \infty$, and $\nu_{\mathfrak{p}} = 0$ for almost all $\mathfrak{p}$.

The replete divisors form a group, which is denoted by $Div(\bar{o})$. It admits a decomposition

$$Div(\bar{o}) \cong Div(o) \times \bigoplus_{\mathfrak{p}|\infty} \mathbb{R}\mathfrak{p}.$$

On the right-hand side, the second factor is endowed with the canonical topology, the first one with the discrete topology. On the product we have the product topology, which makes $Div(\bar{o})$ into a locally compact topological group.

We now study the canonical homomorphism

$$\operatorname{div} : K^* \longrightarrow Div(\bar{o}), \qquad \operatorname{div}(f) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(f)\mathfrak{p}.$$

The elements of the form $\operatorname{div}(f)$ are called **replete principal divisors**.

Remark: The composite of the mapping $\operatorname{div} : K^* \to Div(\bar{o})$ with the mapping

$$Div(\bar{o}) \longrightarrow \prod_{\mathfrak{p}|\infty} \mathbb{R}, \quad \sum_{\mathfrak{p}} \nu_{\mathfrak{p}}\mathfrak{p} \longmapsto (\nu_{\mathfrak{p}} f_{\mathfrak{p}})_{\mathfrak{p}|\infty},$$

is equal, up to sign, to the logarithm map

$$\lambda : K^* \longrightarrow \prod_{\mathfrak{p}|\infty} \mathbb{R}, \quad \lambda(f) = \big(\dots, \log|f|_{\mathfrak{p}}, \dots\big),$$

of Minkowski theory (see chap. I, §7, p. 39, and chap. III, §3, p. 211). By chap. I, (7.3), it maps the unit group o^* onto a complete lattice $\Gamma = \lambda(o^*)$ in trace-zero space $H = \{(x_{\mathfrak{p}}) \in \prod_{\mathfrak{p}|\infty} \mathbb{R} \mid \sum_{\mathfrak{p}|\infty} x_{\mathfrak{p}} = 0\}$.

(1.9) Proposition. *The kernel of $\operatorname{div} : K^* \to Div(\bar{o})$ is the group $\mu(K)$ of roots of unity in K, and its image $\mathcal{P}(\bar{o})$ is a discrete subgroup of $Div(\bar{o})$.*

Proof: By the above remark, the composite of div with the map $Div(\bar{o}) \to \prod_{\mathfrak{p}|\infty} \mathbb{R}$, $\sum_{\mathfrak{p}} \nu_{\mathfrak{p}} \mathfrak{p} \mapsto (\nu_{\mathfrak{p}} f_{\mathfrak{p}})_{\mathfrak{p}|\infty}$, yields, up to sign, the homomorphism $\lambda : K^* \to \prod_{\mathfrak{p}|\infty} \mathbb{R}$. By chap. I, (7.1), the latter fits into the exact sequence

$$1 \longrightarrow \mu(K) \longrightarrow o^* \xrightarrow{\lambda} \Gamma \longrightarrow 0,$$

where Γ is a complete lattice in trace-zero space $H \subseteq \prod_{\mathfrak{p}|\infty} \mathbb{R}$. Therefore $\mu(K)$ is the kernel of div. Since Γ is a lattice, there exists a neighbourhood U of 0 in $\prod_{\mathfrak{p}|\infty} \mathbb{R}$ which contains no element of Γ except 0. Considering the isomorphism $\alpha : \prod_{\mathfrak{p}|\infty} \mathbb{R} \to \bigoplus_{\mathfrak{p}|\infty} \mathbb{R}\mathfrak{p}$, $(\nu_{\mathfrak{p}})_{\mathfrak{p}|\infty} \mapsto \sum_{\mathfrak{p}|\infty} \frac{\nu_{\mathfrak{p}}}{f_{\mathfrak{p}}} \mathfrak{p}$, the set $\{0\} \times \alpha U \subset Div(o) \times \bigoplus_{\mathfrak{p}|\infty} \mathbb{R}\mathfrak{p} = Div(\bar{o})$ is a neighbourhood of 0 in $Div(\bar{o})$ which contains no replete principal divisor except 0. This shows that $\mathcal{P}(\bar{o}) = \operatorname{div}(K^*)$ lies discretely in $Div(\bar{o})$. □

(1.10) Definition. *The factor group*

$$CH^1(\bar{o}) = Div(\bar{o})/\mathcal{P}(\bar{o})$$

is called the **replete divisor class group** (*or* **Arakelov class group**) *of* K.

As $\mathcal{P}(\bar{o})$ is discrete in $Div(\bar{o})$, and is therefore in particular closed, $CH^1(\bar{o})$ becomes a locally compact Hausdorff topological group with respect to the quotient topology. It is the correct analogue of the divisor class group of a function field (see chap. I, § 14). For the latter we introduced the degree map onto the group $\mathbb{Z}$; for $CH^1(\bar{o})$ we obtain a **degree map** onto the group $\mathbb{R}$. It is induced by the continuous homomorphism

$$\deg : Div(\bar{o}) \longrightarrow \mathbb{R}$$

which sends a replete divisor $D = \sum_{\mathfrak{p}} \nu_{\mathfrak{p}} \mathfrak{p}$ to the real number

$$\deg(D) = \sum_{\mathfrak{p}} \nu_{\mathfrak{p}} \log \mathfrak{N}(\mathfrak{p}) = \log\Big(\prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}\Big).$$

From the product formula (1.3), we find for any replete principal divisor $\operatorname{div}(f) \in \mathcal{P}(\bar{o})$ that

$$\deg\big(\operatorname{div}(f)\big) = \sum_{\mathfrak{p}} \log \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}(f)} = \log\Big(\prod_{\mathfrak{p}} |f|_{\mathfrak{p}}^{-1}\Big) = 0.$$

Thus we obtain a well-defined continuous homomorphism

$$\deg : CH^1(\bar{o}) \longrightarrow \mathbb{R}.$$

The **kernel** $CH^1(\bar{o})^0$ of this map is made up from the unit group o^* and the ideal class group $Cl_K \cong CH^1(o)$ of K as follows.

(1.11) Proposition. *Let $\Gamma = \lambda(\mathcal{O}^*)$ denote the complete lattice of units in trace-zero space $H = \{(x_\mathfrak{p}) \in \prod_{\mathfrak{p}|\infty} \mathbb{R} \mid \sum_{\mathfrak{p}|\infty} x_\mathfrak{p} = 0\}$. There is an exact sequence*

$$0 \longrightarrow H/\Gamma \longrightarrow CH^1(\bar{\mathcal{O}})^0 \longrightarrow CH^1(\mathcal{O}) \longrightarrow 0.$$

Proof: Let $Div(\bar{\mathcal{O}})^0$ be the kernel of $\deg : Div(\bar{\mathcal{O}}) \to \mathbb{R}$. Consider the exact sequence

$$0 \longrightarrow \prod_{\mathfrak{p}|\infty} \xrightarrow{\alpha} Div(\bar{\mathcal{O}}) \longrightarrow Div(\mathcal{O}) \longrightarrow 0,$$

where $\alpha((\nu_\mathfrak{p})) = \sum_{\mathfrak{p}|\infty} \frac{\nu_\mathfrak{p}}{f_\mathfrak{p}}\mathfrak{p}$. Restricting to $Div(\bar{\mathcal{O}})^0$ yields the exact commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \lambda(\mathcal{O}^*) & \xrightarrow{\alpha} & \mathcal{P}(\bar{\mathcal{O}}) & \longrightarrow & \mathcal{P}(\mathcal{O}) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & H & \xrightarrow{\alpha} & Div(\bar{\mathcal{O}})^0 & \longrightarrow & Div(\mathcal{O}) & \longrightarrow & 0.
\end{array}$$

Via the snake lemma (see [23], chap. III, § 3, (3.3)), this gives rise to the exact sequence

$$0 \longrightarrow H/\lambda(\mathcal{O}^*) \longrightarrow CH^1(\bar{\mathcal{O}})^0 \longrightarrow CH^1(\mathcal{O}) \longrightarrow 0. \qquad \square$$

The two fundamental facts of algebraic number theory, the finiteness of the class number and Dirichlet's unit theorem, now merge into (and are in fact equivalent to) the simple statement that the kernel $CH^1(\bar{\mathcal{O}})^0$ of the degree map $\deg : CH^1(\bar{\mathcal{O}}) \to \mathbb{R}$ is **compact**.

(1.12) Theorem. *The group $CH^1(\bar{\mathcal{O}})^0$ is compact.*

Proof: This follows immediately from the exact sequence

$$0 \longrightarrow H/\Gamma \longrightarrow CH^1(\bar{\mathcal{O}})^0 \longrightarrow CH^1(\mathcal{O}) \longrightarrow 0.$$

As Γ is a complete lattice in the $\mathbb{R}$-vector space H, the quotient H/Γ is a compact torus. In view of the finiteness of $CH^1(\mathcal{O})$, we obtain $CH^1(\bar{\mathcal{O}})^0$ as the union of the finitely many compact cosets of H/Γ in $CH^1(\bar{\mathcal{O}})^0$. Thus $CH^1(\bar{\mathcal{O}})^0$ itself is compact. $\square$

The correspondence between replete ideals and replete divisors is given by the two mutually inverse mappings

$$\operatorname{div} : J(\bar{o}) \longrightarrow Div(\bar{o}), \qquad \operatorname{div}\Big(\prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}\Big) = \sum_{\mathfrak{p}} -\nu_{\mathfrak{p}}\mathfrak{p},$$

$$Div(\bar{o}) \longrightarrow J(\bar{o}), \qquad \sum_{\mathfrak{p}} \nu_{\mathfrak{p}}\mathfrak{p} \longmapsto \prod_{\mathfrak{p}} \mathfrak{p}^{-\nu_{\mathfrak{p}}}.$$

These are *topological isomorphisms* once we equip

$$J(\bar{o}) = J(o) \times \prod_{\mathfrak{p}|\infty} \mathbb{R}_+^*$$

with the product topology of the discrete topology on $J(o)$ and the canonical topology on $\prod_{\mathfrak{p}|\infty} \mathbb{R}_+^*$. The image of a divisor $D = \sum_{\mathfrak{p}} \nu_{\mathfrak{p}}\mathfrak{p}$ is also written as

$$o(D) = \prod_{\mathfrak{p}} \mathfrak{p}^{-\nu_{\mathfrak{p}}}.$$

The minus sign here is motivated by classical usage in function theory. Replete principal ideals correspond to replete principal divisors in such a way that $P(\bar{o})$ becomes a discrete subgroup of $J(\bar{o})$ by (1.9), and $Pic(\bar{o}) = J(\bar{o})/P(\bar{o})$ is a locally compact Hausdorff topological group. We obtain the following extension of chap. I, (12.14).

(1.13) Proposition. *The mapping* div : $J(\bar{o}) \xrightarrow{\sim} Div(\bar{o})$ *induces a topological isomorphism*

$$\operatorname{div} : Pic(\bar{o}) \xrightarrow{\sim} CH^1(\bar{o}).$$

On the group $J(\bar{o})$ we have the homomorphism $\mathfrak{N} : J(\bar{o}) \to \mathbb{R}_+^*$, and on the group $Div(\bar{o})$ there is the degree map $\deg : Div(\bar{o}) \to \mathbb{R}$. They are obviously related by the formula

$$\deg\big(\operatorname{div}(\mathfrak{a})\big) = -\log \mathfrak{N}(\mathfrak{a}),$$

and we get a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & Pic(\bar{o})^0 \arrow[r] \arrow[d, "\cong"] & Pic(\bar{o}) \arrow[r, "\mathfrak{N}"] \arrow[d, "\operatorname{div}"', "\cong"] & \mathbb{R}_+^* \arrow[r] \arrow[d, "-\log"', "\cong"] & 0 \\
0 \arrow[r] & CH^1(\bar{o})^0 \arrow[r] & CH^1(\bar{o}) \arrow[r, "\deg"] & \mathbb{R} \arrow[r] & 0
\end{tikzcd}$$

with exact rows. (1.12) now yields the

(1.14) Corollary. *The group*

$$Pic(\bar{\mathcal{O}})^0 = \{ [\mathfrak{a}] \in Pic(\bar{\mathcal{O}}) \mid \mathfrak{N}(\mathfrak{a}) = 1 \}$$

is compact.

The preceding isomorphism result (1.13) invites a philosophical reflection. Throughout the historical development of algebraic number theory, a controversy persisted between the followers of Dedekind's ideal-theoretic approach, and the divisor-theoretic method of building up the theory from the valuation-theoretic notion of primes. Both theories are equivalent in the sense of the above isomorphism result, but they are also fundamentally different in nature. The controversy has finally given way to the realization that neither approach is dominant, that each one instead emanates from its own proper world, and that the relation between these worlds is expressed by an important mathematical principle. However, all this becomes evident only in higher dimensional arithmetic algebraic geometry. There, on an algebraic $\mathbb{Z}$-scheme X, one studies on the one hand the totality of all *vector bundles*, and on the other, that of all *irreducible subschemes* of X. From the first, one constructs a series of groups $K_i(X)$ which constitute the subject of algebraic ***K*-theory**. From the second is constructed a series of groups $CH^i(X)$, the subject of **Chow theory**. Vector bundles are by definition locally free $\mathcal{O}_X$-modules. In the special case $X = \operatorname{Spec}(\mathcal{O})$ this includes the fractional ideals. The irreducible subschemes and their formal linear combinations, i.e., the **cycles** of X, are to be considered as generalizations of the primes and divisors. The isomorphism between divisor class group and ideal class group extends to the general setting as a homomorphic relation between the groups $CH^i(X)$ and $K_i(X)$. Thus the initial controversy has been resolved into a seminal mathematical theory (for further reading, see [13]).

Exercise 1 (Strong Approximation Theorem). Let S be a finite set of primes and let $\mathfrak{p}_0$ be another prime of K which does not belong to S. Let $a_\mathfrak{p} \in K$ be given numbers, for $\mathfrak{p} \in S$. Then for every $\varepsilon > 0$, there exists an $x \in K$ such that

$$|x - a_\mathfrak{p}|_\mathfrak{p} < \varepsilon \text{ for } \mathfrak{p} \in S, \text{ and } |x|_\mathfrak{p} \le 1 \text{ for } \mathfrak{p} \notin S \cup \{\mathfrak{p}_0\}.$$

Exercise 2. Let K be totally real, i.e., $K_\mathfrak{p} = \mathbb{R}$ for all $\mathfrak{p}|\infty$. Let T be a proper nonempty subset of $\operatorname{Hom}(K,\mathbb{R})$. Then there exists a unit ε of K satisfying $\tau\varepsilon > 1$ for $\tau \in S$ and $0 < \tau\varepsilon < 1$ for $\tau \notin S$.

Exercise 3. Show that the absolute norm $\mathfrak{N} : Pic(\overline{\mathbb{Z}}) \to \mathbb{R}_+^*$ is an isomorphism.

Exercise 4. Let K and L be number fields, and let $\tau : K \to L$ be a homomorphism.

Given any replete divisor $D = \sum_\mathfrak{P} \nu_\mathfrak{P} \mathfrak{P}$ of L, define a replete divisor of K by the rule

$$\tau_*(D) = \sum_\mathfrak{p} \Big(\sum_{\mathfrak{P}|\mathfrak{p}} \nu_\mathfrak{P} f_{\mathfrak{P}|\mathfrak{p}} \Big) \mathfrak{p},$$

where $f_{\mathfrak{P}|\mathfrak{p}}$ is the inertia degree of $\mathfrak{P}$ over τK and $\mathfrak{P}|\mathfrak{p}$ signifies $\tau\mathfrak{p} = \mathfrak{P}|_{\tau K}$. Show that τ_* induces a homomorphism

$$\tau_* : CH^1(\bar{o}_L) \to CH^1(\bar{o}_K).$$

Exercise 5. Given any replete divisor $D = \sum_{\mathfrak{p}} \nu_{\mathfrak{p}}\mathfrak{p}$ of K, define a replete divisor of L by the rule

$$\tau^*(D) = \sum_{\mathfrak{p}} \sum_{\mathfrak{P}|\mathfrak{p}} \nu_{\mathfrak{p}} e_{\mathfrak{P}|\mathfrak{p}} \mathfrak{P},$$

where $e_{\mathfrak{P}|\mathfrak{p}}$ denotes the ramification index of $\mathfrak{P}$ over K. Show that τ^* induces a homomorphism

$$\tau^* : CH^1(\bar{o}_K) \to CH^1(\bar{o}_L).$$

Exercise 6. Show that $\tau_* \circ \tau^* = [L : K]$ and that

$$\deg(\tau_* D) = \deg(D), \quad \deg(\tau^* D) = [L : K] \deg(D).$$

§ 2. Different and Discriminant

It is our intention to develop a framework for the theory of algebraic number fields which shows the complete analogy with the theory of function fields. This goal leads us naturally to the notions of different and discriminant, as we shall explain in § 3 and § 7. They control the ramification behaviour of an extension of valued fields.

Let $L|K$ be a finite separable field extension, $o \subseteq K$ a Dedekind domain with field of fractions K, and let $\mathcal{O} \subseteq L$ be its integral closure in L. Throughout this section we assume systematically that the residue field extensions $\lambda|\kappa$ of $\mathcal{O}|o$ are separable. The theory of the different originates from the fact that we are given a canonical nondegenerate symmetric bilinear form on the K-vector space L, *viz.*, the **trace form**

$$T(x, y) = Tr(xy)$$

(see chap. I, § 2). It allows us to associate to every fractional ideal $\mathfrak{A}$ of L the **dual** $\mathcal{O}$-module

$${}^*\mathfrak{A} = \left\{ x \in L \;\middle|\; Tr(x\,\mathfrak{A}) \subseteq o \right\}.$$

It is again a fractional ideal. For if $\alpha_1, \dots, \alpha_n \in \mathcal{O}$ is a basis of $L|K$ and $d = \det(Tr(\alpha_i\alpha_j))$ its discriminant, then $ad^*\mathfrak{A} \subseteq \mathcal{O}$ for every nonzero $a \in \mathfrak{A} \cap o$. Indeed, if $x = x_1\alpha_1 + \cdots + x_n\alpha_n \in {}^*\mathfrak{A}$, with $x_i \in K$, then the ax_i satisfy the system of linear equations $\sum_{i=1}^{n} ax_i\, Tr(\alpha_i\alpha_j) = Tr(xa\alpha_j) \in o$. This implies $dax_i \in o$ and thus $dax \in \mathcal{O}$.

The notion of duality is justified by the isomorphism

$$^*\mathfrak{A} \xrightarrow{\sim} \operatorname{Hom}_{\mathcal{O}}(\mathfrak{A}, \mathcal{O}), \quad x \longmapsto \big(y \mapsto Tr(xy)\big)\,.$$

Indeed, since every $\mathcal{O}$-homomorphism $f : \mathfrak{A} \to \mathcal{O}$ extends uniquely to a K-homomorphism $f : L \to K$ in view of $\mathfrak{A}K = L$, we may consider $\operatorname{Hom}_{\mathcal{O}}(\mathfrak{A}, \mathcal{O})$ as a submodule of $\operatorname{Hom}_K(L, K)$, namely, the image of $^*\mathfrak{A}$ with respect to $L \to \operatorname{Hom}_K(L, K)$, $x \mapsto (y \mapsto Tr(xy))$. The module dual to $\mathcal{O}$,

$$^*\mathcal{O} \;\cong\; \operatorname{Hom}_{\mathcal{O}}(\mathcal{O}, \mathcal{O}),$$

will obviously occupy a distinguished place in this theory.

(2.1) Definition. *The fractional ideal*

$$\mathfrak{C}_{\mathcal{O}|\mathcal{o}} = {}^*\mathcal{O} = \big\{x \in L \;\big|\; Tr(x\mathcal{O}) \subseteq \mathcal{o}\big\}$$

is called **Dedekind's complementary module**, *or the* **inverse different**. *Its inverse,*

$$\mathfrak{D}_{\mathcal{O}|\mathcal{o}} = \mathfrak{C}^{-1}_{\mathcal{O}|\mathcal{o}},$$

is called the **different** *of $\mathcal{O}|\mathcal{o}$.*

As $\mathfrak{C}_{\mathcal{O}|\mathcal{o}} \supseteq \mathcal{O}$, the ideal $\mathfrak{D}_{\mathcal{O}|\mathcal{o}} \subseteq \mathcal{O}$ is actually an integral ideal of L. We will frequently denote it by $\mathfrak{D}_{L|K}$, provided the intended subrings $\mathcal{o}, \mathcal{O}$ are evident from the context. In the same way, we write $\mathfrak{C}_{L|K}$ instead of $\mathfrak{C}_{\mathcal{O}|\mathcal{o}}$. The different behaves in the following manner under change of rings $\mathcal{o}$ and $\mathcal{O}$.

(2.2) Proposition.

(i) *For a tower of fields $K \subseteq L \subseteq M$, one has $\mathfrak{D}_{M|K} = \mathfrak{D}_{M|L}\mathfrak{D}_{L|K}$.*

(ii) *For any multiplicative subset S of $\mathcal{o}$, one has $\mathfrak{D}_{S^{-1}\mathcal{O}|S^{-1}\mathcal{o}} = S^{-1}\mathfrak{D}_{\mathcal{O}|\mathcal{o}}$.*

(iii) *If $\mathfrak{P}|\mathfrak{p}$ are prime ideals of $\mathcal{O}$, resp. $\mathcal{o}$, and $\mathcal{O}_{\mathfrak{P}}|\mathcal{o}_{\mathfrak{p}}$ are the associated completions, then*

$$\mathfrak{D}_{\mathcal{O}|\mathcal{o}}\mathcal{O}_{\mathfrak{P}} = \mathfrak{D}_{\mathcal{O}_{\mathfrak{P}}|\mathcal{o}_{\mathfrak{p}}}\,.$$

Proof: (i) Let $A = \mathcal{o} \subseteq K$, and let $B \subseteq L$, $C \subseteq M$ be the integral closure of $\mathcal{o}$ in L, resp. M. It then suffices to show that

$$\mathfrak{C}_{C|A} = \mathfrak{C}_{C|B}\mathfrak{C}_{B|A}\,.$$

The inclusion $\supseteq$ follows from

$$\begin{aligned} Tr_{M|K}(\mathfrak{C}_{C|B}\mathfrak{C}_{B|A}C) &= Tr_{L|K}\, Tr_{M|L}(\mathfrak{C}_{C|B}\mathfrak{C}_{B|A}C) \\ &= Tr_{L|K}\big(\mathfrak{C}_{B|A}\, Tr_{M|L}(\mathfrak{C}_{C|B}C)\big) \subseteq A\,. \end{aligned}$$

In view of $BC = C$, the inclusion $\subseteq$ is derived as follows:

$$Tr_{M|K}(\mathfrak{C}_{C|A}C) = Tr_{L|K}\big(B\, Tr_{M|L}(\mathfrak{C}_{C|A}C)\big) \subseteq A\,,$$

so that $Tr_{M|L}(\mathfrak{C}_{C|A}C) \subseteq \mathfrak{C}_{B|A}$, and thus

$$Tr_{M|L}(\mathfrak{C}_{B|A}^{-1}\mathfrak{C}_{C|A}C) = \mathfrak{C}_{B|A}^{-1}\, Tr_{M|L}(\mathfrak{C}_{C|A}C) \subseteq B\,.$$

This does indeed imply $\mathfrak{C}_{B|A}^{-1}\mathfrak{C}_{C|A} \subseteq \mathfrak{C}_{C|B}$, and so $\mathfrak{C}_{C|A} \subseteq \mathfrak{C}_{C|B}\mathfrak{C}_{B|A}$.

(ii) is trivial.

(iii) By (ii) we may assume that $\mathcal{o}$ is a discrete valuation ring. We show that $\mathfrak{C}_{\mathcal{O}|\mathcal{o}}$ is dense in $\mathfrak{C}_{\mathcal{O}_\mathfrak{P}|\mathcal{o}_\mathfrak{p}}$. In order to do this, we use the formula

$$Tr_{L|K} = \sum_{\mathfrak{P}|\mathfrak{p}} Tr_{L_\mathfrak{P}|K_\mathfrak{p}}$$

(see chap. II, (8.4)). Let $x \in \mathfrak{C}_{\mathcal{O}|\mathcal{o}}$ and $y \in \mathcal{O}_\mathfrak{P}$. The approximation theorem allows us to find an η in L which is close to y with respect to $v_\mathfrak{P}$, and close to 0 with respect to $v_{\mathfrak{P}'}$, for $\mathfrak{P}'|\mathfrak{p}$, $\mathfrak{P}' \neq \mathfrak{P}$. The left-hand side of the equation

$$Tr_{L|K}(x\eta) = Tr_{L_\mathfrak{P}|K_\mathfrak{p}}(x\eta) + \sum_{\mathfrak{P}'\neq\mathfrak{P}} Tr_{L_{\mathfrak{P}'}|K_\mathfrak{p}}(x\eta)$$

then belongs to $\mathcal{o}_\mathfrak{p}$, since $Tr_{L|K}(x\eta) \in \mathcal{o} \subseteq \mathcal{o}_\mathfrak{p}$, but the same is true of the elements $Tr_{L_{\mathfrak{P}'}|K_\mathfrak{p}}(x\eta)$ because they are close to zero with respect to $v_\mathfrak{p}$. Therefore $Tr_{L_\mathfrak{P}|K_\mathfrak{p}}(xy) \in \mathcal{o}_\mathfrak{p}$. This shows that $\mathfrak{C}_{\mathcal{O}|\mathcal{o}} \subseteq \mathfrak{C}_{\mathcal{O}_\mathfrak{P}|\mathcal{o}_\mathfrak{p}}$.

If on the other hand $x \in \mathfrak{C}_{\mathcal{O}_\mathfrak{P}|\mathcal{o}_\mathfrak{p}}$, and if $\xi \in L$ is sufficiently close to x with respect to $v_\mathfrak{P}$, and sufficiently close to 0 with respect to $v_{\mathfrak{P}'}$, for $\mathfrak{P}' \neq \mathfrak{P}$, then $\xi \in \mathfrak{C}_{\mathcal{O}|\mathcal{o}}$. In fact, if $y \in \mathcal{O}$, then $Tr_{L_\mathfrak{P}|K_\mathfrak{p}}(\xi y) \in \mathcal{o}_\mathfrak{p}$, since $Tr_{L_\mathfrak{P}|K_\mathfrak{p}}(xy) \in \mathcal{o}_\mathfrak{p}$. Likewise $Tr_{L_{\mathfrak{P}'}|K_\mathfrak{p}}(\xi y) \in \mathcal{o}_\mathfrak{p}$ for $\mathfrak{P}'|\mathfrak{P}$, because these elements are close to 0. Therefore $Tr_{L|K}(\xi y) \in \mathcal{o}_\mathfrak{p} \cap K = \mathcal{o}$, i.e., $\xi \in \mathfrak{C}_{\mathcal{O}|\mathcal{o}}$. This shows that $\mathfrak{C}_{\mathcal{O}|\mathcal{o}}$ is dense in $\mathfrak{C}_{\mathcal{O}_\mathfrak{P}|\mathcal{o}_\mathfrak{p}}$, in other words, $\mathfrak{C}_{\mathcal{O}|\mathcal{o}}\mathcal{O}_\mathfrak{P} = \mathfrak{C}_{\mathcal{O}_\mathfrak{P}|\mathcal{o}_\mathfrak{p}}$, and so $\mathfrak{D}_{\mathcal{O}|\mathcal{o}}\mathcal{O}_\mathfrak{P} = \mathfrak{D}_{\mathcal{O}_\mathfrak{P}|\mathcal{o}_\mathfrak{p}}$. □

If we put $\mathfrak{D} = \mathfrak{D}_{L|K}$ and $\mathfrak{D}_\mathfrak{P} = \mathfrak{D}_{L_\mathfrak{P}|K_\mathfrak{p}}$, and consider $\mathfrak{D}_\mathfrak{P}$ at the same time as an ideal of $\mathcal{O}$ (i.e., as the ideal $\mathcal{O} \cap \mathfrak{D}_\mathfrak{P}$), then (2.2), (iii) gives us the

(2.3) Corollary. $\mathfrak{D} = \prod_\mathfrak{P} \mathfrak{D}_\mathfrak{P}$.

The name "different" is explained by the following explicit description, which was Dedekind's original way to define it. Let $\alpha \in \mathcal{O}$ and let $f(X) \in o[X]$ be the minimal polynomial of α. We define the **different of the element** α by

$$\delta_{L|K}(\alpha) = \begin{cases} f'(\alpha) & \text{if } L = K(\alpha), \\ 0 & \text{if } L \neq K(\alpha). \end{cases}$$

In the special case where $\mathcal{O} = o[\alpha]$ we then obtain:

(2.4) Proposition. *If $\mathcal{O} = o[\alpha]$, then the different is the principal ideal*

$$\mathfrak{D}_{L|K} = \big(\delta_{L|K}(\alpha)\big).$$

Proof: Let $f(X) = a_0 + a_1X + \cdots + a_nX^n$ be the minimal polynomial of α and

$$\frac{f(X)}{X-\alpha} = b_0 + b_1X + \cdots + b_{n-1}X^{n-1}.$$

The dual basis of $1, \alpha, \ldots, \alpha^{n-1}$ with respect to $Tr(xy)$ is then given by

$$\frac{b_0}{f'(\alpha)}, \ldots, \frac{b_{n-1}}{f'(\alpha)}.$$

For if $\alpha_1, \ldots, \alpha_n$ are the roots of f, then one has

$$\sum_{i=1}^{n} \frac{f(X)}{X-\alpha_i}\,\frac{\alpha_i^r}{f'(\alpha_i)} = X^r, \qquad 0 \le r \le n-1,$$

as the difference of the two sides is a polynomial of degree $\le n-1$ with roots $\alpha_1, \ldots, \alpha_n$, so is identically zero. We may write this equation in the form

$$Tr\Big[\frac{f(X)}{X-\alpha}\,\frac{\alpha^r}{f'(\alpha)}\Big] = X^r.$$

Considering now the coefficient of each of the powers of X, we obtain

$$Tr\Big(\alpha^i\frac{b_j}{f'(\alpha)}\Big) = \delta_{ij}$$

and the claim follows.

As $\mathcal{O} = o + o\alpha + \cdots + o\alpha^{n-1}$, we get

$$\mathfrak{C}_{\mathcal{O}|o} = f'(\alpha)^{-1}(ob_0 + \cdots + ob_{n-1}).$$

From the recursive formulas

$$\begin{aligned} b_{n-1} &= 1, \\ b_{n-2} - \alpha b_{n-1} &= a_{n-1}, \\ \cdots &= \cdots \end{aligned}$$

it follows that

$$b_{n-i} = \alpha^{i-1} + a_{n-1}\alpha^{i-2} + \cdots + a_{n-i+1},$$

so that $\mathcal{o}b_0 + \cdots + \mathcal{o}b_{n-1} = \mathcal{o}[\alpha] = \mathcal{O}$; then $\mathfrak{C}_{\mathcal{O}|\mathcal{o}} = f'(\alpha)^{-1}\mathcal{O}$, and thus $\mathfrak{D}_{L|K} = (f'(\alpha))$. $\square$

The proof shows that the module $^*\mathcal{o}[\alpha] = \{x \in L \mid Tr_{L|K}(x\mathcal{o}[\alpha]) \subseteq \mathcal{o}\}$, which is the dual of the $\mathcal{o}$-module $\mathcal{o}[\alpha]$, always admits the $\mathcal{o}$-basis $\alpha^i/f'(\alpha)$, $i = 0, \ldots, n-1$. We exploit this for the following characterization of the different in the general case where $\mathcal{O}$ need not be monogenous.

(2.5) Theorem. *The different $\mathfrak{D}_{L|K}$ is the ideal generated by all differents of elements $\delta_{L|K}(\alpha)$ for $\alpha \in \mathcal{O}$.*

Proof: Let $\alpha \in \mathcal{O}$ such that $L = K(\alpha)$, and let $f(X)$ be the minimal polynomial of α. In order to show that $f'(\alpha) \in \mathfrak{D}_{L|K}$, we consider the "conductor" $\mathfrak{f} = \mathfrak{f}_{\mathcal{o}[\alpha]} = \{x \in L \mid x\mathcal{O} \subseteq \mathcal{o}[\alpha]\}$ of $\mathcal{o}[\alpha]$ (see chap. I, § 12, p. 79). On putting $b = f'(\alpha)$, we have for $x \in L$:

$$\begin{aligned} x \in \mathfrak{f} &\iff x\mathcal{O} \subseteq \mathcal{o}[\alpha] &&\iff b^{-1}x\mathcal{O} \subseteq b^{-1}\mathcal{o}[\alpha] = {}^*\mathcal{o}[\alpha] \\ &\iff Tr(b^{-1}x\mathcal{O}) \subseteq \mathcal{o} &&\iff b^{-1}x \in \mathfrak{D}_{L|K}^{-1} \iff x \in b\mathfrak{D}_{L|K}^{-1}. \end{aligned}$$

Therefore $(f'(\alpha)) = \mathfrak{f}_{\mathcal{o}[\alpha]}\mathfrak{D}_{L|K}$, so in particular, $f'(\alpha) \in \mathfrak{D}_{L|K}$.

$\mathfrak{D}_{L|K}$ thus divides all the differents of elements $\delta_{L|K}(\alpha)$. In order to prove that $\mathfrak{D}_{L|K}$ is in fact the greatest common divisor of all $\delta_{L|K}(\alpha)$, it suffices to show that, for every prime ideal $\mathfrak{P}$, there exists an $\alpha \in \mathcal{O}$ such that $L = K(\alpha)$ and $v_{\mathfrak{P}}(\mathfrak{D}_{L|K}) = v_{\mathfrak{P}}(f'(\alpha))$.

We think of L as embedded into the separable closure $\overline{K}_{\mathfrak{p}}$ of $K_{\mathfrak{p}}$ in such a way that the absolute value $|\ |$ of $\overline{K}_{\mathfrak{p}}$ defines the prime $\mathfrak{P}$.

By chap. II, (10.4), we find an element β in the valuation ring $\mathcal{O}_{\mathfrak{P}}$ of the completion $L_{\mathfrak{P}}$ satisfying $\mathcal{O}_{\mathfrak{P}} = \mathcal{o}_{\mathfrak{p}}[\beta]$, and the proof *loc. cit.* shows that, for every element $\alpha \in \mathcal{O}_{\mathfrak{P}}$ which is sufficiently close to β, one also has $\mathcal{O}_{\mathfrak{P}} = \mathcal{o}_{\mathfrak{p}}[\alpha]$. From (2.2), (iii) and (2.4), it follows that

$$v_{\mathfrak{P}}(\mathfrak{D}_{L|K}) = v_{\mathfrak{P}}(\mathfrak{D}_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}) = v_{\mathfrak{P}}\big(\delta_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}(\alpha)\big).$$

It therefore suffices to show that we can find an element α in $\mathcal{O}$ such that $L = K(\alpha)$ and

$$v_{\mathfrak{P}}\big(\delta_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}(\alpha)\big) = v_{\mathfrak{P}}\big(\delta_{L|K}(\alpha)\big).$$

For this, let $\sigma_2, \ldots, \sigma_r : L \to \overline{K}_{\mathfrak{p}}$ be K-embeddings giving the primes $\mathfrak{P}_i | \mathfrak{p}$ different from $\mathfrak{P}$. Let $a \in \mathcal{o}_{\mathfrak{p}}$ be an element such that

$$(*) \qquad |\tau\beta - a| = 1 \quad \text{for all} \quad \tau \in G_{\mathfrak{p}} = G(\overline{K}_{\mathfrak{p}}|K_{\mathfrak{p}}).$$

(Choose $a = 1$, resp. $a = 0$, according as the residue classes $\tau\beta \bmod \mathfrak{P}$ which are conjugate over $\mathcal{O}_\mathfrak{p}/\mathfrak{p}$ are zero or not.) Using the Chinese remainder theorem, we now pick an $\alpha \in \mathcal{O}$ such that $|\alpha - \beta|$ and $|\sigma_i\alpha - a|$, for $i = 2, \dots, r$, are very small. We may even assume that $L = K(\alpha)$ (if not, modify by $\alpha + \pi^\nu\gamma$, $\pi \in \mathfrak{p}$, for ν big, $\gamma \in \mathcal{O}$, $L = K(\gamma)$; for suitable $\nu \neq \mu$, one then finds $K(\alpha + \pi^\nu\gamma) = K(\alpha + \pi^\mu\gamma) = K(\gamma)$). Since α is close to β, we have $\mathcal{O}_\mathfrak{P} = \mathcal{O}_\mathfrak{p}[\alpha]$. Now

$$\delta_{L_\mathfrak{P}|K_\mathfrak{p}}(\alpha) = \prod_{\tau \neq 1} (\alpha - \tau\alpha),$$

where τ runs through the $K_\mathfrak{p}$-embeddings $L_\mathfrak{P} \to \overline{K}_\mathfrak{p}$ different from 1. Furthermore,

$$\delta_{L|K}(\alpha) = \prod_{\sigma \neq 1} (\alpha - \sigma\alpha) = \prod_{\tau \neq 1} (\alpha - \tau\alpha) \prod_{i=2}^{r} \prod_{j} (\alpha - \tau_{ij}\sigma_i\alpha),$$

where σ runs over the K-embeddings different from 1, and the τ_{ij} are certain elements in $G_\mathfrak{p}$. But now

$$|\alpha - \tau_{ij}\sigma_i\alpha| = |\tau_{ij}^{-1}\alpha - \sigma_i\alpha| = |\tau_{ij}^{-1}\alpha - a + a - \sigma_i\alpha| = 1,$$

since $|a - \sigma_i\alpha|$ is very small, and $\tau_{ij}^{-1}\alpha$ is very close to $\tau_{ij}^{-1}\beta$ (see $(*)$). Therefore $v_\mathfrak{P}(\delta_{L|K}(\alpha)) = v_\mathfrak{P}(\prod_{\tau \neq 1}(\alpha - \tau\alpha)) = v_\mathfrak{P}(\delta_{L_\mathfrak{P}|K_\mathfrak{p}}(\alpha))$, as required. □

The different characterizes the ramification behaviour of the extension $L|K$ as follows.

(2.6) Theorem. *A prime ideal $\mathfrak{P}$ of L is ramified over K if and only if $\mathfrak{P} | \mathfrak{D}_{L|K}$.*

Let $\mathfrak{P}^s$ be the maximal power of $\mathfrak{P}$ dividing $\mathfrak{D}_{L|K}$, and let e be the ramification index of $\mathfrak{P}$ over K. Then one has

$$s = e - 1, \qquad \text{if } \mathfrak{P} \text{ is tamely ramified,}$$

$$e \le s \le e - 1 + v_\mathfrak{P}(e), \qquad \text{if } \mathfrak{P} \text{ is wildly ramified.}$$

Proof: By (2.2), (iii), we may assume that $\mathcal{O}$ is a complete discrete valuation ring with maximal ideal $\mathfrak{p}$. Then, by chap. II, (10.4), we have $\mathcal{O} = \mathcal{o}[\alpha]$ for a suitable $\alpha \in \mathcal{O}$. Let $f(X)$ be the minimal polynomial of α. (2.4) says that $s = v_\mathfrak{P}(f'(\alpha))$. Assume $L|K$ is unramified. Then $\bar{\alpha} = \alpha \bmod \mathfrak{P}$ is a simple zero of $\bar{f}(X) = f(X) \bmod \mathfrak{p}$, so that $f'(\alpha) \in \mathcal{O}^*$ and thus $s = 0 = e - 1$.

By (2.2), (i) and chap. II, (7.5), we may now pass to the maximal unramified extension and assume that $L|K$ is totally ramified. Then α may be chosen to be a prime element of $\mathcal{O}$. In this case the minimal polynomial

$$f(X) = a_0X^e + a_1X^{e-1} + \cdots + a_e\,, \quad a_0 = 1\,,$$

is an Eisenstein polynomial. Let us look at the derivative

$$f'(\alpha) = ea_0\alpha^{e-1} + (e-1)a_1\alpha^{e-2} + \cdots + a_{e-1}\,.$$

For $i = 0, \dots, e-1$, we find

$$v_{\mathfrak{P}}\big((e-i)a_i\alpha^{e-i-1}\big) = ev_{\mathfrak{p}}(e-i) + ev_{\mathfrak{p}}(a_i) + e - i - 1 \equiv -i-1 \bmod e\,,$$

so that the individual terms of $f'(\alpha)$ have distinct valuations. Therefore

$$s = v_{\mathfrak{P}}\big(f'(\alpha)\big) = \min_{0\le i<e}\big\{v_{\mathfrak{P}}((e-i)a_i\alpha^{e-i-1})\big\}\,.$$

If now $L|K$ is tamely ramified, i.e., if $v_{\mathfrak{p}}(e) = 0$, then the minimum is obviously equal to $e-1$, and for $v_{\mathfrak{p}}(e) \ge 1$, we deduce that $e \le s \le v_{\mathfrak{P}}(e) + e - 1$. □

The geometric significance of the different, and thus also the way it fits into higher dimensional algebraic geometry, is brought out by the following characterization, which is due to *E. Kähler*. For an arbitrary extension $B|A$ of commutative rings, consider the homomorphism

$$\mu : B \otimes_A B \longrightarrow B\,, \quad x \otimes y \longmapsto xy\,,$$

whose kernel we denote by I. Then

$$\Omega^1_{B|A} := I/I^2 = I \otimes_{B\otimes B} B$$

is a $B \otimes B$-module, and hence in particular also a B-module, via the embedding $B \to B \otimes B$, $b \mapsto b \otimes 1$. It is called the **module of differentials** of $B|A$, and its elements are called **Kähler differentials**. If we put

$$dx = x \otimes 1 - 1 \otimes x \bmod I^2\,,$$

then we obtain a mapping

$$d : B \longrightarrow \Omega^1_{B|A}$$

satisfying

$$d(xy) = x\,dy + y\,dx\,,$$

$$da = 0 \quad \text{for} \quad a \in A\,.$$

Such a map is called a **derivation** of $B|A$. One can show that d is universal among all derivations of $B|A$ with values in B-modules. $\Omega^1_{B|A}$ consists of the linear combinations $\sum y_i\,dx_i$. The link with the different is now this.

(2.7) Proposition. *The different $\mathfrak{D}_{\mathcal{O}|\mathfrak{o}}$ is the* **annihilator** *of the $\mathcal{O}$-module $\Omega^1_{\mathcal{O}|\mathfrak{o}}$, i.e.,*

$$\mathfrak{D}_{\mathcal{O}|\mathfrak{o}} = \{x \in \mathcal{O} \mid x\,dy = 0 \quad \textit{for all} \quad y \in \mathcal{O}\}\,.$$

Proof: For greater notational clarity, let us put $\mathcal{O} = B$ and $\mathfrak{o} = A$. If A' is any commutative A-algebra and $B' = B \otimes_A A'$, then it is easy to see that $\Omega^1_{B'|A'} = \Omega^1_{B|A} \otimes_A A'$. Thus the module of differentials is preserved under localization and completion, and we may therefore assume that A is a complete discrete valuation ring. Then we find by chap. II, (10.4), that $B = A[x]$, and if $f(X) \in A[X]$ is the minimal polynomial of x, then $\Omega_{B|A}$ is generated by dx (exercise 3). The annihilator of dx is $f'(x)$. On the other hand, by (2.4) we have $\mathfrak{D}_{B|A} = (f'(x))$. This proves the claim. □

A most intimate connection holds between the different and the discriminant of $\mathcal{O}|\mathfrak{o}$. The latter is defined as follows.

(2.8) Definition. *The* **discriminant** *$\mathfrak{d}_{\mathcal{O}|\mathfrak{o}}$ is the ideal of $\mathfrak{o}$ which is generated by the discriminants $d(\alpha_1, \dots, \alpha_n)$ of all the bases $\alpha_1, \dots, \alpha_n$ of $L|K$ which are contained in $\mathcal{O}$.*

We will frequently write $\mathfrak{d}_{L|K}$ instead of $\mathfrak{d}_{\mathcal{O}|\mathfrak{o}}$. If $\alpha_1, \dots, \alpha_n$ is an integral basis of $\mathcal{O}|\mathfrak{o}$, then $\mathfrak{d}_{L|K}$ is the principal ideal generated by $d(\alpha_1, \dots, \alpha_n) = d_{L|K}$, because all other bases contained in $\mathcal{O}$ are transforms of the given one by matrices with entries in $\mathfrak{o}$. The discriminant is obtained from the different by taking the norm $N_{L|K}$ (see § 1).

(2.9) Theorem. *The following relation exists between the discriminant and the different:*

$$\mathfrak{d}_{L|K} = N_{L|K}(\mathfrak{D}_{L|K})\,.$$

Proof: If S is a multiplicative subset of $\mathfrak{o}$, then clearly $\mathfrak{d}_{S^{-1}\mathcal{O}|S^{-1}\mathfrak{o}} = S^{-1}\mathfrak{d}_{\mathcal{O}|\mathfrak{o}}$ and $\mathfrak{D}_{S^{-1}\mathcal{O}|S^{-1}\mathfrak{o}} = S^{-1}\mathfrak{D}_{\mathcal{O}|\mathfrak{o}}$. We may therefore assume that $\mathfrak{o}$ is a discrete valuation ring. Then, since $\mathfrak{o}$ is a principal ideal domain, so is $\mathcal{O}$ (see chap. I, § 3, exercise 4), and it admits an integral basis $\alpha_1, \dots, \alpha_n$ (see chap. I, (2.10)). So we have $\mathfrak{d}_{L|K} = (d(\alpha_1, \dots, \alpha_n))$. Dedekind's complementary module $\mathfrak{C}_{\mathcal{O}|\mathfrak{o}}$ is generated by the dual basis $\alpha_1', \dots, \alpha_n'$ which is characterized by $Tr_{L|K}(\alpha_i\alpha_j') = \delta_{ij}$. On the other hand, $\mathfrak{C}_{\mathcal{O}|\mathfrak{o}}$ is a principal ideal (β) and admits the $\mathfrak{o}$-basis $\beta\alpha_1, \dots, \beta\alpha_n$ of discriminant

$$d(\beta\alpha_1, \dots, \beta\alpha_n) = N_{L|K}(\beta)^2 d(\alpha_1, \dots, \alpha_n)\,.$$

But $(N_{L|K}(\beta)) = N_{L|K}(\mathfrak{C}_{\mathcal{O}|o}) = N_{L|K}(\mathfrak{D}_{L|K}^{-1}) = N_{L|K}(\mathfrak{D}_{L|K})^{-1}$, and $(d(\alpha_1, \dots, \alpha_n)) = \mathfrak{d}_{L|K}$. One has $d(\alpha_1, \dots, \alpha_n) = \det((\sigma_i\alpha_j))^2$, $d(\alpha_1', \dots, \alpha_n') = \det((\sigma_i\alpha_j'))^2$, for $\sigma_i \in \operatorname{Hom}_K(L, \overline{K})$, and $Tr(\alpha_i\alpha_j') = \delta_{ij}$. Then $d(\alpha_1, \dots, \alpha_n) \cdot d(\alpha_1', \dots, \alpha_n') = 1$. Combining these yields

$$\begin{aligned}\mathfrak{d}_{L|K}^{-1} &= (d(\alpha_1, \dots, \alpha_n)^{-1}) = (d(\alpha_1', \dots, \alpha_n')) = (d(\beta\alpha_1, \dots, \beta\alpha_n)) \\ &= N_{L|K}(\mathfrak{D}_{L|K})^{-2}\mathfrak{d}_{L|K}\end{aligned}$$

and hence $N_{L|K}(\mathfrak{D}_{L|K}) = \mathfrak{d}_{L|K}$. □

(2.10) Corollary. *For a tower of fields $K \subseteq L \subseteq M$, one has*

$$\mathfrak{d}_{M|K} = \mathfrak{d}_{L|K}^{[M:L]} N_{L|K}(\mathfrak{d}_{M|L}).$$

Proof: Applying to $\mathfrak{D}_{M|K} = \mathfrak{D}_{M|L}\mathfrak{D}_{L|K}$ the norm $N_{M|K} = N_{L|K} \circ N_{M|L}$, (1.6) gives

$$\mathfrak{d}_{M|K} = N_{L|K}(\mathfrak{d}_{M|L})N_{L|K}(\mathfrak{D}_{L|K}^{[M:L]}) = N_{L|K}(\mathfrak{d}_{M|L})\mathfrak{d}_{L|K}^{[M:L]}. \qquad \square$$

Putting $\mathfrak{d} = \mathfrak{d}_{L|K}$ and $\mathfrak{d}_{\mathfrak{P}} = \mathfrak{d}_{L_{\mathfrak{P}}|K_{\mathfrak{p}}}$ and viewing $\mathfrak{d}_{\mathfrak{P}}$ also as the ideal $\mathfrak{d}_{\mathfrak{P}} \cap o$ of K, the product formula (2.3) for the different, together with theorem (2.9), yields:

(2.11) Corollary. $\mathfrak{d} = \prod_{\mathfrak{P}} \mathfrak{d}_{\mathfrak{P}}$.

The extension $L|K$ is called **unramified** if all prime ideals $\mathfrak{p}$ of K are unramified. This amounts to requiring that all primes of K be unramified. In fact, the infinite primes are always to be regarded as unramified because $e_{\mathfrak{P}|\mathfrak{p}} = 1$.

(2.12) Corollary. *A prime ideal $\mathfrak{p}$ of K is ramified in L if and only if $\mathfrak{p}|\mathfrak{d}$. In particular, the extension $L|K$ is unramified if the discriminant $\mathfrak{d} = (1)$.*

Combining this result with Minkowski theory leads to two important theorems on unramified extensions of number fields which belong to the classical body of algebraic number theory. The first of these results is the following.

(2.13) Theorem. *Let K be an algebraic number field and let S be a finite set of primes of K. Then there exist only finitely many extensions $L|K$ of given degree n which are unramified outside of S.*

Proof: If $L|K$ is an extension of degree n which is unramified outside of S, then, by (2.12) and (2.6), its discriminant $\mathfrak{d}_{L|K}$ is one of the finite number of divisors of the ideal $\mathfrak{a} = \prod_{\substack{\mathfrak{p}\in S \\ \mathfrak{p}\nmid\infty}} \mathfrak{p}^{n(1+n)}$. It therefore suffices to show that there are only finitely many extensions $L|K$ of degree n with given discriminant. We may assume without loss of generality that $K = \mathbb{Q}$. For if $L|K$ is an extension of degree n with discriminant $\mathfrak{d}$, then $L|\mathbb{Q}$ is an extension of degree $m = n[K:\mathbb{Q}]$ with discriminant $(d) = \mathfrak{d}_{K|\mathbb{Q}}^n N_{K|\mathbb{Q}}(\mathfrak{d})$. Finally, the discriminant of $L(\sqrt{-1})|\mathbb{Q}$ differs from the discriminant of $L|\mathbb{Q}$ only by a constant factor. So we are reduced to proving that there exist only finitely many fields $K|\mathbb{Q}$ of degree n containing $\sqrt{-1}$ with a given discriminant d. Such a field K has only complex embeddings $\tau : K \to \mathbb{C}$. Choose one of them: τ_0. In the Minkowski space

$$K_{\mathbb{R}} = \Big[\prod_{\tau}\mathbb{C}\Big]^+$$

(see chap. I, §5) consider the convex, centrally symmetric subset

$$X = \Big\{ (z_\tau) \in K_{\mathbb{R}} \;\Big|\; |\mathrm{Im}(z_{\tau_0})| < C\sqrt{|d|},\\ |\mathrm{Re}(z_{\tau_0})| < 1,\; |z_\tau| < 1 \text{ for } \tau \neq \tau_0, \bar{\tau}_0 \Big\},$$

where C is an arbitrarily big constant which depends only on n. For a convenient choice of C, the volume will satisfy

$$\mathrm{vol}(X) > 2^n\sqrt{|d|} = 2^n \,\mathrm{vol}(\mathcal{O}_K),$$

where $\mathrm{vol}(\mathcal{O}_K)$ is the volume of a fundamental mesh of the lattice $j\mathcal{O}_K$ in $K_{\mathbb{R}}$ – see chap. I, (5.2). By Minkowski's lattice point theorem (chap. I, (4.4)), we thus find $\alpha \in \mathcal{O}_K$, $\alpha \neq 0$, such that $j\alpha = (\tau\alpha) \in X$, that is,

$$(*) \quad \big|\mathrm{Im}(\tau_0\alpha)\big| < C\sqrt{|d|},\; \big|\mathrm{Re}(\tau_0\alpha)\big| < 1, \quad |\tau\alpha| < 1 \quad \text{for} \quad \tau \neq \tau_0, \bar{\tau}_0.$$

This α is a primitive element of K, i.e., one has $K = \mathbb{Q}(\alpha)$. Indeed, $|N_{K|\mathbb{Q}}(\alpha)| = \prod_\tau |\tau\alpha| \geq 1$ implies $|\tau_0\alpha| > 1$; thus $\mathrm{Im}(\tau_0\alpha) \neq 0$ so that the conjugates $\tau_0\alpha$ and $\bar{\tau}_0\alpha$ of α have to be distinct. Since $|\tau\alpha| < 1$ for $\tau \neq \tau_0, \bar{\tau}_0$, one has $\tau_0\alpha \neq \tau\alpha$ for all $\tau \neq \tau_0$. This implies $K = \mathbb{Q}(\alpha)$, because if $\mathbb{Q}(\alpha) \subsetneqq K$ then the restriction $\tau_0|_{\mathbb{Q}(\alpha)}$ would admit an extension τ different from τ_0, contradicting $\tau_0\alpha \neq \tau\alpha$.

Since the conjugates $\tau\alpha$ of α are subject to the conditions $(*)$, which only depend on d and n, the coefficients of the minimal polynomial of α

are bounded once d and n are fixed. Thus every field $K|\mathbb{Q}$ of degree n with discriminant d is generated by one of the finitely many lattice points α in the bounded region X. Therefore there are only finitely many fields with given degree and discriminant. □

The second theorem alluded to above is in fact a strengthening of the first. It follows from the following *bound on the discriminant.*

(2.14) Proposition. *The discriminant of an algebraic number field K of degree n satisfies*

$$|d_K|^{1/2} \geq \frac{n^n}{n!}\Big(\frac{\pi}{4}\Big)^{n/2}.$$

Proof: In Minkowski space $K_{\mathbb{R}} = \big[\prod_\tau \mathbb{C}\big]^+$, $\tau \in \operatorname{Hom}(K,\mathbb{C})$, consider the convex, centrally symmetric subset

$$X_t = \Big\{ (z_\tau) \in K_{\mathbb{R}} \;\Big|\; \sum_\tau |z_\tau| \leq t \Big\}.$$

Its volume is

$$\operatorname{vol}(X_t) = 2^r \pi^s \frac{t^n}{n!}\,.$$

Leaving aside the proof of this formula for the moment (which incidentally was exercise 2 of chap. I, § 5), we deduce the proposition from Minkowski's lattice point theorem (chap. I, (4.4)) as follows. Consider in $K_{\mathbb{R}}$ the lattice $\Gamma = j\mathcal{O}$ defined by $\mathcal{O}$. By chap. I, (5.2), the volume of a fundamental mesh is $\operatorname{vol}(\Gamma) = \sqrt{|d_K|}$. The inequality

$$\operatorname{vol}(X_t) > 2^n \operatorname{vol}(\Gamma)$$

therefore holds if and only if $2^r \pi^s \frac{t^n}{n!} > 2^n \sqrt{|d_K|}$, or equivalently if

$$t^n = n!\Big(\frac{4}{\pi}\Big)^s \sqrt{|d_K|} + \varepsilon,$$

for some $\varepsilon > 0$. If this is the case, there exists an $a \in \mathcal{O}$, $a \neq 0$, such that $ja \in X_t$. As this holds for all $\varepsilon > 0$, and since X_t contains only finitely many lattice points, it continues to hold for $\varepsilon = 0$. Applying now the inequality between arithmetic and geometric means,

$$\frac{1}{n}\sum_\tau |z_\tau| \geq \Big(\prod_\tau |z_\tau|\Big)^{1/n},$$

we obtain the desired result:

$$\begin{aligned} 1 \leq \big|N_{K|\mathbb{Q}}(a)\big| &= \prod_\tau |\tau a| \leq \frac{1}{n^n}\Big(\sum_\tau |\tau a|\Big)^n < \frac{t^n}{n^n} = \frac{n!}{n^n}\Big(\frac{4}{\pi}\Big)^s \sqrt{|d_K|} \\ &\leq \frac{n!}{n^n}\Big(\frac{4}{\pi}\Big)^{n/2} \sqrt{|d_K|}. \end{aligned}$$

Given this, it remains to prove the following lemma. □

(2.15) Lemma. *In Minkowski space* $K_{\mathbb{R}} = \left[\prod_\tau \mathbb{C}\right]^+$, *the domain*

$$X_t = \Big\{ (z_\tau) \in K_{\mathbb{R}} \;\Big|\; \sum_\tau |z_\tau| < t \Big\}$$

has volume

$$\operatorname{vol}(X_t) = 2^r \pi^s \frac{t^n}{n!} \cdot$$

Proof: $\operatorname{vol}(X_t)$ is 2^s times the Lebesgue volume $\operatorname{Vol}(f(X_t))$ of the image $f(X_t)$ under the mapping chap. I, (5.1),

$$f : K_{\mathbb{R}} \longrightarrow \prod_\tau \mathbb{R}, \quad (z_\tau) \longmapsto (x_\tau),$$

where $x_\rho = z_\rho$, $x_\sigma = \operatorname{Re}(z_\sigma)$, $x_{\bar\sigma} = \operatorname{Im}(z_\sigma)$. Substituting x_i, $i = 1, \dots, r$, instead of x_ρ, and y_j, z_j, $j = 1, \dots, s$, instead of $x_\sigma, x_{\bar\sigma}$, we see that $f(X_t)$ is described by the inequality

$$|x_1| + \cdots + |x_r| + 2\sqrt{y_1^2 + z_1^2} + \cdots + 2\sqrt{y_s^2 + z_s^2} < t.$$

The factor 2 occurs because $|z_{\bar\sigma}| = |\bar z_\sigma| = |z_\sigma|$. Passing to polar coordinates $y_j = u_j \cos\theta_j$, $z_j = u_j \sin\theta_j$, where $0 \le \theta_j \le 2\pi$, $0 \le u_j$, one sees that $\operatorname{Vol}(f(X_t))$ is computed by the integral

$$I(t) = \int u_1 \cdots u_s \, dx_1 \cdots dx_r \, du_1 \cdots du_s \, d\theta_1 \cdots d\theta_s$$

extended over the domain

$$|x_1| + \cdots + |x_r| + 2u_1 + \cdots + 2u_s \le t.$$

Restricting this domain of integration to $x_i \ge 0$, the integral gets divided by 2^r. Substituting $2u_j = w_j$ gives

$$I(t) = 2^r 4^{-s} (2\pi)^s I_{r,s}(t),$$

where the integral

$$I_{r,s}(t) = \int w_1 \cdots w_s \, dx_1 \cdots dx_r \, dw_1 \cdots dw_s$$

has to be taken over the domain $x_i \ge 0$, $w_j \ge 0$ and

$$(*) \qquad x_1 + \cdots + x_r + w_1 + \cdots + w_s \le t.$$

Clearly $I_{r,s}(t) = t^{r+2s} I_{r,s}(1) = t^n I_{r,s}(1)$. Writing $x_2 + \cdots + x_r + w_1 + \cdots + w_s \le t - x_1$ instead of $(*)$, Fubini's theorem yields

$$\begin{aligned} I_{r,s}(1) &= \int_0^1 I_{r-1,s}(1 - x_1)\, dx_1 = \int_0^1 (1 - x_1)^{n-1} dx_1 \cdot I_{r-1,s}(1) \\ &= \frac{1}{n} I_{r-1,s}(1). \end{aligned}$$

By induction, this implies that

$$I_{r,s}(1) = \frac{1}{n(n-1)\cdots(n-r+1)} I_{0,s}(1).$$

In the same way, one gets

$$I_{0,s}(1) = \int_0^1 w_1(1-w_1)^{2s-2}dw_1 I_{0,s-1}(1),$$

and, doing the integration, induction shows that

$$I_{0,s}(1) = \frac{1}{(2s)!} I_{0,0}(1) = \frac{1}{(2s)!}.$$

Together, this gives $I_{r,s}(1) = \frac{1}{n!}$ and therefore indeed

$$\mathrm{vol}(X_t) = 2^s \mathrm{Vol}\big(f(X_t)\big) = 2^s 2^r 4^{-s}(2\pi)^s t^n I_{r,s}(1) = \frac{2^r \pi^s}{n!} t^n. \qquad \square$$

If we combine Stirling's formula,

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\frac{\theta}{12n}}, \quad 0 < \theta < 1,$$

with the inequality (2.14), we obtain the inequality

$$|d_K| > \left(\frac{\pi}{4}\right)^{2s} \frac{1}{2\pi n} e^{2n - \frac{1}{6n}}.$$

This shows that the absolute value of the discriminant of an algebraic number field tends to infinity with the degree. In the proof of (2.13) we saw that there are only finitely many number fields with bounded degree and discriminant. So now, since the degree is bounded if the discriminant is, we may strengthen (2.13), obtaining

(2.16) Hermite's Theorem. *There exist only finitely many number fields with bounded discriminant.*

The expression $a_n = \frac{n^n}{n!}\left(\frac{\pi}{4}\right)^{n/2}$ satisfies

$$\frac{a_{n+1}}{a_n} = \left(\frac{\pi}{4}\right)^{1/2}\left(1+\frac{1}{n}\right)^n > 1,$$

i.e., $a_{n+1} > a_n$. Since $a_2 = \frac{\pi}{2} > 1$, (2.14) yields

(2.17) Minkowski's Theorem. *The discriminant of a number field K different from $\mathbb{Q}$ is $\neq \pm 1$.*

Combining this result with corollary (2.12), we obtain the

(2.18) Theorem. *The field $\mathbb{Q}$ does not admit any unramified extensions.*

These last theorems are of fundamental importance for number theory. Their significance is seen especially clearly in the light of higher dimensional analogues. For instance, let us replace the finite field extensions $L|K$ of a number field K by all smooth complete (i.e., proper) algebraic curves defined over K of a fixed genus g. If $\mathfrak{p}$ is a prime ideal of K, then for any such curve X, one may define the "reduction mod $\mathfrak{p}$". This is a curve defined over the residue class field of $\mathfrak{p}$. One says that X has *good reduction* at the prime $\mathfrak{p}$ if its reduction $\bmod \mathfrak{p}$ is again a smooth curve. This corresponds to an extension $L|K$ being unramified. In analogy to Hermite's theorem, the Russian mathematician *I.S. Šafarevič* formulated the conjecture that there exist only finitely many smooth complete curves of genus g over K with good reduction outside a fixed finite set of primes S. This conjecture was proved in 1983 by the mathematician *Gerd Faltings* (see [35]). The impact of this result can be gauged by the non-expert from the fact that it was the basis for *Faltings*'s proof of the famous **Mordell Conjecture**:

Every algebraic equation

$$f(x,y) = 0$$

of genus $g > 1$ with coefficients in K admits only **finitely many solutions** in K.

A 1-dimensional analogue of Minkowski's theorem (2.18) was proved in 1985 by the French mathematician *J.-M. Fontaine*: over the field $\mathbb{Q}$, there are no smooth proper curves with good reduction $\bmod p$ for all prime numbers p (see [39]).

Exercise 1. Let $d(\alpha) = d(1, \alpha, \dots, \alpha^{n-1})$, for an element $\alpha \in \mathcal{O}$ such that $L = K(\alpha)$. Show that $\mathfrak{d}_{L|K}$ is generated by all discriminants of elements $d(\alpha)$ if o is a complete discrete valuation ring and the residue field extension $\lambda|\kappa$ is separable. In other words, $\mathfrak{d}_{L|K}$ equals the gcd of all discriminants of individual elements. This fails to hold in general. Counterexample: $K = \mathbb{Q}$, $L = \mathbb{Q}(\alpha)$, $\alpha^3 - \alpha^2 - 2\alpha - 8 = 0$. (See [60], chap. III, §25, p. 443. The untranslatable German catch phrase for this phenomenon is: there are *"außerwesentliche Diskriminantenteiler"*.)

Exercise 2. Let $L|K$ be a Galois extension of henselian fields with separable residue field extension $\lambda|\kappa$, and let G_i, $i \geq 0$, be the i-th ramification group. Then, if $\mathfrak{D}_{L|K} = \mathfrak{P}^s$, one has

$$s = \sum_{i=0}^{\infty}(\#G_i - 1).$$

Hint: If $\mathcal{O} = o[x]$ (see chap. II, (10.4)), then $s = v_L(\delta_{L|K}(x)) = \sum_{\substack{\sigma\in G\\ \sigma\neq 1}} v_L(x - \sigma x)$.

Exercise 3. The module of differentials $\Omega^1_{\mathcal{O}|o}$ is generated by a single element dx, $x \in \mathcal{O}$, and there is an exact sequence of $\mathcal{O}$-modules

$$0 \to \mathfrak{D}_{\mathcal{O}|o} \to \mathcal{O} \to \Omega^1_{\mathcal{O}|o} \to 0.$$

Exercise 4. For a tower $M \supseteq L \supseteq K$ of algebraic number fields there is an exact sequence of o_M-modules

$$0 \to \Omega^1_{L|K} \otimes o_M \to \Omega^1_{M|K} \to \Omega^1_{M|L} \to 0.$$

Exercise 5. If ζ is a primitive p^n-th root of unity, then

$$\mathfrak{D}_{\mathbb{Q}_p(\zeta)|\mathbb{Q}_p} = \mathfrak{P}^{p^{n-1}(pn-n-1)}.$$

§ 3. Riemann-Roch

The notion of replete divisor introduced into our development of number theory in § 1 is a terminology reminiscent of the function-theoretic model. We now have to ask the question to what extent this point of view does justice to our goal to also couch the number-theoretic *results* in a geometric function-theoretic fashion, and conversely to give arithmetic significance to the classical theorems of function theory. Among the latter, the Riemann-Roch theorem stands out as the most important representative. If number theory is to proceed in a geometric manner, it must work towards finding an adequate way to incorporate this result as well. This is the task we are now going to tackle.

First recall the classical situation in function theory. There the basic data is a compact Riemann surface X with the sheaf o_X of holomorphic functions. To each divisor $D = \sum_{P\in X} \nu_P P$ on X corresponds a **line bundle** $o(D)$, i.e., an o_X-module which is locally free of rank 1. If U is an open subset of X and $K(U)$ is the ring of meromorphic functions on U, then the vector space $o(D)(U)$ of sections of the sheaf $o(D)$ over U is given as

$$o(D)(U) = \big\{ f \in K(U) \bigm| \operatorname{ord}_P(f) \geq -\nu_P \text{ for all } P \in U \big\}.$$

The Riemann-Roch problem is to calculate the dimension

$$\ell(D) = \dim H^0\big(X, o(D)\big)$$

of the vector space of global sections

$$H^0\big(X, \mathcal{O}(D)\big) = \mathcal{O}(D)(X).$$

In its first version the Riemann-Roch theorem does not provide a formula for $H^0(X, \mathcal{O}(D))$ itself, but for the **Euler-Poincaré characteristic**

$$\chi\big(\mathcal{O}(D)\big) = \dim H^0\big(X, \mathcal{O}(D)\big) - \dim H^1\big(X, \mathcal{O}(D)\big).$$

The formula reads

$$\chi\big(\mathcal{O}(D)\big) = \deg(D) + 1 - g,$$

where g is the **genus** of X. For the divisor $D = 0$, one has $\mathcal{O}(D) = \mathcal{O}_X$ and $\deg(D) = 0$, so that $\chi(\mathcal{O}_X) = 1 - g$; then this equation may also be replaced by

$$\chi(\mathcal{O}(D)) = \deg(D) + \chi(\mathcal{O}_X).$$

The classical Riemann-Roch formula

$$\ell(D) - \ell(\mathcal{K} - D) = \deg(D) + 1 - g$$

is then obtained by using SERRE duality, which states that $H^1(X, \mathcal{O}(D))$ is dual to $H^0(X, \omega \otimes \mathcal{O}(-D))$, where $\omega = \Omega^1_X$ is the so-called **canonical module** of X, and $\mathcal{K} = \operatorname{div}(\omega)$ is the associated divisor (see for instance [51], chap. III, 7.12.1 and chap. IV, 1.1.3).

In order to mimic this state of affairs in number theory, let us recall the explanations of chap. I, §14 and chap. III, §1. We endow the places $\mathfrak{p}$ of an algebraic number field K with the rôle of points of a space X which should be conceived of as the analogue of a *compact* Riemann surface. The elements $f \in K^*$ will be given the rôle of "meromorphic functions" on this space X. The order of the pole, resp. zero of f at the point $\mathfrak{p} \in X$, for $\mathfrak{p} \nmid \infty$, is defined to be the integer $v_\mathfrak{p}(f)$, and for $\mathfrak{p}|\infty$ it is the real number $v_\mathfrak{p}(f) = -\log|\tau f|$. In this way we associate to each $f \in K^*$ the replete divisor

$$\operatorname{div}(f) = \sum_{\mathfrak{p}} v_\mathfrak{p}(f)\mathfrak{p} \in Div(\bar{\mathcal{O}}).$$

More precisely, for a given divisor $D = \sum_\mathfrak{p} \nu_\mathfrak{p}\mathfrak{p}$, we are interested in the replete ideal

$$\mathcal{O}(D) = \prod_{\mathfrak{p}} \mathfrak{p}^{-\nu_\mathfrak{p}}$$

and the set

$$\begin{aligned} H^0(\mathcal{O}(D)) &= \big\{ f \in K^* \,\big|\, \operatorname{div}(f) \geq -D \big\} \\ &= \big\{ f \in \mathcal{O}(D)_\mathfrak{f} \,\big|\, 0 \neq |f|_\mathfrak{p} \leq \mathfrak{N}(\mathfrak{p})^{\nu_\mathfrak{p}} \text{ for } \mathfrak{p}|\infty \big\}, \end{aligned}$$

where the relation $D' \geq D$ between divisors $D' = \sum_{\mathfrak{p}} \nu'_{\mathfrak{p}}\mathfrak{p}$ and $D = \sum_{\mathfrak{p}} \nu_{\mathfrak{p}}\mathfrak{p}$ is simply defined to mean $\nu'_{\mathfrak{p}} \geq \nu_{\mathfrak{p}}$ for all $\mathfrak{p}$. Note that $H^0(\mathcal{O}(D))$ is no longer a vector space. An analogue of $H^1(X, \mathcal{O}(D))$ is completely missing. Instead of attacking directly the problem of measuring the size of $H^0(\mathcal{O}(D))$, we proceed as in the function-theoretic model by looking at the "Euler-Poincaré characteristic" of the replete ideal $\mathcal{O}(D)$. Before defining this, we want to establish the relation between the Minkowski space $K_{\mathbb{R}} = \left[\prod_\tau \mathbb{C}\right]^+$, $\tau \in \operatorname{Hom}(K, \mathbb{C})$, and the product $\prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}$. The reader will allow us to explain this simple situation in the following sketch.

We have the correspondences

$$\begin{array}{lcll} \rho\colon K \to \mathbb{R} & \longmapsto & \mathfrak{p} & \text{real prime}, \ \rho = \rho_{\mathfrak{p}} : K_{\mathfrak{p}} \xrightarrow{\sim} \mathbb{R}, \\[2ex] \sigma, \overline{\sigma}\colon K \to \mathbb{C} & \longmapsto & \mathfrak{p} & \text{complex prime}, \ \sigma = \sigma_{\mathfrak{p}} : K_{\mathfrak{p}} \xrightarrow{\sim} \mathbb{C}. \end{array}$$

There are the following isomorphisms

$$\begin{array}{ll} K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} K_{\mathbb{R}}, & a \otimes x \longmapsto \big((\tau a)x\big)_\tau, \\[2ex] K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} \prod\limits_{\mathfrak{p}|\infty} K_{\mathfrak{p}}, & a \otimes x \longmapsto \big((\tau_{\mathfrak{p}} a)x\big)_{\mathfrak{p}|\infty}, \end{array}$$

$\tau_{\mathfrak{p}}$ being the canonical embedding $K \to K_{\mathfrak{p}}$ (see chap. II, (8.3)). They fit into the commutative diagram

$$\begin{array}{ccccccc} K \otimes \mathbb{R} & \cong & K_{\mathbb{R}} & = & \prod\limits_{\rho} \mathbb{R} & \times & \prod\limits_{\sigma} [\mathbb{C} \times \mathbb{C}]^+ \\ \Big\| & & \wr\Big\uparrow & & \wr\Big\uparrow \Pi\rho_{\mathfrak{p}} & & \wr\Big\uparrow \Pi[\sigma_{\mathfrak{p}} \times \overline{\sigma}_{\mathfrak{p}}] \\ K \otimes \mathbb{R} & \cong & \prod\limits_{\mathfrak{p}|\infty} K_{\mathfrak{p}} & = & \prod\limits_{\mathfrak{p}\text{ real}} K_{\mathfrak{p}} & \times & \prod\limits_{\mathfrak{p}\text{ complex}} K_{\mathfrak{p}}, \end{array}$$

where the arrow on the right is given by $a \mapsto (\sigma a, \overline{\sigma} a)$. Via this isomorphism, we identify $K_{\mathbb{R}}$ with $\prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}$:

$$K_{\mathbb{R}} = \prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}.$$

The scalar product $\langle x, y\rangle = \sum_\tau x_\tau \overline{y}_\tau$ on $K_{\mathbb{R}}$ is then transformed into

$$\langle x, y\rangle = \sum_{\mathfrak{p}\text{ real}} x_{\mathfrak{p}} y_{\mathfrak{p}} + \sum_{\mathfrak{p}\text{ complex}} (x_{\mathfrak{p}}\overline{y}_{\mathfrak{p}} + \overline{x}_{\mathfrak{p}} y_{\mathfrak{p}}).$$

The Haar measure μ on $K_{\mathbb{R}}$ which is determined by $\langle x, y\rangle$ becomes the product measure

$$\mu = \prod_{\mathfrak{p}|\infty} \mu_{\mathfrak{p}},$$

where

$$\mu_{\mathfrak{p}} = \text{Lebesgue measure on } K_{\mathfrak{p}} = \mathbb{R}, \text{ if } \mathfrak{p} \text{ real},$$
$$\mu_{\mathfrak{p}} = 2\,\text{Lebesgue measure on } K_{\mathfrak{p}} = \mathbb{C}, \text{ if } \mathfrak{p} \text{ complex}.$$

Indeed, the system $1/\sqrt{2}$, $i/\sqrt{2}$ is an orthonormal basis with respect to the scalar product $x\bar{y} + \bar{x}y$ on $K_{\mathfrak{p}} = \mathbb{C}$. Hence the square $Q = \{z = x + iy \mid 0 \le x, y \le 1/\sqrt{2}\}$ has volume $\mu_{\mathfrak{p}}(Q) = 1$, but Lebesgue volume 1/2.

Finally, the logarithm map

$$\ell : \big[\prod_\tau \mathbb{C}^*\big]^+ \longrightarrow \big[\prod_\tau \mathbb{R}\big]^+, \quad x \longmapsto \big(\log|x_\tau|\big)_\tau$$

studied in Minkowski theory is transformed into the mapping

$$\ell : \prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}^* \longrightarrow \prod_{\mathfrak{p}|\infty} \mathbb{R}, \quad x \longmapsto \big(\log|x_{\mathfrak{p}}|_{\mathfrak{p}}\big),$$

for one has the commutative diagram

$$\begin{array}{ccc}
K_{\mathbb{R}}^* & \xrightarrow{\ \ell\ } & \big[\prod_\tau \mathbb{R}\big]^+ \\
\downarrow & & \downarrow \\
\prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}^* & \xrightarrow{\ \ell\ } & \prod_{\mathfrak{p}|\infty} \mathbb{R},
\end{array}$$

where the arrow on the right,

$$\big[\prod_\tau \mathbb{R}\big]^+ = \prod_\rho \mathbb{R} \times \prod_\sigma [\mathbb{R} \times \mathbb{R}]^+ \longrightarrow \prod_{\mathfrak{p}|\infty} \mathbb{R},$$

is defined by $x \mapsto x$ for $\rho \leftrightarrow \mathfrak{p}$, and by $(x,x) \mapsto 2x$ for $\sigma \leftrightarrow \mathfrak{p}$. This isomorphism takes the trace map $x \mapsto \sum_\tau x_\tau$ on $\big[\prod_\tau \mathbb{R}\big]^+$ into the trace map $x \mapsto \sum_{\mathfrak{p}|\infty} x_{\mathfrak{p}}$ on $\prod_{\mathfrak{p}|\infty} \mathbb{R}$, and hence the trace-zero space $H = \big\{x \in \big[\prod_\tau \mathbb{R}\big]^+ \mid \sum_\tau x_\tau = 0\big\}$ into the trace-zero space

$$H = \Big\{x \in \prod_{\mathfrak{p}|\infty} \mathbb{R} \;\Big|\; \sum_{\mathfrak{p}|\infty} x_{\mathfrak{p}} = 0\Big\}.$$

In this way we have translated all necessary invariants of the Minkowski space $K_{\mathbb{R}}$ to the product $\prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}$.

To a given replete ideal

$$\mathfrak{a} = \mathfrak{a}_f \cdot \mathfrak{a}_\infty = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{\nu_{\mathfrak{p}}} \times \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_{\mathfrak{p}}}$$

we now associate the following complete lattice $j\mathfrak{a}$ in $K_{\mathbb{R}}$. The fractional ideal $\mathfrak{a}_f \subseteq K$ is mapped by the embedding $j : K \to K_{\mathbb{R}}$ onto a complete lattice $j\mathfrak{a}_f$ of $K_{\mathbb{R}} = \prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}$. By componentwise multiplication, $\mathfrak{a}_\infty = \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_\mathfrak{p}} = (\dots, e^{\nu_\mathfrak{p}}, \dots)_{\mathfrak{p}|\infty}$ yields an isomorphism

$$\mathfrak{a}_\infty : K_{\mathbb{R}} \to K_{\mathbb{R}}, \quad (x_\mathfrak{p})_{\mathfrak{p}|\infty} \mapsto (e^{\nu_\mathfrak{p}} x_\mathfrak{p})_{\mathfrak{p}|\infty},$$

with determinant

$$(*) \qquad \det(\mathfrak{a}_\infty) = \prod_{\mathfrak{p}|\infty} e^{\nu_\mathfrak{p} f_\mathfrak{p}} = \prod_{\mathfrak{p}|\infty} \mathfrak{N}(\mathfrak{p})^{\nu_\mathfrak{p}} = \mathfrak{N}(\mathfrak{a}_\infty).$$

The image of the lattice $j\mathfrak{a}_f$ under this map is a complete lattice

$$j\mathfrak{a} := \mathfrak{a}_\infty j\mathfrak{a}_f.$$

Let $\mathrm{vol}(\mathfrak{a})$ denote the **volume of a fundamental mesh** of $j\mathfrak{a}$ with respect to the canonical measure. By $(*)$, we then have

$$\mathrm{vol}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{a}_\infty)\,\mathrm{vol}(\mathfrak{a}_f).$$

(3.1) Definition. *If $\mathfrak{a}$ is a replete ideal of K, then the real number*

$$\chi(\mathfrak{a}) = -\log \mathrm{vol}(\mathfrak{a})$$

will be called the **Euler-Minkowski characteristic** *of $\mathfrak{a}$.*

The reason for this terminology will become clear in § 8.

(3.2) Proposition. *The Euler-Minkowski characteristic $\chi(\mathfrak{a})$ only depends on the class of $\mathfrak{a}$ in $Pic(\bar{\mathcal{O}}) = J(\bar{\mathcal{O}})/P(\bar{\mathcal{O}})$.*

Proof: Let $[a] = [a]_f \cdot [a]_\infty = (a) \times [a]_\infty$ be a replete principal ideal. Then one has

$$[a]\mathfrak{a} = a\mathfrak{a}_f \times [a]_\infty \mathfrak{a}_\infty.$$

The lattice $j(a\mathfrak{a}_f)$ is the image of the lattice $j\mathfrak{a}_f$ under the linear map $ja : K_{\mathbb{R}} \to K_{\mathbb{R}}$, $(x_\mathfrak{p})_{\mathfrak{p}|\infty} \mapsto (ax_\mathfrak{p})_{\mathfrak{p}|\infty}$. The absolute value of the determinant of this mapping is obviously given by

$$\left|\det(ja)\right| = \prod_{\mathfrak{p}|\infty} |a|_\mathfrak{p} = \prod_{\mathfrak{p}|\infty} \mathfrak{N}(\mathfrak{p})^{-v_\mathfrak{p}(a)} = \mathfrak{N}([a]_\infty)^{-1}.$$

For the canonical measure, we therefore have

$$\mathrm{vol}(a\mathfrak{a}_f) = \mathfrak{N}([a]_\infty)^{-1}\,\mathrm{vol}(\mathfrak{a}_f).$$

Taken together with $(*)$, this yields

$$\mathrm{vol}([a]\mathfrak{a}) = \mathfrak{N}([a]_\infty \mathfrak{a}_\infty)\,\mathrm{vol}(a\mathfrak{a}_f) = \mathfrak{N}(\mathfrak{a}_\infty)\,\mathrm{vol}(\mathfrak{a}_f) = \mathrm{vol}(\mathfrak{a}),$$

so that $\chi([a]\mathfrak{a}) = \chi(\mathfrak{a})$. □

The explicit evaluation of the Euler-Minkowski characteristic results from a result of Minkowski theory, *viz.*, proposition (5.2) of chap. I.

(3.3) Proposition. *For every replete ideal $\mathfrak{a}$ of K one has*

$$\mathrm{vol}(\mathfrak{a}) = \sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}).$$

Proof: Multiplying by a replete principal ideal $[a]$ we may assume, as $\mathrm{vol}([a]\mathfrak{a}) = \mathrm{vol}(\mathfrak{a})$ and $\mathfrak{N}([a]\mathfrak{a}) = \mathfrak{N}(\mathfrak{a})$, that $\mathfrak{a}_f$ is an integral ideal of K. By chap. I, (5.2) the volume of a fundamental mesh of $\mathfrak{a}_f$ is given by

$$\mathrm{vol}(\mathfrak{a}_f) = \sqrt{|d_K|}(\mathcal{O} : \mathfrak{a}_f).$$

Hence

$$\mathrm{vol}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{a}_\infty)\,\mathrm{vol}(\mathfrak{a}_f) = \mathfrak{N}(\mathfrak{a}_\infty)\sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}_f) = \sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}). \qquad \square$$

In view of the commutative diagram in §1, p. 192, we will now introduce the **degree** of the replete ideal $\mathfrak{a}$ to be the real number

$$\deg(\mathfrak{a}) = -\log \mathfrak{N}(\mathfrak{a}) = \deg\big(\mathrm{div}(\mathfrak{a})\big).$$

Observing that

$$\chi(\mathcal{O}) = -\log\sqrt{|d_K|},$$

we deduce from proposition (3.3) the first version of the Riemann-Roch theorem:

(3.4) Proposition. *For every replete ideal of K we have the formula*

$$\chi(\mathfrak{a}) = \deg(\mathfrak{a}) + \chi(\mathcal{O}).$$

In function theory there is the following relationship between the Euler-Poincaré characteristic and the genus g of the Riemann surface X in question:

$$\chi(\mathcal{O}) = \dim H^0(X, \mathcal{O}_X) - \dim H^1(X, \mathcal{O}_X) = 1 - g.$$

There is no immediate analogue of $H^1(X, \mathcal{O}_X)$ in arithmetic. However, there is an analogue of $H^0(X, \mathcal{O}_X)$. For each replete ideal $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_\mathfrak{p}}$ of the number field K, we define

$$H^0(\mathfrak{a}) = \big\{ f \in K^* \mid v_\mathfrak{p}(f) \geq \nu_\mathfrak{p} \text{ for all } \mathfrak{p} \big\}.$$

This is a finite set because $jH^0(\mathfrak{a})$ lies in the part of the lattice $j\mathfrak{a}_f \subseteq K_\mathbb{R}$ which is bounded by the conditions $|f|_\mathfrak{p} \leq e^{-\nu_\mathfrak{p} f_\mathfrak{p}}$, $\mathfrak{p}|\infty$. As the analogue of the dimension, we put $\ell(\mathfrak{a}) = 0$ if $H^0(\mathfrak{a}) = \emptyset$, and in all other cases

$$\ell(\mathfrak{a}) := \log \frac{\#H^0(\mathfrak{a})}{\mathrm{vol}(W)},$$

where the normalizing factor $\operatorname{vol}(W)$ is the volume of the set

$$W = \left\{ (z_\tau) \in K_{\mathbb{R}} = \left[\prod_\tau \mathbb{C}\right]^+ \mid |z_\tau| \le 1 \right\}.$$

This volume is given explicitly by

$$\operatorname{vol}(W) = 2^r (2\pi)^s,$$

where r, resp. s, is the number of real, resp. complex, primes of K (see the proof of chap. I, (5.3)). In particular, one has

$$H^0(\mathcal{O}) = \mu(K), \quad \text{so that} \quad \ell(\mathcal{O}) = \log \frac{\#\mu(K)}{2^r (2\pi)^s},$$

because $|f|_\mathfrak{p} \le 1$ for all $\mathfrak{p}$, and $\prod_\mathfrak{p} |f|_\mathfrak{p} = 1$ implies $|f|_\mathfrak{p} = 1$ for all $\mathfrak{p}$, so that $H^0(\mathcal{O})$ is a finite subgroup of K^* and thus must consist of all roots of unity. This normalization leads us necessarily to the following definition of the genus of a number field, which had already been proposed *ad hoc* by the French mathematician André Weil in 1939 (see [138]).

(3.5) Definition. *The* **genus** *of a number field K is defined to be the real number*

$$g = \ell(\mathcal{O}) - \chi(\mathcal{O}) = \log \frac{\#\mu(K)\sqrt{|d_K|}}{2^r (2\pi)^s}.$$

Observe that the genus of the field of rational numbers $\mathbb{Q}$ is 0. Using this definition, the Riemann-Roch formula (3.4) takes the following shape:

(3.6) Proposition. *For every replete ideal $\mathfrak{a}$ of K one has*

$$\chi(\mathfrak{a}) = \deg(\mathfrak{a}) + \ell(\mathcal{O}) - g.$$

The analogue of the strong Riemann-Roch formula

$$\ell(D) = \deg(D) + 1 - g + \ell(\mathcal{K} - D),$$

hinges on the following deep theorem of Minkowski theory, which is due to Serge Lang and which reflects an arithmetic analogue of Serre duality. As usual, let r, resp. s, denote the number of real, resp. complex, primes, and $n = [K : \mathbb{Q}]$.

(3.7) Theorem (S. Lang). *For replete ideals $\mathfrak{a} = \prod_\mathfrak{p} \mathfrak{p}^{\nu_\mathfrak{p}} \in J(\bar{\mathcal{O}})$, one has*

$$\#H^0(\mathfrak{a}^{-1}) = \frac{2^r (2\pi)^s}{\sqrt{|d_K|}} \mathfrak{N}(\mathfrak{a}) + O\left(\mathfrak{N}(\mathfrak{a})^{1-\frac{1}{n}}\right)$$

if $\mathfrak{N}(\mathfrak{a}) \to \infty$. Here, as usual, $O(t)$ denotes a function such that $O(t)/t$ remains bounded as $t \to \infty$.

For the proof of the theorem we need the following

(3.8) Lemma. *Let* $\mathfrak{a}_1, \dots, \mathfrak{a}_h$ *be fractional ideals representing the classes of the finite ideal class group* $Pic(\mathcal{O})$. *Let* c *be a positive constant and*

$$\mathfrak{A}_i = \Big\{ \mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_\mathfrak{p}} \;\Big|\; \mathfrak{a}_f = \mathfrak{a}_i,\ \mathfrak{N}(\mathfrak{p})^{\nu_\mathfrak{p}} \le c\,\mathfrak{N}(\mathfrak{a})^{f_\mathfrak{p}/n} \text{ for } \mathfrak{p}|\infty \Big\}.$$

Then the constant c *can be chosen in such a way that*

$$J(\bar{\mathcal{O}}) = \bigcup_{i=1}^{h} \mathfrak{A}_i P(\bar{\mathcal{O}}).$$

Proof: Let $\mathfrak{B}_i = \{\mathfrak{a} \in J(\bar{\mathcal{O}}) \mid \mathfrak{a}_f = \mathfrak{a}_i\}$. Multiplying by a suitable replete principal ideal $[a]$, every $\mathfrak{a} \in J(\bar{\mathcal{O}})$ may be transformed into a replete ideal $\mathfrak{a}' = \mathfrak{a}[a]$ such that $\mathfrak{a}'_f = \mathfrak{a}_i$ for some i. Consequently, one has $J(\bar{\mathcal{O}}) = \bigcup_{i=1}^{h} \mathfrak{B}_i P(\bar{\mathcal{O}})$. It therefore suffices to show that $\mathfrak{B}_i \subseteq \mathfrak{A}_i P(\bar{\mathcal{O}})$ for $i = 1, \dots, h$, if the constant c is chosen conveniently. To do this, let $\mathfrak{a} = \mathfrak{a}_i \mathfrak{a}_\infty \in \mathfrak{B}_i$, $\mathfrak{a}_\infty = \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_\mathfrak{p}} \in \prod_{\mathfrak{p}|\infty} \mathbb{R}_+^*$. Then we find for the replete ideal

$$\mathfrak{a}'_\infty = \mathfrak{a}_\infty \mathfrak{N}(\mathfrak{a}_\infty)^{-\frac{1}{n}} = \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu'_\mathfrak{p}},$$

where $\nu'_\mathfrak{p} = \nu_\mathfrak{p} - \frac{1}{n}\sum_{\mathfrak{q}|\infty} f_\mathfrak{q}\nu_\mathfrak{q}$, that $\mathfrak{N}(\mathfrak{a}'_\infty) = 1$, and thus $\sum_{\mathfrak{p}|\infty} f_\mathfrak{p}\nu'_\mathfrak{p} = 0$. The vector

$$(\dots, f_\mathfrak{p}\nu'_\mathfrak{p}, \dots) \in \prod_{\mathfrak{p}|\infty} \mathbb{R}$$

therefore lies in the trace-zero space $H = \{(x_\mathfrak{p}) \in \prod_{\mathfrak{p}|\infty} \mathbb{R} \mid \sum_{\mathfrak{p}|\infty} x_\mathfrak{p} = 0\}$. Inside it we have – see chap. I, (7.3) – the complete unit lattice $\lambda(\mathcal{O}^*)$. Thus there exists a lattice point $\lambda(u) = (\dots, -f_\mathfrak{p} v_\mathfrak{p}(u), \dots)_{\mathfrak{p}|\infty}$, $u \in \mathcal{O}^*$, such that

$$\big| f_\mathfrak{p}\nu'_\mathfrak{p} - f_\mathfrak{p} v_\mathfrak{p}(u) \big| \le f_\mathfrak{p} c_0$$

with a constant c_0 depending only on the lattice $\lambda(\mathcal{O}^*)$. This implies

$$\nu_\mathfrak{p} - v_\mathfrak{p}(u) = \nu'_\mathfrak{p} + \frac{1}{n}\sum_{\mathfrak{q}|\infty} f_\mathfrak{q}\nu_\mathfrak{q} - v_\mathfrak{p}(u) \le \frac{1}{n}\log\mathfrak{N}(\mathfrak{a}_\infty) + c_0 = \frac{1}{n}\log\mathfrak{N}(\mathfrak{a}) + c_1$$

with $c_1 = c_0 - \frac{1}{n}\log\mathfrak{N}(\mathfrak{a}_i)$. Putting now $\mathfrak{b} = \mathfrak{a}[u^{-1}] = \prod_\mathfrak{p} \mathfrak{p}^{n_\mathfrak{p}}$, we get $\mathfrak{b}_f = \mathfrak{a}_i$. This is because $[u]_f = (u) = (1)$ and

$$f_\mathfrak{p} n_\mathfrak{p} = f_\mathfrak{p}\big(\nu_\mathfrak{p} - v_\mathfrak{p}(u)\big) \le \frac{f_\mathfrak{p}}{n}\log\mathfrak{N}(\mathfrak{a}) + nc_1,$$

so that $\mathfrak{N}(\mathfrak{p})^{n_\mathfrak{p}} \le e^{nc_1}\mathfrak{N}(\mathfrak{a})^{f_\mathfrak{p}/n}$ for $\mathfrak{p}|\infty$; then $\mathfrak{b} \in \mathfrak{A}_i$, so that $\mathfrak{a} = \mathfrak{b}[u] \in \mathfrak{A}_i P(\bar{\mathcal{O}})$, where $c = e^{nc_1}$. □

Proof of (3.7): As $O(t) = O(t) - 1$, we may replace $H^0(\mathfrak{a}^{-1})$ by

$$\overline{H}^0(\mathfrak{a}^{-1}) = H^0(\mathfrak{a}^{-1}) \cup \{0\} = \left\{ f \in \mathfrak{a}_{\mathfrak{f}}^{-1} \mid |f|_{\mathfrak{p}} \leq \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}} \text{ for } \mathfrak{p}|\infty \right\}.$$

We have to show that there are constants C, C' such that

$$(*) \qquad \left| \#\overline{H}^0(\mathfrak{a}^{-1}) - \frac{2^r(2\pi)^s}{\sqrt{|d_K|}} \mathfrak{N}(\mathfrak{a}) \right| \leq C \mathfrak{N}(\mathfrak{a})^{1-\frac{1}{n}}$$

for all $\mathfrak{a} \in J(\bar{\mathcal{O}})$ satisfying $\mathfrak{N}(\mathfrak{a}) \geq C'$. For $a \in K^*$, the set $\overline{H}^0(\mathfrak{a}^{-1})$ is mapped bijectively via $x \mapsto ax$ onto the set $\overline{H}^0([a]\mathfrak{a}^{-1})$. The numbers $\#\overline{H}^0(\mathfrak{a}^{-1})$ and $\mathfrak{N}(\mathfrak{a})$ thus depend only on the class $\mathfrak{a} \bmod P(\bar{\mathcal{O}})$. As by the preceding lemma $J(\bar{\mathcal{O}}) = \bigcup_{i=1}^{h} \mathfrak{A}_i P(\bar{\mathcal{O}})$, it suffices to show $(*)$ for $\mathfrak{a}$ ranging over the set $\mathfrak{A}_i$.

For this, we shall use the identification of Minkowski space

$$K_{\mathbb{R}} = \prod_{\mathfrak{p}|\infty} K_{\mathfrak{p}}$$

with its canonical measure. Since $\mathfrak{a}_{\mathfrak{f}} = \mathfrak{a}_i$ for $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}} \in \mathfrak{A}_i$, we have

$$\overline{H}^0(\mathfrak{a}^{-1}) = \left\{ f \in \mathfrak{a}_i^{-1} \mid |f|_{\mathfrak{p}} \leq \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}} \text{ for } \mathfrak{p}|\infty \right\}.$$

We therefore have to count the lattice points in $\Gamma = j\mathfrak{a}_i^{-1} \subseteq K_{\mathbb{R}}$ which fall into the domain

$$P_{\mathfrak{a}} = \prod_{\mathfrak{p}|\infty} D_{\mathfrak{p}}$$

where $D_{\mathfrak{p}} = \{x \in K_{\mathfrak{p}} \mid |x|_{\mathfrak{p}} \leq \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}\}$. Let F be a fundamental mesh of Γ. We consider the sets

$$X = \left\{ \gamma \in \Gamma \mid (F + \gamma) \cap P_{\mathfrak{a}} \neq \emptyset \right\},$$

$$Y = \left\{ \gamma \in \Gamma \mid F + \gamma \subseteq \mathring{P}_{\mathfrak{a}} \right\},$$

$$X \smallsetminus Y = \left\{ \gamma \in \Gamma \mid (F + \gamma) \cap \partial P_{\mathfrak{a}} \neq \emptyset \right\}.$$

As $Y \subseteq \Gamma \cap P_{\mathfrak{a}} = \overline{H}^0(\mathfrak{a}^{-1}) \subseteq X$ and as $\bigcup_{\gamma \in Y}(F+\gamma) \subseteq P_{\mathfrak{a}} \subseteq \bigcup_{\gamma \in X}(F+\gamma)$, one has

$$\#Y \leq \#\overline{H}^0(\mathfrak{a}^{-1}) \leq \#X$$

as well as

$$\#Y \operatorname{vol}(F) \leq \operatorname{vol}(P_{\mathfrak{a}}) \leq \#X \operatorname{vol}(F).$$

This implies

$$\left| \#\overline{H}^0(\mathfrak{a}^{-1}) - \frac{\operatorname{vol}(P_{\mathfrak{a}})}{\operatorname{vol}(F)} \right| \leq \#X - \#Y = \#(X \smallsetminus Y).$$

For the set $P_{\mathfrak{a}} = \prod_{\mathfrak{p}|\infty} D_{\mathfrak{p}}$, we now have

$$\operatorname{vol}(P_{\mathfrak{a}}) = \prod_{\mathfrak{p} \text{ real}} 2\mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}} \prod_{\mathfrak{p} \text{ complex}} 2\pi\mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}} = 2^r(2\pi)^s\mathfrak{N}(\mathfrak{a}_\infty)$$

(observe here that, under the identification $K_{\mathfrak{p}} = \mathbb{C}$, one has the equation $|x|_{\mathfrak{p}} = |x|^2$). For the fundamental mesh F, (3.3) yields

$$\operatorname{vol}(F) = \sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}_f^{-1}).$$

From this we get

$$\left|\#\overline{H}^0(\mathfrak{a}^{-1}) - \frac{2^r(2\pi)^s}{\sqrt{|d_K|}}\mathfrak{N}(\mathfrak{a})\right| \le \#(X \setminus Y).$$

Having obtained this inequality, it suffices to show that there exist constants C, C' such that

$$\#(X \setminus Y) = \#\big\{\gamma \in \Gamma \mid (F+\gamma) \cap \partial P_{\mathfrak{a}} \neq \emptyset\big\} \le C\,\mathfrak{N}(\mathfrak{a})^{1-\frac{1}{n}},$$

for all $\mathfrak{a} \in \mathfrak{A}_i$ with $\mathfrak{N}(\mathfrak{a}) \ge C'$. We choose $C' = 1$ and find the constant C in the remainder of the proof. We parametrize the set $P_{\mathfrak{a}} = \prod_{\mathfrak{p}|\infty} D_{\mathfrak{p}}$ via the mapping

$$\varphi : I^n \to P_{\mathfrak{a}},$$

where $I = [0,1]$, which is given by

$$I \longrightarrow D_{\mathfrak{p}}, \quad t \longmapsto 2\alpha_{\mathfrak{p}}(t - \tfrac{1}{2}), \qquad \text{if } \mathfrak{p} \text{ real},$$

$$I^2 \longrightarrow D_{\mathfrak{p}}, \quad (\rho,\theta) \longmapsto \sqrt{\alpha_{\mathfrak{p}}}(\rho\cos 2\pi\theta, \rho\sin 2\pi\theta), \quad \text{if } \mathfrak{p} \text{ complex},$$

where $\alpha_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}$. We bound the norm $\|d\varphi(x)\|$ of the derivative $d\varphi(x) : \mathbb{R}^n \to \mathbb{R}^n$ ($x \in I^n$). If $d\varphi(x) = (a_{ik})$, then $\|d\varphi(x)\| \le n \max |a_{ik}|$. Every partial derivative of φ is now bounded by $2\alpha_{\mathfrak{p}}$, resp. $2\pi\sqrt{\alpha_{\mathfrak{p}}}$. Since $\mathfrak{a} \in \mathfrak{A}_i$, we have that $\alpha_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}} \le c\mathfrak{N}(\mathfrak{a})^{f_{\mathfrak{p}}/n}$, for all $\mathfrak{p}|\infty$. It follows that

$$\|d\varphi(x)\| \le 2\pi n \max \alpha_{\mathfrak{p}}^{1/f_{\mathfrak{p}}} \le c_1\mathfrak{N}(\mathfrak{a})^{1/n}.$$

The mean value theorem implies that

$$(**) \qquad \|\varphi(x) - \varphi(y)\| \le c_1\mathfrak{N}(\mathfrak{a})^{1/n}\|x - y\|,$$

where $\|\ \|$ is the euclidean norm. The boundary of $P_{\mathfrak{a}}$,

$$\partial P_{\mathfrak{a}} = \bigcup_{\mathfrak{p}} \Big[\partial D_{\mathfrak{p}} \times \prod_{\mathfrak{q}\neq\mathfrak{p}} D_{\mathfrak{q}}\Big],$$

is parametrized by a finite number of boundary cubes I^{n-1} of I^n. We subdivide every edge of I^{n-1} into $m = [\mathfrak{N}(\mathfrak{a})^{1/n}] \ge C' = 1$ segments of equal length and obtain for I^{n-1} a decomposition into m^{n-1} small cubes of diameter $\leq (n-1)^{1/2}/m$. From (∗∗), the image of such a small cube under φ has a diameter $\leq \frac{(n-1)^{1/2}}{m} c_1 \mathfrak{N}(\mathfrak{a})^{1/n} \leq (n-1)^{1/2} c_1 \frac{m+1}{m} \leq (n-1)^{1/2} c_1 2 =: c_2$. The number of translates $F+\gamma$, $\gamma \in \Gamma$, meeting a domain of diameter $\leq c_2$ is bounded by a constant c_3 which depends only on c_2 and the fundamental mesh F. The image of a small cube under φ thus meets at most c_3 translates $F+\gamma$. Since there are precisely $m^{n-1} = [\mathfrak{N}(\mathfrak{a})^{1/n}]^{n-1}$ cubes in $\varphi(I^{n-1})$, we see that $\varphi(I^{n-1})$ meets at most $c_3[\mathfrak{N}(\mathfrak{a})^{1/n}]^{n-1} \leq c_3 \mathfrak{N}(\mathfrak{a})^{1-\frac{1}{n}}$ translates, and since the boundary $\partial P_{\mathfrak{a}}$ is covered by at most $2n$ such parts $\varphi(I^{n-1})$, we do indeed find that

$$\#\big\{\gamma \in \Gamma \mid (F+\gamma) \cap \partial P_{\mathfrak{a}} \neq \emptyset\big\} \leq C\, \mathfrak{N}(\mathfrak{a})^{1-\frac{1}{n}}$$

for all $\mathfrak{a} \in \mathfrak{A}_i$ with $\mathfrak{N}(\mathfrak{a}) \geq 1$, where $C = 2nc_3$ is a constant which is independent of $\mathfrak{a} \in \mathfrak{A}_i$, as required. □

From the theorem we have just proved, we now obtain the strong version of the Riemann-Roch theorem. We want to formulate it in the language of divisors. Let $D = \sum_{\mathfrak{p}} \nu_{\mathfrak{p}} \mathfrak{p}$ be a replete divisor of K,

$$H^0(D) = H^0\big(\mathcal{O}(D)\big) = \big\{ f \in K^* \mid v_{\mathfrak{p}}(f) \geq -\nu_{\mathfrak{p}} \big\},$$

$$\ell(D) = \ell(\mathcal{O}(D)) = \log \frac{\# H^0(D)}{\operatorname{vol}(W)} \quad \text{and} \quad \chi(D) = \chi(\mathcal{O}(D)).$$

We call the number

$$i(D) = \ell(D) - \chi(D)$$

the **index of specialty** of D and get the

(3.9) Theorem (Riemann-Roch). *For every replete divisor $D \in Div(\bar{\mathcal{O}})$ we have the formula*

$$\ell(D) = \deg(D) + \ell(\mathcal{O}) - g + i(D).$$

The index of specialty $i(D)$ satisfies

$$i(D) = O\big(e^{-\frac{1}{n}\deg(D)}\big),$$

in particular, $i(D) \to 0$ for $\deg(D) \to \infty$.

Proof: The formula for $\ell(D)$ follows from $\chi(D) = \deg(D) + \ell(\mathcal{O}) - g$ and $\chi(D) = \ell(D) - i(D)$. Putting $\mathfrak{a}^{-1} = \mathcal{O}(D)$, we find by (3.7) that

$$\frac{\# H^0(\mathfrak{a}^{-1})}{2^r (2\pi)^s} = \frac{\mathfrak{N}(\mathfrak{a})}{\sqrt{|d_K|}} \big(1 + \varphi(\mathfrak{a})\, \mathfrak{N}(\mathfrak{a})^{-1/n}\big),$$

for some function $\varphi(\mathfrak{a})$ which remains bounded as $\mathfrak{N}(\mathfrak{a}) \to \infty$, so that $\deg(D) = -\log \mathfrak{N}(\mathfrak{a}^{-1}) = \log \mathfrak{N}(\mathfrak{a}) \to \infty$. Taking logarithms and observing that $\log(1 + O(t)) = O(t)$ and $\mathfrak{N}(\mathfrak{a})^{-1/n} = \exp(-\frac{1}{n}\deg D)$, we obtain

$$\begin{aligned}\ell(D) = \ell(\mathfrak{a}^{-1}) &= -\log\big(\sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}^{-1})\big) + O\big(\mathfrak{N}(\mathfrak{a})^{-1/n}\big)\\ &= \chi(D) + O(e^{-\frac{1}{n}\deg D}).\end{aligned}$$

Hence $i(D) = \ell(D) - \chi(D) = O(e^{-\frac{1}{n}\deg D})$. □

To conclude this section, let us study the variation of the Euler-Minkowski characteristic and of the genus when we change the field K. Let $L|K$ be a finite extension and $\mathcal{o}$, resp. $\mathcal{O}$, the ring of integers of K, resp. L. In §2 we considered **Dedekind's complementary module**

$$\mathfrak{C}_{L|K} = \big\{x \in L \mid Tr(x\mathcal{O}) \subseteq \mathcal{o}\big\} \cong \mathrm{Hom}_{\mathcal{o}}(\mathcal{O}, \mathcal{o}).$$

It is a fractional ideal in L whose inverse is the different $\mathfrak{D}_{L|K}$. From (2.6), it is divisible only by the prime ideals of L which are ramified over K.

(3.10) Definition. *The fractional ideal*

$$\omega_K = \mathfrak{C}_{K|\mathbb{Q}} \cong \mathrm{Hom}_{\mathbb{Z}}(\mathcal{o}, \mathbb{Z})$$

is called the **canonical module** *of the number field K.*

By (2.2) we have the

(3.11) Proposition. *The canonical modules of L and K satisfy the relation*

$$\omega_L = \mathfrak{C}_{L|K}\,\omega_K\,.$$

The canonical module ω_K is related to the Euler-Minkowski characteristic $\chi(\mathcal{o})$ and the genus g of K in the following way, by formula (3.3):

$$\mathrm{vol}(\mathcal{o}) = \sqrt{|d_K|}.$$

(3.12) Proposition. $\deg\omega_K = -2\chi(\mathcal{o}) = 2g - 2\ell(\mathcal{o})$.

Proof: By (2.9) we know that $N_{K|\mathbb{Q}}(\mathfrak{D}_{K|\mathbb{Q}})$ is the discriminant ideal $\mathfrak{d}_{K|\mathbb{Q}} = (d_K)$, and therefore by (1.6), (iii):

$$\mathfrak{N}(\omega_K) = \mathfrak{N}(\mathfrak{D}_{K|\mathbb{Q}})^{-1} = \mathfrak{N}(\mathfrak{d}_{K|\mathbb{Q}})^{-1} = |d_K|^{-1},$$

so that, as $\mathrm{vol}(\mathcal{o}) = \sqrt{|d_K|}$, we have indeed

$$\deg\omega_K = -\log\mathfrak{N}(\omega_K) = \log|d_K| = 2\log\mathrm{vol}(\mathcal{o}) = -2\chi(\mathcal{o}) = 2g - 2\ell(\mathcal{o})\,.$$

□

As for the genus, we now obtain the following analogue of the **Riemann-Hurwitz formula** of function theory.

(3.13) Proposition. *Let $L|K$ be a finite extension and g_L, resp. g_K, the genus of L, resp. K. Then one has*

$$g_L - \ell(\mathcal{O}_L) = [L:K]\big(g_K - \ell(\mathcal{O}_K)\big) + \frac{1}{2}\deg \mathfrak{D}_{L|K}\,.$$

In particular, in the case of an unramified extension $L|K$:

$$\chi(\mathcal{O}_L) = [L:K]\chi(\mathcal{O}_K)\,.$$

Proof: Since $\omega_L = \mathfrak{D}_{L|K}\omega_K$, one has

$$\mathfrak{N}(\omega_L) = \mathfrak{N}(i_{L|K}\omega_K)\,\mathfrak{N}(\mathfrak{D}_{L|K}) = \mathfrak{N}(\omega_K)^{[L:K]}\,\mathfrak{N}(\mathfrak{D}_{L|K}),$$

so that

$$\deg \omega_L = [L:K]\deg\omega_K + \deg \mathfrak{D}_{L|K}\,.$$

Thus the proposition follows from (3.12). □

The Riemann-Hurwitz formula tells us in particular that, in the decision we took in § 1, we really had no choice but to consider the extension $\mathbb{C}|\mathbb{R}$ as *unramified.* In fact, in function theory the module corresponding by analogy to the ideal $\mathfrak{D}_{L|K}$ takes account of precisely the branch points of the covering of Riemann surfaces in question. In order to obtain the same phenomenon in number theory we are forced to declare all the infinite primes $\mathfrak{P}$ of L unramified, since they do not occur in the ideal $\mathfrak{D}_{L|K}$.

Thus the fact that $\mathbb{C}|\mathbb{R}$ is unramified appears to be forced by nature itself. Investigating the matter a little more closely, however, this turns out not to be the case. It is rather a consequence of a well-hidden initial choice that we made. In fact, in chap. I, § 5, we equipped the Minkowski space

$$K_{\mathbb{R}} = \Big[\prod_\tau \mathbb{C}\Big]^+$$

with the "canonical metric"

$$\langle x, y\rangle = \sum_\tau x_\tau \bar{y}_\tau.$$

Replacing it, for instance, by the "Minkowski metric"

$$(x, y) = \sum_\tau \alpha_\tau x_\tau \bar{y}_\tau\,,$$

$\alpha_\tau = 1$ if $\tau = \bar{\tau}$, $\alpha_\tau = \frac{1}{2}$ if $\tau \neq \bar{\tau}$, changes the whole picture. The Haar measures on $K_{\mathbb{R}}$ belonging to $\langle\, ,\rangle$ and $(\, ,)$ are related as follows:

$$\mathrm{vol}_{\text{canonical}}(X) = 2^s \,\mathrm{vol}_{\text{Minkowski}}(X).$$

Distinguishing the invariants of Riemann-Roch theory with respect to the Minkowski measure by a tilde, we get the relations

$$\tilde{\chi}(\mathfrak{a}) = \chi(\mathfrak{a}) + \log 2^s, \quad \tilde{\ell}(\mathfrak{a}) = \ell(\mathfrak{a}) + \log 2^s$$

(the latter in case that $H^0(\mathfrak{a}) \neq \emptyset$), whereas the genus remains unchanged. Substituting this into the Riemann-Hurwitz formula (3.13) preserves its shape only if one enriches $\mathfrak{C}_{L|K}$ into a replete ideal in which all infinite primes $\mathfrak{P}$ such that $L_\mathfrak{P} \neq K_\mathfrak{p}$ occur. This forces us to consider the extension $\mathbb{C}|\mathbb{R}$ as *ramified*, to put $\tilde{e}_{\mathfrak{P}|\mathfrak{p}} = [L_\mathfrak{P} : K_\mathfrak{p}]$, $\tilde{f}_{\mathfrak{P}|\mathfrak{p}} = 1$, and in particular

$$\tilde{e}_\mathfrak{p} = [K_\mathfrak{p} : \mathbb{R}], \quad \tilde{f}_\mathfrak{p} = 1.$$

The following modifications ensue from this. For an infinite prime $\mathfrak{p}$ one has to define

$$\tilde{v}_\mathfrak{p}(a) = -\tilde{e}_\mathfrak{p} \log|\tau a|, \quad \mathfrak{p}^\nu = e^{\nu/\tilde{e}_\mathfrak{p}}, \quad \tilde{\mathfrak{N}}(\mathfrak{p}) = e.$$

The absolute norm as well as the degree of a replete ideal $\mathfrak{a}$ remain unaltered:

$$\tilde{\mathfrak{N}}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{a}), \quad \widetilde{\deg}(\mathfrak{a}) = -\log \tilde{\mathfrak{N}}(\mathfrak{a}) = \deg(\mathfrak{a}).$$

The canonical module ω_K however has to be changed:

$$\tilde{\omega}_K = \omega_K \prod_{\mathfrak{p}\text{ complex}} \mathfrak{p}^{2\log 2},$$

in order for the equation

$$\deg \tilde{\omega}_K = -2\tilde{\chi}(\mathcal{O}) = 2g - 2\tilde{\ell}(\mathcal{O})$$

to hold. By the same token, the ideal $\mathfrak{C}_{L|K}$ has to be replaced by the replete ideal

$$\tilde{\mathfrak{C}}_{L|K} = \mathfrak{C}_{L|K} \prod_{\substack{\mathfrak{P}|\infty \\ e_{\tilde{\mathfrak{P}}|\tilde{\mathfrak{p}}} \neq 1}} \mathfrak{P}^{2\log 2}$$

so that

$$\tilde{\omega}_L = \tilde{\mathfrak{C}}_{L|K}\, i_{L|K}(\tilde{\omega}_K).$$

In the same way as in (3.13), this yields the Riemann-Hurwitz formula

$$g_L - \tilde{\ell}(\mathcal{O}_L) = [L:K]\big(g_K - \tilde{\ell}(\mathcal{O}_K)\big) + \frac{1}{2}\deg \tilde{\mathfrak{C}}_{L|K}.$$

In view of this sensitivity to the chosen metric on Minkowski space $K_{\mathbb{R}}$, the mathematician *Uwe Jannsen* proposes as analogues of the function fields not just number fields K by themselves but number fields equipped with a metric of the type

$$\langle x, y\rangle_K = \sum_\tau \alpha_\tau x_\tau \bar{y}_\tau ,$$

$\alpha_\tau > 0$, $\alpha_\tau = \alpha_{\bar\tau}$, on $K_{\mathbb{R}}$. Let these new objects be called **metrized number fields**. This idea does indeed do justice to the situation in question in a very precise manner, and it is of fundamental importance for algebraic number theory. We denote metrized number fields $(K, \langle\ ,\ \rangle_K)$ as $\widetilde{K}$ and attach to them the following invariants. Let

$$\langle x, y\rangle_K = \sum_\tau \alpha_\tau x_\tau \bar{y}_\tau .$$

Let $\mathfrak{p} = \mathfrak{p}_\tau$ be the infinite prime corresponding to $\tau : K \to \mathbb{C}$. We then put $\alpha_{\mathfrak{p}} = \alpha_\tau$. At the same time, we also use the letter $\mathfrak{p}$ for the positive real number

$$\mathfrak{p} = e^{\alpha_{\mathfrak{p}}} \in \mathbb{R}_+^*,$$

which we interpret as the replete ideal $(1) \times (1, \cdots, 1, e^{\alpha_{\mathfrak{p}}}, 1, \cdots, 1) \in J(\mathcal{O}) \times \prod\limits_{\mathfrak{p}|\infty} \mathbb{R}_+^*$. We put

$$e_{\mathfrak{p}} = 1/\alpha_{\mathfrak{p}} \quad \text{and} \quad f_{\mathfrak{p}} = \alpha_{\mathfrak{p}}[K_{\mathfrak{p}} : \mathbb{R}],$$

and we define the valuation $v_{\mathfrak{p}}$ of K^* associated to $\mathfrak{p}$ by

$$v_{\mathfrak{p}}(a) = -e_{\mathfrak{p}} \log |\tau a| .$$

Further, we put

$$\mathfrak{N}(\mathfrak{p}) = e^{f_{\mathfrak{p}}} \quad \text{and} \quad |a|_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{-v_{\mathfrak{p}}(a)},$$

so that again $|a|_{\mathfrak{p}} = |\tau a|$ if $\mathfrak{p}$ is real, and $|a|_{\mathfrak{p}} = |\tau a|^2$ if $\mathfrak{p}$ is complex. For every replete ideal $\mathfrak{a}$ of K, there is a unique representation $\mathfrak{a} = \prod\limits_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$, which gives the absolute norm $\mathfrak{N}(\mathfrak{a}) = \prod\limits_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}$, and the degree

$$\deg_{\widetilde{K}}(\mathfrak{a}) = -\log\, \mathfrak{N}(\mathfrak{a}).$$

The *canonical module* of $\widetilde{K}$ is defined to be the replete ideal

$$\omega_{\widetilde{K}} = \omega_K \cdot \omega_\infty \in J(\bar{\mathcal{O}}) = J(\mathcal{O}) \times \prod_{\mathfrak{p}|\infty} \mathbb{R}_+^*,$$

where ω_K is the inverse of the different $\mathfrak{D}_{K|\mathbb{Q}}$ of $K|\mathbb{Q}$, and

$$\omega_\infty = (\alpha_{\mathfrak{p}}^{-1})_{\mathfrak{p}|\infty} \in \prod_{\mathfrak{p}|\infty} \mathbb{R}_+^* .$$

The Riemann-Roch theory may be transferred without any problem, using the definitions given above, to metrized number fields $\widetilde{K} = (K, \langle\ ,\ \rangle_K)$. Distinguishing their invariants by the suffix $\widetilde{K}$ yields the relations

$$\mathrm{vol}_{\widetilde{K}}(X) = \prod_\tau \sqrt{\alpha_\tau}\, \mathrm{vol}(X),$$

because $T_\alpha : (K_{\mathbb{R}}, \langle\ ,\ \rangle_K) \to (K_{\mathbb{R}}, \langle\ ,\ \rangle)$, $(x_\tau) \mapsto (\sqrt{\alpha_\tau}\, x_\tau)$, is an isometry with determinant $\prod_\tau \sqrt{\alpha_\tau}$, and therefore

$$\chi_{\widetilde{K}}(\mathcal{O}_K) = -\log \mathrm{vol}_{\widetilde{K}}(\mathcal{O}_K) \quad = \chi(\mathcal{O}_K) - \log \prod_\tau \sqrt{\alpha_\tau},$$

$$\ell_{\widetilde{K}}(\mathcal{O}_K) = \quad \log \frac{\# H^0(\mathcal{O}_K)}{\mathrm{vol}_{\widetilde{K}}(W)} \quad = \ell(\mathcal{O}_K) - \log \prod_\tau \sqrt{\alpha_\tau}.$$

The genus

$$g_{\widetilde{K}} = \ell_{\widetilde{K}}(\mathcal{O}_K) - \chi_{\widetilde{K}}(\mathcal{O}_K) = \ell(\mathcal{O}_K) - \chi(\mathcal{O}_K) = \log \frac{\#\mu(K)\sqrt{|d_K|}}{2^r (2\pi)^s}$$

does not depend on the choice of metric.

Just as in function theory, there is then no longer one smallest field, but $\mathbb{Q}$ is replaced by the continuous family of metrized fields $(\mathbb{Q}, \alpha xy)$, $\alpha \in \mathbb{R}_+^*$, all of which have genus $g = 0$. One even has the

(3.14) Proposition. *The metrized fields $(\mathbb{Q}, \alpha xy)$ are the only metrized number fields of genus 0.*

Proof: We have

$$g_{\widetilde{K}} = \log \frac{\#\mu(K)\sqrt{|d_K|}}{2^r (2\pi)^s} = 0 \quad \iff \quad \#\mu(K)\sqrt{|d_K|} = 2^r (2\pi)^s .$$

Since π is transcendental, one has $s = 0$, i.e., K is totally real. Thus $\#\mu(K) = 2$ so that $|d_K| = 4^{n-1}$, where $n = r = [K : \mathbb{Q}]$. In view of the bound (2.14) on the discriminant

$$|d_K|^{1/2} \geq \frac{n^n}{n!}\Big(\frac{\pi}{4}\Big)^{n/2},$$

this can only happen if $n \leq 6$. But for this case one has sharper bounds, due to *Odlyzko* (see [111], table 2):

n	$=$	3	4	5	6
$\lvert d_K\rvert^{1/n}$	$\geq$	3,09	4,21	5,30	6,35

This is not compatible with $|d_K|^{1/n} = 4^{\frac{n-1}{n}}$, so we may conclude that $n \leq 2$. But there is no real quadratic field with discriminant $|d_K| = 4$ (see chap. I, §2, exercise 4). Hence $n = 1$, so that $K = \mathbb{Q}$. $\square$

An *extension of metrized number fields* is a pair $\widetilde{K} = (K, \langle\ ,\ \rangle_K)$, $\widetilde{L} = (L, \langle\ ,\ \rangle_L)$, such that $K \subseteq L$ and the metrics

$$\langle x, y\rangle_K = \sum_\tau \alpha_\tau x_\tau \bar{y}_\tau, \quad \langle x, y\rangle_L = \sum_\sigma \beta_\sigma x_\sigma \bar{y}_\sigma$$

satisfy the relation $\alpha_\tau \geq \beta_\sigma$ whenever $\tau = \sigma|_K$. If $\mathfrak{P}|\mathfrak{p}$ are infinite primes of $L|K$, $\mathfrak{P}$ belonging to σ and $\mathfrak{p}$ to $\tau = \sigma|_K$, we define the *ramification index* and *inertia degree* by

$$e_{\mathfrak{P}|\mathfrak{p}} = \alpha_\tau/\beta_\sigma \quad \text{and} \quad f_{\mathfrak{P}|\mathfrak{p}} = \beta_\sigma/\alpha_\tau[L_{\mathfrak{P}} : K_{\mathfrak{p}}].$$

Thus the fundamental identity

$$\sum_{\mathfrak{P}|\mathfrak{p}} e_{\mathfrak{P}|\mathfrak{p}} f_{\mathfrak{P}|\mathfrak{p}} = [L : K]$$

is preserved. Also $\mathfrak{P}$ is unramified if and only if $\alpha_\tau = \beta_\sigma$. For "replete prime ideals" $\mathfrak{p} = e^{\alpha_\tau}$, $\mathfrak{P} = e^{\beta_\sigma}$, we put

$$i_{L|K}(\mathfrak{p}) = \prod_{\mathfrak{P}|\mathfrak{p}} \mathfrak{P}^{e_{\mathfrak{P}|\mathfrak{p}}}, \quad N_{L|K}(\mathfrak{P}) = \mathfrak{p}^{f_{\mathfrak{P}|\mathfrak{p}}}.$$

Finally we define the *different* of $\widetilde{L}|\widetilde{K}$ to be the replete ideal

$$\mathfrak{D}_{\widetilde{L}|\widetilde{K}} = \mathfrak{D}_{L|K} \cdot \mathfrak{D}_\infty \in J(\bar{\mathcal{O}}_L) = J(\mathcal{O}_L) \times \prod_{\mathfrak{P}|\infty} \mathbb{R}_+^*,$$

where $\mathfrak{D}_{L|K}$ is the different of $L|K$ and

$$\mathfrak{D}_\infty = (\beta_{\mathfrak{P}}/\alpha_{\mathfrak{p}})_{\mathfrak{P}|\infty} \in \prod_{\mathfrak{P}|\infty} \mathbb{R}_+^*,$$

where $\beta_{\mathfrak{P}} = \beta_\sigma$ and $\alpha_{\mathfrak{p}} = \alpha_\tau$ ($\mathfrak{P}$ belongs to σ and $\mathfrak{p}$ to $\tau = \sigma|_K$). With this convention, we obtain the general *Riemann-Hurwitz formula*

$$g_{\widetilde{L}} - \ell_{\widetilde{L}}(\mathcal{O}_L) = [L : K]\big(g_{\widetilde{K}} - \ell_{\widetilde{K}}(\mathcal{O}_K)\big) - \tfrac{1}{2}\deg \mathfrak{D}_{\widetilde{L}|\widetilde{K}}.$$

If we consider only number fields endowed with the Minkowski metric, then $L_{\mathfrak{P}} \neq K_{\mathfrak{p}}$ is always ramified. In this way the convention found in many textbooks is no longer incompatible with the customs introduced in the present book.

§ 4. Metrized $\mathcal{O}$-Modules

The Riemann-Roch theory which was presented in the preceding section in the case of replete ideals is embedded in a much more far-reaching theory which treats finitely generated $\mathcal{O}$-modules. It is only in this setting that the theory displays its true nature, and becomes susceptible to the most comprehensive generalization. This theory is subject to a formalism which has been constructed by *Alexander Grothendieck* for higher dimensional algebraic varieties, and which we will now develop for number fields. In doing so, our principal attention will be focused as before on the kind of compactification which is accomplished by taking into account the infinite places. The effect is that a leading rôle is claimed by linear algebra – for which we refer to [15]. Our treatment is based on a course on "Arakelov Theory and Grothendieck-Riemann-Roch" taught by *Günter Tamme*. There, however, proofs were not given directly, as we will do here, but usually deduced as special cases from the general abstract theory.

Let K be an algebraic number field and $\mathcal{O}$ the ring of integers of K. For the passage from K to $\mathbb{R}$ and $\mathbb{C}$, we start by considering the ring

$$K_{\mathbb{C}} = K \otimes_{\mathbb{Q}} \mathbb{C}. \tag{1}$$

It admits the following two further interpretations, between which we will freely go back and forth in the sequel without further explanation. The set

$$X(\mathbb{C}) = \mathrm{Hom}(K, \mathbb{C})$$

induces a canonical decomposition of rings

$$K_{\mathbb{C}} \cong \bigoplus_{\sigma \in X(\mathbb{C})} \mathbb{C}, \quad a \otimes z \longmapsto \bigoplus_{\sigma \in X(\mathbb{C})} z\sigma a. \tag{2}$$

Alternatively, the right-hand side may be viewed as the set $\mathbb{C}^{X(\mathbb{C})} = \mathrm{Hom}(X(\mathbb{C}), \mathbb{C})$ of all functions $x : X(\mathbb{C}) \to \mathbb{C}$, i.e.,

$$K_{\mathbb{C}} \cong \mathrm{Hom}\big(X(\mathbb{C}), \mathbb{C}\big). \tag{3}$$

The field K is embedded in $K_{\mathbb{C}}$ via

$$K \longrightarrow K \otimes_{\mathbb{Q}} \mathbb{C}, \quad a \longmapsto a \otimes 1,$$

and we identify it with its image. In the interpretation (2), the image of $a \in K$ appears as the tuple $\bigoplus_\sigma \sigma a$ of conjugates of a, and in the interpretation (3) as the function $x(\sigma) = \sigma a$.

We denote the generator of the Galois group $G(\mathbb{C}|\mathbb{R})$ by F_∞, or simply by F. This underlines the fact that it has a position analogous to the Frobenius automorphism $F_p \in G(\overline{\mathbb{F}}_p|\mathbb{F}_p)$, in accordance with our decision of §1 to view the extension $\mathbb{C}|\mathbb{R}$ as unramified. F induces an involution F on $K_{\mathbb{C}}$

which, in the representation $K_{\mathbb{C}} = \mathrm{Hom}(X(\mathbb{C}), \mathbb{C})$ for $x : X(\mathbb{C}) \to \mathbb{C}$, is given by

$$(Fx)(\sigma) = \overline{x(\overline{\sigma})}.$$

F is an automorphism of the $\mathbb{R}$-algebra $K_{\mathbb{C}}$. It is called the **Frobenius correspondence**. Sometimes we also consider, besides F, the involution $z \mapsto \bar{z}$ on $K_{\mathbb{C}}$ which is given by

$$\bar{z}(\sigma) = \overline{z(\sigma)}.$$

We call it the **conjugation**. Finally, we call an element $x \in K_{\mathbb{C}}$, that is, a function $x : X(\mathbb{C}) \to \mathbb{C}$, **positive** (written $x > 0$) if it takes real values, and if $x(\sigma) > 0$ for all $\sigma \in X(\mathbb{C})$.

By convention every $\mathcal{O}$-module considered in the sequel will be supposed to be *finitely generated*. For every such $\mathcal{O}$-module M, we put

$$M_{\mathbb{C}} = M \otimes_{\mathbb{Z}} \mathbb{C}.$$

This is a module over the ring $K_{\mathbb{C}} = \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{C}$, and viewing $\mathcal{O}$ as a subring of $K_{\mathbb{C}}$ – as we agreed above – we may also write

$$M_{\mathbb{C}} = M \otimes_{\mathcal{O}} K_{\mathbb{C}}$$

as $M \otimes_{\mathbb{Z}} \mathbb{C} = M \otimes_{\mathcal{O}} (\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{C})$. The involution $x \mapsto Fx$ on $K_{\mathbb{C}}$ induces the involution

$$F(a \otimes x) = a \otimes Fx$$

on $M_{\mathbb{C}}$. In the representation $M_{\mathbb{C}} = M \otimes_{\mathbb{Z}} \mathbb{C}$, one clearly has

$$F(a \otimes z) = a \otimes \bar{z}.$$

(4.1) Definition. *A* **hermitian metric** *on the $K_{\mathbb{C}}$-module $M_{\mathbb{C}}$ is a sesquilinear mapping*

$$\langle\ ,\ \rangle_M : M_{\mathbb{C}} \times M_{\mathbb{C}} \longrightarrow K_{\mathbb{C}},$$

i.e., a $K_{\mathbb{C}}$-linear form $\langle x, y\rangle_M$ in the first variable satisfying

$$\overline{\langle x, y\rangle_M} = \langle y, x\rangle_M,$$

such that one has $\langle x, x\rangle_M > 0$ for $x \neq 0$.

The metric $\langle\ ,\ \rangle_M$ is called **F-invariant** *if we have furthermore*

$$F\langle x, y\rangle_M = \langle Fx, Fy\rangle_M.$$

This notion may be immediately reduced to the usual notion of a hermitian metric if we view the $K_{\mathbb{C}}$-module $M_{\mathbb{C}}$, by means of the decomposition $K_{\mathbb{C}} = \bigoplus_\sigma \mathbb{C}$, as a direct sum

$$M_{\mathbb{C}} = M \otimes_{\mathcal{O}} K_{\mathbb{C}} = \bigoplus_{\sigma \in X(\mathbb{C})} M_\sigma$$

of $\mathbb{C}$-vector spaces

$$M_\sigma = M \otimes_{\mathcal{O},\sigma} \mathbb{C}.$$

The hermitian metric $\langle\ ,\ \rangle_M$ then splits into the direct sum

$$\langle x, y\rangle_M = \bigoplus_{\sigma \in X(\mathbb{C})} \langle x_\sigma, y_\sigma\rangle_{M_\sigma}$$

of hermitian scalar products $\langle\ ,\ \rangle_{M_\sigma}$ on the $\mathbb{C}$-vector spaces M_σ. In this interpretation, the F-invariance of $\langle x, y\rangle_M$ amounts to the commutativity of the diagrams

$$\begin{tikzcd}
M_\sigma \times M_\sigma \arrow[r, "\langle\ ,\ \rangle_{M_\sigma}"] \arrow[d, "F\times F"'] & \mathbb{C} \arrow[d, "F"] \\
M_{\bar\sigma} \times M_{\bar\sigma} \arrow[r, "\langle\ ,\ \rangle_{M_{\bar\sigma}}"'] & \mathbb{C}.
\end{tikzcd}$$

(4.2) Definition. *A* **metrized** $\mathcal{O}$**-module** *is a finitely generated $\mathcal{O}$-module M with an F-invariant hermitian metric on $M_{\mathbb{C}}$.*

Example 1: Every fractional ideal $\mathfrak{a} \subseteq K$ of $\mathcal{O}$, in particular $\mathcal{O}$ itself, may be equipped with the **trivial metric**

$$\langle x, y\rangle = x\bar{y} = \bigoplus_{\sigma \in X(\mathbb{C})} x_\sigma \bar{y}_\sigma$$

on $\mathfrak{a} \otimes_{\mathbb{Z}} \mathbb{C} = K \otimes_{\mathbb{Q}} \mathbb{C} = K_{\mathbb{C}}$. All the F-invariant hermitian metrics on $\mathfrak{a}$ are obtained as

$$\alpha\langle x, y\rangle = \alpha x\bar{y} = \bigoplus_\sigma \alpha(\sigma) x_\sigma \bar{y}_\sigma\,,$$

where $\alpha \in K_{\mathbb{C}}$ varies over the functions $\alpha : X(\mathbb{C}) \to \mathbb{R}_+^*$ such that $\alpha(\bar\sigma) = \alpha(\sigma)$.

Example 2: Let $L|K$ be a finite extension and $\mathfrak{A}$ a fractional ideal of L, which we view as an $\mathcal{O}$-module M. If $Y(\mathbb{C}) = \operatorname{Hom}(L, \mathbb{C})$, we have the restriction map $Y(\mathbb{C}) \to X(\mathbb{C})$, $\tau \mapsto \tau|_K$, and we write $\tau|\sigma$ if $\sigma = \tau|_K$. For the complexification $M_{\mathbb{C}} = \mathfrak{A} \otimes_{\mathbb{Z}} \mathbb{C} = L_{\mathbb{C}}$, we obtain the decomposition

$$M_{\mathbb{C}} = \bigoplus_{\tau \in Y(\mathbb{C})} \mathbb{C} = \bigoplus_{\sigma \in X(\mathbb{C})} M_\sigma\,,$$

where $M_\sigma = \bigoplus_{\tau|\sigma} \mathbb{C}$. M is turned into a metrized $\mathcal{O}$-module by fixing the **standard metrics**

$$\langle x, y\rangle_{M_\sigma} = \sum_{\tau|\sigma} x_\tau \bar{y}_\tau$$

on the $[L : K]$-dimensional $\mathbb{C}$-vector spaces M_σ.

If M and M' are metrized $\mathcal{O}$-modules, then so is their direct sum $M \oplus M'$, the tensor product $M \otimes M'$, the dual $\check{M} = \mathrm{Hom}_{\mathcal{O}}(M, \mathcal{O})$ and the n-th exterior power $\bigwedge^n M$. In fact, we have that

$$(M \oplus M')_{\mathbb{C}} = M_{\mathbb{C}} \oplus M'_{\mathbb{C}}, \quad (M \otimes_{\mathcal{O}} M')_{\mathbb{C}} = M_{\mathbb{C}} \otimes_{K_{\mathbb{C}}} M'_{\mathbb{C}},$$
$$\check{M}_{\mathbb{C}} = \mathrm{Hom}_{K_{\mathbb{C}}}(M_{\mathbb{C}}, K_{\mathbb{C}}), \quad (\textstyle\bigwedge^n M)_{\mathbb{C}} = \bigwedge^n{}_{K_{\mathbb{C}}} M_{\mathbb{C}},$$

and the metrics on these $K_{\mathbb{C}}$-modules are given by

$$\langle x \oplus x', y \oplus y'\rangle_{M\oplus M'} = \langle x, y\rangle_M + \langle x', y'\rangle_{M'}, \qquad \text{resp.}$$
$$\langle x \otimes x', y \otimes y'\rangle_{M\otimes M'} = \langle x, y\rangle_M \cdot \langle x', y'\rangle_{M'}, \qquad \text{resp.}$$
$$\langle \check{x}, \check{y}\rangle_{\check{M}} = \overline{\langle x, y\rangle_M}, \qquad \text{resp.}$$
$$\big\langle x_1 \wedge \ldots \wedge x_n, y_1 \wedge \ldots \wedge y_n\big\rangle_{\bigwedge^n M} = \det\big(\langle x_i, y_j\rangle_M\big).$$

Here $\check{x}$, in the case of the module $\check{M}_{\mathbb{C}}$, denotes the homomorphism $\check{x} = \langle\ , x\rangle_M : M_{\mathbb{C}} \to K_{\mathbb{C}}$.

Among all $\mathcal{O}$-modules M the **projective** ones play a special rôle. They are defined by the condition that for every exact sequence of $\mathcal{O}$-modules $F' \to F \to F''$ the sequence

$$\mathrm{Hom}_{\mathcal{O}}(M, F') \longrightarrow \mathrm{Hom}_{\mathcal{O}}(M, F) \longrightarrow \mathrm{Hom}_{\mathcal{O}}(M, F'')$$

is also exact. This is equivalent to any of the following conditions (the last two, because $\mathcal{O}$ is a Dedekind domain). For the proof, we refer the reader to standard textbooks of commutative algebra (see for instance [90], chap. IV, §3, or [16], chap. 7, §4).

(4.3) Proposition. *For any finitely generated $\mathcal{O}$-module M the following conditions are equivalent:*

(i) *M is projective,*

(ii) *M is a direct summand of a finitely generated free $\mathcal{O}$-module,*

(iii) *M is locally free, i.e., $M \otimes_{\mathcal{O}} \mathcal{O}_{\mathfrak{p}}$ is a free $\mathcal{O}_{\mathfrak{p}}$-module for any prime ideal $\mathfrak{p}$,*

(iv) *M is torsion free, i.e., the map $M \to M$, $x \mapsto ax$, is injective for all nonzero $a \in \mathcal{O}$,*

(v) *$M \cong \mathfrak{a} \oplus \mathcal{O}^n$ for some ideal $\mathfrak{a}$ of $\mathcal{O}$ and some integer $n \geq 0$.*

In order to distinguish them from projective $\mathcal{O}$-modules, we will henceforth call arbitrary finitely generated $\mathcal{O}$-modules *coherent*. The **rank** of a coherent $\mathcal{O}$-module M is defined to be the dimension

$$\mathrm{rk}(M) = \dim_K(M \otimes_{\mathcal{O}} K).$$

The projective $\mathcal{O}$-modules L of rank 1 are called **invertible** $\mathcal{O}$-modules, because for them $L \otimes_{\mathcal{O}} \check{L} \to \mathcal{O}$, $a \otimes \check{a} \mapsto \check{a}(a)$, is an isomorphism. The invertible $\mathcal{O}$-modules are either fractional ideals of K, or isomorphic to a fractional ideal as $\mathcal{O}$-modules. Indeed, if L is projective of rank 1 and $\alpha \in L$, $\alpha \neq 0$, then, by (4.3), (iv), mapping

$$L \longrightarrow L \otimes_{\mathcal{O}} K = K(\alpha \otimes 1), \quad x \longmapsto f(x)(\alpha \otimes 1),$$

gives an injective $\mathcal{O}$-module homomorphism $L \to K$, $x \mapsto f(x)$, onto a fractional ideal $\mathfrak{a} \subseteq K$.

To see the connection with the Riemann-Roch theory of the last section, which we are about to generalize, we observe that every replete ideal

$$\mathfrak{a} = \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{\nu_{\mathfrak{p}}} \prod_{\mathfrak{p} \mid \infty} \mathfrak{p}^{\nu_{\mathfrak{p}}} = \mathfrak{a}_f \mathfrak{a}_\infty$$

of K defines an invertible, metrized $\mathcal{O}$-module. In fact, the identity $\mathfrak{a}_\infty = \prod_{\mathfrak{p} \mid \infty} \mathfrak{p}^{\nu_{\mathfrak{p}}}$ yields the function

$$\alpha : X(\mathbb{C}) \longrightarrow \mathbb{R}_+, \quad \alpha(\sigma) = e^{2\nu_{\mathfrak{p}_\sigma}},$$

where $\mathfrak{p}_\sigma$ denotes as before the infinite place defined by $\sigma : K \to \mathbb{C}$. Since $\mathfrak{p}_{\bar{\sigma}} = \mathfrak{p}_\sigma$, one has $\alpha(\bar{\sigma}) = \alpha(\sigma)$, and we obtain on the complexification

$$\mathfrak{a}_{f\mathbb{C}} = \mathfrak{a}_f \otimes_{\mathbb{Z}} \mathbb{C} = K_{\mathbb{C}}$$

the F-invariant hermitian metric

$$\langle x, y \rangle_{\mathfrak{a}} = \alpha x \bar{y} = \bigoplus_{\sigma \in X(\mathbb{C})} e^{2\nu_{\mathfrak{p}_\sigma}} x_\sigma \bar{y}_\sigma$$

(see example 1, p. 227). We denote the metrized $\mathcal{O}$-module thus obtained by $L(\mathfrak{a})$.

The ordinary fractional ideals, i.e., the replete ideals $\mathfrak{a}$ such that $\mathfrak{a}_\infty = 1$, and in particular $\mathcal{O}$ itself, are thus equipped with the *trivial* metric $\langle x, y \rangle_{\mathfrak{a}} = \langle x, y \rangle = x\bar{y}$.

(4.4) Definition. *Two metrized $\mathcal{O}$-modules M and M' are called* **isometric** *if there exists an isomorphism*

$$f : M \longrightarrow M'$$

of $\mathcal{O}$-modules which induces an isometry $f_{\mathbb{C}} : M_{\mathbb{C}} \to M'_{\mathbb{C}}$.

(4.5) Proposition.

(i) *Two replete ideals* $\mathfrak{a}$ *and* $\mathfrak{b}$ *define isometric metrized* $\mathcal{O}$*-modules* $L(\mathfrak{a})$ *and* $L(\mathfrak{b})$ *if and only if they differ by a replete principal ideal* $[a]$: $\mathfrak{a} = \mathfrak{b}[a]$.

(ii) *Every invertible metrized* $\mathcal{O}$*-module is isometric to an* $\mathcal{O}$*-module* $L(\mathfrak{a})$.

(iii) $L(\mathfrak{a}\mathfrak{b}) \cong L(\mathfrak{a}) \otimes_{\mathcal{O}} L(\mathfrak{b}),\ L(\mathfrak{a}^{-1}) = \check{L}(\mathfrak{a})$.

Proof: (i) Let $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$, $\mathfrak{b} = \prod_{\mathfrak{p}} \mathfrak{p}^{\mu_{\mathfrak{p}}}$, $[a] = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(a)}$, and let

$$\alpha(\sigma) = e^{2\nu_{\mathfrak{p}_\sigma}}, \quad \beta(\sigma) = e^{2\mu_{\mathfrak{p}_\sigma}}, \quad \gamma(\sigma) = e^{2v_{\mathfrak{p}_\sigma}(a)}.$$

If $\mathfrak{a} = \mathfrak{b}[a]$, then $\nu_{\mathfrak{p}} = \mu_{\mathfrak{p}} + v_{\mathfrak{p}}(a)$; thus $\alpha = \beta\gamma$, and $\mathfrak{a}_f = \mathfrak{b}_f(a)$. The $\mathcal{O}$-module isomorphism $\mathfrak{b}_f \to \mathfrak{a}_f$, $x \mapsto ax$, takes the form $\langle\ ,\ \rangle_{\mathfrak{b}}$ to the form $\langle\ ,\ \rangle_{\mathfrak{a}}$. Indeed, viewing a as embedded in $K_{\mathbb{C}}$, we find $a = \bigoplus_\sigma \sigma a$ and

$$a\bar{a} = \bigoplus_\sigma e^{-2v_{\mathfrak{p}_\sigma}(a)} = \gamma^{-1},$$

because $v_{\mathfrak{p}_\sigma}(a) = -\log|\sigma a|$, so that

$$\langle ax, ay\rangle_{\mathfrak{a}} = \alpha\langle ax, ay\rangle = \alpha\gamma^{-1}\langle x, y\rangle = \beta\langle x, y\rangle = \langle x, y\rangle_{\mathfrak{b}}.$$

Therefore $\mathfrak{b}_f \to \mathfrak{a}_f$, $x \mapsto ax$, gives an isometry $L(\mathfrak{a}) \cong L(\mathfrak{b})$.

Conversely, let $g : L(\mathfrak{b}) \to L(\mathfrak{a})$ be an isometry. Then the $\mathcal{O}$-module homomorphism

$$g : \mathfrak{b}_f \to \mathfrak{a}_f$$

is given as multiplication by some element $a \in \mathfrak{b}_f^{-1}\mathfrak{a}_f \cong \mathrm{Hom}_{\mathcal{O}}(\mathfrak{b}_f, \mathfrak{a}_f)$. The identity

$$\beta\langle x, y\rangle = \langle x, y\rangle_{\mathfrak{b}} = \langle g(x), g(y)\rangle_{\mathfrak{a}} = \alpha\langle ax, ay\rangle = \alpha\gamma^{-1}\langle x, y\rangle$$

then implies that $\alpha = \beta\gamma$, so that $\nu_{\mathfrak{p}} = \mu_{\mathfrak{p}} + v_{\mathfrak{p}}(a)$ for all $\mathfrak{p}|\infty$. In view of $\mathfrak{a}_f = \mathfrak{b}_f(a)$, this yields $\mathfrak{a} = \mathfrak{b}[a]$.

(ii) Let L be an invertible metrized $\mathcal{O}$-module. As mentioned before, we have an isomorphism

$$g : L \longrightarrow \mathfrak{a}_f$$

for the underlying $\mathcal{O}$-module onto a fractional ideal $\mathfrak{a}_f$. The isomorphism $g_{\mathbb{C}} : L_{\mathbb{C}} \to \mathfrak{a}_{f\mathbb{C}} = K_{\mathbb{C}}$ gives us the F-invariant hermitian metric $h(x, y) = \langle g_{\mathbb{C}}^{-1}(x), g_{\mathbb{C}}^{-1}(y)\rangle_L$ on $K_{\mathbb{C}}$. It is of the form

$$h(x, y) = \alpha x\bar{y}$$

for some function $\alpha : X(\mathbb{C}) \to \mathbb{R}_+^*$ such that $\alpha(\bar{\sigma}) = \alpha(\sigma)$. Putting now $\alpha(\sigma) = e^{2\nu_{\mathfrak{p}_\sigma}}$, with $\nu_{\mathfrak{p}_\sigma} \in \mathbb{R}$, makes $\mathfrak{a}_f$ with the metric h into the metrized

$\mathcal{O}$-module $L(\mathfrak{a})$ associated to the replete ideal $\mathfrak{a} = \mathfrak{a}_f \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_\mathfrak{p}}$, and L is isometric to $L(\mathfrak{a})$.

(iii) Let $\mathfrak{a} = \prod_\mathfrak{p} \mathfrak{p}^{\nu_\mathfrak{p}}$, $\mathfrak{b} = \prod_\mathfrak{p} \mathfrak{p}^{\mu_\mathfrak{p}}$, $\alpha(\sigma) = e^{2\nu_{\mathfrak{p}\sigma}}$, $\beta(\sigma) = e^{2\mu_{\mathfrak{p}\sigma}}$. The isomorphism

$$\mathfrak{a}_f \otimes_\mathcal{O} \mathfrak{b}_f \longrightarrow \mathfrak{a}_f\mathfrak{b}_f, \quad a \otimes b \longmapsto ab,$$

between the $\mathcal{O}$-modules underlying $L(\mathfrak{a}) \otimes_\mathcal{O} L(\mathfrak{b})$ and $L(\mathfrak{a}\mathfrak{b})$ then yields, as $\langle ab, a'b'\rangle_{\mathfrak{a}\mathfrak{b}} = \alpha\beta\, ab\, \overline{a'b'} = \alpha\langle a, a'\rangle\beta\langle b, b'\rangle = \langle a, a'\rangle_\mathfrak{a}\langle b, b'\rangle_\mathfrak{b}$, an isometry $L(\mathfrak{a}) \otimes_\mathcal{O} L(\mathfrak{b}) \cong L(\mathfrak{a}\mathfrak{b})$.

The $\mathcal{O}$-module $\operatorname{Hom}_\mathcal{O}(\mathfrak{a}_f, \mathcal{O})$ underlying $\check{L}(\mathfrak{a})$ is isomorphic, via the isomorphism

$$g : \mathfrak{a}_f^{-1} \longrightarrow \operatorname{Hom}_\mathcal{O}(\mathfrak{a}_f, \mathcal{O}), \quad a \longmapsto \big(g(a) : x \mapsto ax\big),$$

to the fractional ideal $\mathfrak{a}_f^{-1}$. For the induced $K_\mathbb{C}$-isomorphism

$$g_\mathbb{C} : K_\mathbb{C} \longrightarrow \operatorname{Hom}_{K_\mathbb{C}}(K_\mathbb{C}, K_\mathbb{C})$$

we have

$$g_\mathbb{C}(x)(y) = xy = \alpha^{-1}\alpha xy = \alpha^{-1}\langle y, \bar{x}\rangle_{L(\mathfrak{a})},$$

so that $g_\mathbb{C}(x) = \alpha^{-1}\check{\bar{x}}$, and thus

$$\begin{aligned}\big\langle g_\mathbb{C}(x), g_\mathbb{C}(y)\big\rangle_{\check{L}(\mathfrak{a})} &= \alpha^{-2}\langle \check{\bar{x}}, \check{\bar{y}}\rangle_{\check{L}(\mathfrak{a})} = \alpha^{-2}\,\overline{\langle \bar{x}, \bar{y}\rangle}_{L(\mathfrak{a})} \\ &= \alpha^{-1}x\bar{y} = \langle x, y\rangle_{L(\mathfrak{a}^{-1})}.\end{aligned}$$

Thus g gives an isometry $\check{L}(\mathfrak{a}) \cong L(\mathfrak{a}^{-1})$. □

(4.6) Definition. *A* **short exact sequence**

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

of metrized $\mathcal{O}$-modules is by definition a short exact sequence of the underlying $\mathcal{O}$-modules which splits isometrically, i.e., in the sequence

$$0 \longrightarrow M'_\mathbb{C} \xrightarrow{\alpha_\mathbb{C}} M_\mathbb{C} \xrightarrow{\beta_\mathbb{C}} M''_\mathbb{C} \longrightarrow 0,$$

$M'_\mathbb{C}$ is mapped isometrically onto $\alpha_\mathbb{C}M'_\mathbb{C}$, and the orthogonal complement $(\alpha_\mathbb{C}M'_\mathbb{C})^\perp$ is mapped isometrically onto $M''_\mathbb{C}$.

The homomorphisms α, β in a short exact sequence of metrized $\mathcal{O}$-modules are called an **admissible monomorphism**, resp. **epimorphism**.

To each projective metrized $\mathcal{O}$-module M is associated its **determinant** $\det M$, an invertible metrized $\mathcal{O}$-module. The determinant is the highest exterior power of M, i.e.,

$$\det M = \bigwedge^n M, \quad n = \operatorname{rk}(M).$$

(4.7) Proposition. *If* $0 \to M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \to 0$ *is a short exact sequence of projective metrized* $\mathcal{O}$*-modules, we have a canonical isometry*

$$\det M' \otimes_{\mathcal{O}} \det M'' \cong \det M\,.$$

Proof: Let $n' = \mathrm{rk}(M')$ and $n'' = \mathrm{rk}(M'')$. We obtain an isomorphism

$$\kappa : \det M' \otimes_{\mathcal{O}} \det M'' \xrightarrow{\sim} \det M$$

of projective $\mathcal{O}$-modules of rank 1 by mapping

$$(m_1' \wedge \ldots \wedge m_{n'}') \otimes (m_1'' \wedge \ldots \wedge m_{n''}'') \mapsto \alpha m_1' \wedge \ldots \wedge \alpha m_{n'}' \wedge \tilde{m}_1'' \wedge \ldots \wedge \tilde{m}_{n''}'',$$

where $\tilde{m}_1'', \ldots, \tilde{m}_{n''}''$ are preimages of $m_1'', \ldots, m_{n''}''$ under $\beta : M \to M''$. This mapping does not depend on the choice of the preimages, for if, say, $\tilde{m}_1'' + \alpha m_{n'+1}'$, where $m_{n'+1}' \in M'$, is another preimage of m_1'', then

$$\begin{aligned}\alpha m_1' \wedge \ldots \wedge \alpha m_{n'}' \wedge (\tilde{m}_1'' + \alpha m_{n'+1}') \wedge \tilde{m}_2'' \wedge \ldots \wedge \tilde{m}_{n''}''&\\ = \alpha m_1' \wedge \ldots \wedge \alpha m_{n'}' \wedge \tilde{m}_1'' \wedge \ldots \wedge \tilde{m}_{n''}''&\end{aligned}$$

since $\alpha m_1' \wedge \ldots \wedge \alpha m_{n'}' \wedge \alpha m_{n'+1}' = 0$. We show that the $\mathcal{O}$-module isomorphism κ is an isometry. According to the rules of multilinear algebra it induces an isomorphism

$$\kappa : \det M_{\mathbb{C}}' \otimes_{K_{\mathbb{C}}} \det M_{\mathbb{C}}'' \xrightarrow{\sim} \det M_{\mathbb{C}}$$

of $K_{\mathbb{C}}$-modules. Let $x_i', y_i' \in M_{\mathbb{C}}'$, $i = 1, \ldots, n'$, and $x_j, y_j \in \alpha M'^{\perp}_{\mathbb{C}}$, $j = 1, \ldots, n''$, and furthermore

$$x' = \textstyle\bigwedge_i x_i', \quad y' = \bigwedge_i y_i'; \quad x = \bigwedge_j x_j, \quad y = \bigwedge_j y_j\,.$$

Then we have

$$\begin{aligned}\langle \kappa(x' \otimes \beta x), \kappa(y' \otimes \beta y)\rangle_{\det M} &= \langle \alpha x' \wedge x, \alpha y' \wedge y\rangle_{\det M}\\ &= \det\left(\begin{array}{c|c} \langle x_i', y_k'\rangle_{M'} & 0 \\ \hline 0 & \langle \beta x_j, \beta y_\ell\rangle_{M''}\end{array}\right)\\ &= \det\big(\langle x_i', y_k'\rangle_{M'}\big)\det\big(\langle \beta x_j, \beta y_\ell\rangle_{M''}\big)\\ &= \langle x', y'\rangle_{\det M'} \langle \beta x, \beta y\rangle_{\det M''}\\ &= \langle x' \otimes \beta x, y' \otimes \beta y\rangle_{\det M' \otimes \det M''}\,.\end{aligned}$$

Thus κ is an isometry. $\square$

Exercise 1. If M,N,L are metrized $\mathcal{O}$-modules, then one has canonical isometries

$$M\otimes_{\mathcal{O}} N \cong N\otimes_{\mathcal{O}} M, \quad (M\otimes_{\mathcal{O}} N)\otimes_{\mathcal{O}} L \cong M\otimes_{\mathcal{O}}(N\otimes_{\mathcal{O}} L),$$

$$M\otimes_{\mathcal{O}}(N\oplus L) \cong (M\otimes_{\mathcal{O}} N)\oplus(M\otimes_{\mathcal{O}} L).$$

Exercise 2. For any two projective metrized $\mathcal{O}$-modules M,N, one has

$$\bigwedge^n(M\oplus N) \cong \bigoplus_{i+j=n}\bigwedge^i M\otimes\bigwedge^j N.$$

Exercise 3. For any two projective metrized $\mathcal{O}$-modules M,N, one has

$$\det(M\otimes_{\mathcal{O}} N) \cong (\det M)^{\otimes \operatorname{rk}(N)}\otimes_{\mathcal{O}}(\det N)^{\otimes \operatorname{rk}(M)}.$$

Exercise 4. If M is a projective metrized $\mathcal{O}$-module of rank n, and $p\geq 0$, then there is a canonical isometry

$$\det(\bigwedge^p M) \cong (\det M)^{\otimes\binom{n-1}{p-1}}.$$

§ 5. Grothendieck Groups

We will now manufacture two abelian groups from the collection of all metrized $\mathcal{O}$-modules, resp. the collection of all projective metrized $\mathcal{O}$-modules. We denote by $\{M\}$ the isometry class of a metrized $\mathcal{O}$-module M and form the free abelian group

$$F_0(\bar{\mathcal{O}}) = \bigoplus_{\{M\}}\mathbb{Z}\{M\}, \quad \text{resp.} \quad F^0(\bar{\mathcal{O}}) = \bigoplus_{\{M\}}\mathbb{Z}\{M\},$$

on the isometry classes of projective, resp. coherent, metrized $\mathcal{O}$-modules. In this group, we consider the subgroup

$$R_0(\bar{\mathcal{O}})\subseteq F_0(\bar{\mathcal{O}}), \quad \text{resp.} \quad R^0(\bar{\mathcal{O}})\subseteq F^0(\bar{\mathcal{O}}),$$

generated by all elements $\{M'\}-\{M\}+\{M''\}$ which arise from a short exact sequence

$$0\longrightarrow M'\longrightarrow M\longrightarrow M''\longrightarrow 0$$

of projective, resp. coherent, metrized $\mathcal{O}$-modules.

(5.1) Definition. *The quotient groups*

$$K_0(\bar{\mathcal{O}}) = F_0(\bar{\mathcal{O}})/R_0(\bar{\mathcal{O}}), \quad \textit{resp.} \quad K^0(\bar{\mathcal{O}}) = F^0(\bar{\mathcal{O}})/R^0(\bar{\mathcal{O}})$$

are called the **replete** (*or* **compactified**) **Grothendieck groups** *of* $\mathcal{O}$. *If M is a metrized $\mathcal{O}$-module, then $[M]$ denotes the class it defines in $K_0(\bar{\mathcal{O}})$, resp. $K^0(\bar{\mathcal{O}})$.*

The construction of the Grothendieck groups is such that a short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

of metrized $\mathcal{O}$-modules becomes an additive decomposition in the group:

$$[M] = [M'] + [M''].$$

In particular, one has

$$[M' \oplus M''] = [M'] + [M''].$$

The tensor product even induces a ring structure on $K_0(\bar{\mathcal{O}})$, and $K^0(\bar{\mathcal{O}})$ then becomes a $K_0(\bar{\mathcal{O}})$-module: extending the product

$$\{M\}\{M'\} := \{M \otimes_{\mathcal{O}} M'\}$$

by linearity, and observing that $N \otimes M \cong M \otimes N$ and $(M \otimes N) \otimes L \cong M \otimes (N \otimes L)$, we find right away that $F^0(\bar{\mathcal{O}})$ is a commutative ring and $F_0(\bar{\mathcal{O}})$ is a subring. Furthermore the subgroups $R_0(\bar{\mathcal{O}}) \subseteq F_0(\bar{\mathcal{O}})$ and $R^0(\bar{\mathcal{O}}) \subseteq F^0(\bar{\mathcal{O}})$ turn out to be $F_0(\bar{\mathcal{O}})$-submodules. For if

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is a short exact sequence of coherent metrized $\mathcal{O}$-modules, and N is a projective metrized $\mathcal{O}$-module, then it is clear that

$$0 \longrightarrow N \otimes M' \longrightarrow N \otimes M \longrightarrow N \otimes M'' \longrightarrow 0$$

is a short exact sequence of metrized $\mathcal{O}$-modules as well, so that, along with a generator $\{M'\} - \{M\} + \{M''\}$, the element

$$\{N\}\big(\{M'\} - \{M\} + \{M''\}\big) = \{N \otimes M'\} - \{N \otimes M\} + \{N \otimes M''\}$$

will also belong to $R_0(\bar{\mathcal{O}})$, resp. $R^0(\bar{\mathcal{O}})$. This is why $K_0(\bar{\mathcal{O}}) = F_0(\bar{\mathcal{O}})/R_0(\bar{\mathcal{O}})$ is a ring and $K^0(\bar{\mathcal{O}}) = F^0(\bar{\mathcal{O}})/R^0(\bar{\mathcal{O}})$ is a $K_0(\bar{\mathcal{O}})$-module.

Associating to the class $[M]$ of a projective $\mathcal{O}$-module M in $K_0(\bar{\mathcal{O}})$ its class in $K^0(\bar{\mathcal{O}})$ (which again is denoted by $[M]$), defines a homomorphism

$$K_0(\bar{\mathcal{O}}) \longrightarrow K^0(\bar{\mathcal{O}}).$$

It is called the **Poincaré homomorphism**. We will show next that the Poincaré homomorphism is an isomorphism. The proof is based on the following two lemmas.

(5.2) Lemma. *All coherent metrized $\mathcal{O}$-modules M admit a "metrized projective resolution", i.e., a short exact sequence*

$$0 \longrightarrow E \longrightarrow F \longrightarrow M \longrightarrow 0$$

of metrized $\mathcal{O}$-modules in which E and F are projective.

Proof: If $\alpha_1, \dots, \alpha_n$ is a system of generators of M, and F is the free $\mathcal{O}$-module $F = \mathcal{O}^n$, then

$$F \longrightarrow M, \quad (x_1, \dots, x_n) \longmapsto \sum_{i=1}^{n} x_i \alpha_i ,$$

is a surjective $\mathcal{O}$-module homomorphism. Its kernel E is torsion free, and hence a projective $\mathcal{O}$-module by (4.3). In the exact sequence

$$0 \longrightarrow E_{\mathbb{C}} \longrightarrow F_{\mathbb{C}} \xrightarrow{f} M_{\mathbb{C}} \longrightarrow 0,$$

we choose a section $s : M_{\mathbb{C}} \to F_{\mathbb{C}}$ of f, so that $F_{\mathbb{C}} = E_{\mathbb{C}} \oplus sM_{\mathbb{C}}$. We obtain a metric on $F_{\mathbb{C}}$ by transferring the metric of $M_{\mathbb{C}}$ to $sM_{\mathbb{C}}$, and by choosing any metric on $E_{\mathbb{C}}$. This makes $0 \to E \to F \to M \to 0$ into a short exact sequence of metrized $\mathcal{O}$-modules in which E and F are projective. □

In a diagram of metrized projective resolutions of M

$$\begin{array}{ccccccccc} 0 & \longrightarrow & E & \longrightarrow & F & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & E' & \longrightarrow & F' & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

the resolution in the top line will be called *dominant* if the vertical arrows are admissible epimorphisms.

(5.3) Lemma. *Let*

$$0 \longrightarrow E' \longrightarrow F' \xrightarrow{f'} M \longrightarrow 0, \qquad 0 \longrightarrow E'' \longrightarrow F'' \xrightarrow{f''} M \longrightarrow 0$$

be two metrized projective resolutions of the metrized $\mathcal{O}$-module M. Then, taking the $\mathcal{O}$-module

$$F = F' \times_M F'' = \left\{ (x', x'') \in F' \times F'' \,\middle|\, f'(x') = f''(x'') \right\}$$

and the mapping $f : F \to M$, $(x', x'') \mapsto f'(x') = f''(x'')$, one obtains a third metrized projective resolution

$$0 \longrightarrow E \longrightarrow F \longrightarrow M \longrightarrow 0$$

with kernel $E = E' \times E''$ which dominates both given ones.

Proof: Since $F' \oplus F''$ is projective, so is F, being the kernel of the homomorphism $F' \oplus F'' \xrightarrow{f'-f''} M$. Thus E is also projective, being the kernel of $F \to M$. We consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & E'_{\mathbb{C}} \arrow[r] & F'_{\mathbb{C}} \arrow[r, shift left, "f'"] & M_{\mathbb{C}} \arrow[l, dashed, shift left, "s'"] \arrow[r] & 0 \\
0 \arrow[r] & E_{\mathbb{C}} \arrow[r] \arrow[u] \arrow[d] & F_{\mathbb{C}} \arrow[r, shift left, "f"] \arrow[u] \arrow[d] & M_{\mathbb{C}} \arrow[l, dashed, shift left, "s"] \arrow[r] \arrow[u, equal] \arrow[d, equal] & 0 \\
0 \arrow[r] & E''_{\mathbb{C}} \arrow[r] & F''_{\mathbb{C}} \arrow[r, shift left, "f''"] & M_{\mathbb{C}} \arrow[l, dashed, shift left, "s''"] \arrow[r] & 0,
\end{tikzcd}$$

where the vertical arrows are induced by the surjective projections

$$F \xrightarrow{\pi'} F', \quad F \xrightarrow{\pi''} F''.$$

The canonical isometries

$$s' : M_{\mathbb{C}} \longrightarrow s'M_{\mathbb{C}}, \quad s'' : M_{\mathbb{C}} \longrightarrow s''M_{\mathbb{C}}$$

give a section

$$s : M_{\mathbb{C}} \longrightarrow F_{\mathbb{C}}, \quad sx = (s'x, s''x),$$

of F which transfers the metric on $M_{\mathbb{C}}$ to a metric on $sM_{\mathbb{C}}$. $E_{\mathbb{C}} = E'_{\mathbb{C}} \times E''_{\mathbb{C}}$ carries the sum of the metrics of $E'_{\mathbb{C}}$, $E''_{\mathbb{C}}$, so that $F_{\mathbb{C}} = E_{\mathbb{C}} \oplus sM_{\mathbb{C}}$ also receives a metric, and

$$0 \longrightarrow E \longrightarrow F \longrightarrow M \longrightarrow 0$$

becomes a metrized projective resolution of M. It is trivial that the projections $E \to E'$, and $E \to E''$ are admissible epimorphisms, and it remains to show this for the projections $\pi' : F \to F'$, $\pi'' : F \to F''$. But we clearly have the exact sequence of $\mathcal{O}$-modules

$$0 \longrightarrow E'' \xrightarrow{i} F = F' \times_M F'' \xrightarrow{\pi'} F' \longrightarrow 0,$$

where $ix'' = (0, x'')$. As the restriction of the metric of F to $E = E' \times E''$ is the sum of the metrics on E' and E'', we see that $i : E''_{\mathbb{C}} \to iE''_{\mathbb{C}}$ is an isometry. The orthogonal complement of $iE''_{\mathbb{C}}$ in $F_{\mathbb{C}}$ is the space

$$F'_{\mathbb{C}} \times_{M_{\mathbb{C}}} s''M_{\mathbb{C}} = \left\{ (x', s''a) \in F'_{\mathbb{C}} \times s''M_{\mathbb{C}} \mid f'(x') = a \right\}.$$

Indeed, on the one hand it is clearly mapped bijectively onto $F'_{\mathbb{C}}$, and on the other hand it is orthogonal to $iE''_{\mathbb{C}}$. For if we write $x' = s'a + e'$, with $e' \in E'_{\mathbb{C}}$, then

$$(x', s''a) = sa + (e', 0),$$

where $(e',0) \in E_{\mathbb{C}}$ and we find that, for all $x'' \in E''_{\mathbb{C}}$,

$$\big\langle ix'', (x', s''a)\big\rangle_F = \big\langle (0, x''), sa\big\rangle_F + \big\langle (0, x''), (e', 0)\big\rangle_E = 0.$$

Finally, the projection $F'_{\mathbb{C}} \times_{M_{\mathbb{C}}} s''M_{\mathbb{C}} \to F'_{\mathbb{C}}$ is an isometry, for if $(x', s''a)$, $(y', s''b) \in F'_{\mathbb{C}} \times_{M_{\mathbb{C}}} s''M_{\mathbb{C}}$ and $x' = s'a + e'$, $y' = s'b + d'$, with $e', d' \in E'_{\mathbb{C}}$, then we get

$$(x', s''a) = sa + (e', 0), \quad (y', s''b) = sb + (d', 0)$$

and

$$\begin{aligned}\big\langle (x', s''a), (y', s''b)\big\rangle_F &= \big\langle sa, sb\big\rangle_F + \big\langle sa, (d', 0)\big\rangle_F + \big\langle (e', 0), sb\big\rangle_F \\ &\quad + \big\langle (e', 0), (d', 0)\big\rangle_E \\ &= \langle a, b\rangle_M + \langle e', d'\rangle_{E'} = \langle s'a, s'b\rangle_{F'} + \langle e', d'\rangle_{E'} \\ &= \langle s'a + e', s'b + d'\rangle_{F'} = \langle x', y'\rangle_{F'}.\end{aligned}$$

□

(5.4) Theorem. *The Poincaré homomorphism*

$$K_0(\bar{\mathcal{O}}) \longrightarrow K^0(\bar{\mathcal{O}})$$

is an isomorphism.

Proof: We define a mapping

$$\pi : F^0(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{O}})$$

by choosing, for every coherent metrized $\mathcal{O}$-module M, a metrized projective resolution

$$0 \longrightarrow E \longrightarrow F \longrightarrow M \longrightarrow 0$$

and associating to the class $\{M\}$ in $F^0(\bar{\mathcal{O}})$ the difference $[F] - [E]$ of the classes $[F]$ and $[E]$ in $K_0(\bar{\mathcal{O}})$. To see that this mapping is well-defined let us first consider a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & E & \longrightarrow & F & \longrightarrow & M & \longrightarrow & 0 \\ & & \big\downarrow{\scriptstyle\alpha} & & \big\downarrow{\scriptstyle\beta} & & \Vert & & \\ 0 & \longrightarrow & E' & \longrightarrow & F' & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

of two metrized projective resolutions of M, with the top one dominating the bottom one. Then $E \to F$ induces an isometry $\ker(\alpha) \xrightarrow{\sim} \ker(\beta)$, so that we have the following identity in $K_0(\bar{\mathcal{O}})$:

$$[F] - [E] = [F'] + [\ker(\beta)] - [E'] - [\ker(\alpha)] = [F'] - [E'].$$

If now $0 \to E' \to F' \to M \to 0$, and $0 \to E'' \to F'' \to M \to 0$ are two arbitrary metrized projective resolutions of M, then by (5.3) we find a third one, $0 \to E \to F \to M \to 0$, dominating both, such that

$$[F'] - [E'] = [F] - [E] = [F''] - [E''].$$

This shows that the map $\pi : F^0(\bar{\mathcal{O}}) \to K_0(\bar{\mathcal{O}})$ is well-defined. We now show that it factorizes via $K^0(\bar{\mathcal{O}}) = F^0(\bar{\mathcal{O}})/R^0(\bar{\mathcal{O}})$. Let $0 \to M' \to M \xrightarrow{\alpha} M'' \to 0$ be a short exact sequence of metrized coherent $\mathcal{O}$-modules. By (5.2), we can pick a metrized projective resolution $0 \to E \to F \xrightarrow{f} M \to 0$. Then clearly $0 \to E'' \to F \xrightarrow{f''} M'' \to 0$ is a short exact sequence of metrized $\mathcal{O}$-modules as well, where we write $f'' = \alpha \circ f$ and $E'' = \ker(f'')$. We thus get the commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & E & \longrightarrow & F & \xrightarrow{f} & M & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \alpha} & & \\
0 & \longrightarrow & E'' & \longrightarrow & F & \xrightarrow{f''} & M'' & \longrightarrow & 0
\end{array}$$

and the snake lemma gives the exact sequence of $\mathcal{O}$-modules

$$0 \longrightarrow E \longrightarrow E'' \xrightarrow{f} M' \longrightarrow 0.$$

It is actually a short exact sequence of metrized $\mathcal{O}$-modules, for $E_{\mathbb{C}}^{\perp}$ is mapped isometrically by f onto M, so that $E''^{\perp}_{\mathbb{C}} \subseteq E_{\mathbb{C}}^{\perp}$ is mapped isometrically by f onto $M''_{\mathbb{C}}$. We therefore obtain in $K_0(\bar{\mathcal{O}})$ the identity

$$\pi\{M'\} - \pi\{M\} + \pi\{M''\} = [E''] - [E] - \big([F] - [E]\big) + [F] - [E''] = 0.$$

It shows that $\pi : F^0(\bar{\mathcal{O}}) \to K_0(\bar{\mathcal{O}})$ does indeed factorize via a homomorphism

$$K^0(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{O}}).$$

It is the inverse of the Poincaré homomorphism because the composed maps

$$K_0(\bar{\mathcal{O}}) \longrightarrow K^0(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{O}}) \quad \text{and} \quad K^0(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{O}}) \longrightarrow K^0(\bar{\mathcal{O}})$$

are the identity homomorphisms. Indeed, if $0 \to E \to F \to M \to 0$ is a projective resolution of M, and M is projective, resp. coherent, then in $K_0(\bar{\mathcal{O}})$, resp. $K^0(\bar{\mathcal{O}})$, one has the identity $[M] = [F] - [E]$. □

The preceding theorem shows that the Grothendieck group $K_0(\bar{\mathcal{O}})$ does not just accommodate all projective metrized $\mathcal{O}$-modules, but in fact all coherent metrized $\mathcal{O}$-modules. This fact has fundamental significance. For when dealing with projective modules, one is led very quickly to non-projective modules, for instance, to the residue class rings $\mathcal{O}/\mathfrak{a}$. The corresponding classes in $K^0(\bar{\mathcal{O}})$, however, can act out their important rôles only inside the ring $K_0(\bar{\mathcal{O}})$, because only this ring can be immediately subjected to a more advanced theory.

The following relationship holds between the Grothendieck ring $K_0(\bar{\mathcal{O}})$ and the replete Picard group $Pic(\bar{\mathcal{O}})$, which was introduced in § 1.

(5.5) Proposition. *Associating to a replete ideal $\mathfrak{a}$ of K the metrized $\mathcal{O}$-module $L(\mathfrak{a})$ yields a homomorphism*

$$Pic(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{O}})^*, \quad [\mathfrak{a}] \longmapsto [L(\mathfrak{a})],$$

into the unit group of the ring $K_0(\bar{\mathcal{O}})$.

Proof: The correspondence $[\mathfrak{a}] \mapsto [L(\mathfrak{a})]$ is independent of the choice of a replete ideal $\mathfrak{a}$ inside the class $[\mathfrak{a}] \in Pic(\bar{\mathcal{O}})$. Indeed, if $\mathfrak{b}$ is another representative, then we have $\mathfrak{a} = \mathfrak{b}[a]$, for some replete principal ideal $[a]$, and the metrized $\mathcal{O}$-modules $L(\mathfrak{a})$ and $L(\mathfrak{b})$ are isometric by (4.5), (i), so that $[L(\mathfrak{a})] = [L(\mathfrak{b})]$. The correspondence is a multiplicative homomorphism as

$$[L(\mathfrak{a}\mathfrak{b})] = [L(\mathfrak{a}) \otimes_{\mathcal{O}} L(\mathfrak{b})] = [L(\mathfrak{a})][L(\mathfrak{b})]. \qquad \square$$

In the sequel, we simply denote the class of a metrized invertible $\mathcal{O}$-module $L(\mathfrak{a})$ in $K_0(\bar{\mathcal{O}})$ by $[\mathfrak{a}]$. In particular, to the replete ideal $\mathcal{O} = \prod_{\mathfrak{p}} \mathfrak{p}^0$ corresponds the class $\mathbf{1} = [\mathcal{O}]$ of the $\mathcal{O}$-module $\mathcal{O}$ equipped with the trivial metric.

(5.6) Proposition. *$K_0(\bar{\mathcal{O}})$ is generated as an additive group by the elements $[\mathfrak{a}]$.*

Proof: Let M be a projective metrized $\mathcal{O}$-module. By (4.3), the underlying $\mathcal{O}$-module admits as quotient a fractional ideal $\mathfrak{a}_{\mathfrak{f}}$, i.e., we have an exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow \mathfrak{a}_{\mathfrak{f}} \longrightarrow 0$$

of $\mathcal{O}$-modules. This becomes an exact sequence of metrized $\mathcal{O}$-modules once we restrict the metric from M to N and choose on $\mathfrak{a}_{\mathfrak{f}}$ the metric which is transferred via the isomorphism $N_{\mathbb{C}}^{\perp} \cong \mathfrak{a}_{\mathfrak{f}\mathbb{C}}$. Thus $\mathfrak{a}_{\mathfrak{f}}$ becomes the metrized $\mathcal{O}$-module $L(\mathfrak{a})$ corresponding to the replete ideal $\mathfrak{a}$ of K, so that we get the identity $[M] = [N] + [\mathfrak{a}]$ in $K_0(\bar{\mathcal{O}})$. Induction on the rank shows that for every projective metrized $\mathcal{O}$-module M, there is a decomposition

$$[M] = [\mathfrak{a}_1] + \cdots + [\mathfrak{a}_r]. \qquad \square$$

The elements $[\mathfrak{a}]$ in $K_0(\bar{\mathcal{O}})$ satisfy the following remarkable relation.

(5.7) Proposition. *For any two replete ideals $\mathfrak{a}$ and $\mathfrak{b}$ of K we have in $K_0(\bar{\mathcal{O}})$ the equation*

$$([\mathfrak{a}] - 1)([\mathfrak{b}] - 1) = 0.$$

Proof (*Tamme*): For every function $\alpha : X(\mathbb{C}) \to \mathbb{C}$ let us consider on the $K_{\mathbb{C}}$-module $K_{\mathbb{C}} = \bigoplus_{\sigma \in X(\mathbb{C})} \mathbb{C}$ the form

$$\alpha x \bar{y} = \bigoplus_{\sigma} \alpha(\sigma) x_\sigma \bar{y}_\sigma.$$

For every matrix $A = \begin{pmatrix} \alpha & \gamma \\ \delta & \beta \end{pmatrix}$ of such functions, we consider on the $K_{\mathbb{C}}$-module $K_{\mathbb{C}} \oplus K_{\mathbb{C}}$ the form

$$\langle x \oplus y, x' \oplus y' \rangle_A = \alpha x \bar{x}' + \gamma x \bar{y}' + \delta y \bar{x}' + \beta y \bar{y}'.$$

$\alpha x \bar{y}$, resp. $\langle\ ,\ \rangle_A$, is an F-invariant metric on $K_{\mathbb{C}}$, resp. on $K_{\mathbb{C}} \oplus K_{\mathbb{C}}$, if and only if α is F-invariant (i.e., $\overline{\alpha(\sigma)} = \alpha(\bar{\sigma})$) and $\alpha(\sigma) \in \mathbb{R}_+^*$, resp. if all the functions $\alpha, \beta, \gamma, \delta$ are F-invariant, $\alpha(\sigma), \beta(\sigma) \in \mathbb{R}_+^*$ and $\delta = \bar{\gamma}$, and if moreover $\det A = \alpha\beta - \gamma\bar{\gamma} > 0$. We now assume this in what follows.

Let $\mathfrak{a}$ and $\mathfrak{b}$ be fractional ideals of K. We have to prove the formula

$$[\mathfrak{a}] + [\mathfrak{b}] = [\mathfrak{a}\mathfrak{b}] + 1.$$

We may assume that $\mathfrak{a}_{\mathfrak{f}}$ and $\mathfrak{b}_{\mathfrak{f}}$ are *integral* ideals *relatively prime* to one another, because if necessary we may pass to replete ideals $\mathfrak{a}' = \mathfrak{a}[a]$, $\mathfrak{b}' = \mathfrak{b}[b]$ with corresponding ideals $\mathfrak{a}'_{\mathfrak{f}} = \mathfrak{a}_{\mathfrak{f}} a$, $\mathfrak{b}'_{\mathfrak{f}} = \mathfrak{b}_{\mathfrak{f}} b$ without changing the classes $[\mathfrak{a}], [\mathfrak{b}], [\mathfrak{a}\mathfrak{b}]$ in $K_0(\bar{\mathcal{O}})$. We denote the $\mathcal{O}$-module $\mathfrak{a}_{\mathfrak{f}}$, when metrized by $\alpha x \bar{y}$, by $(\mathfrak{a}_{\mathfrak{f}}, \alpha)$, and the $\mathcal{O}$-module $\mathfrak{a}_{\mathfrak{f}} \oplus \mathfrak{b}_{\mathfrak{f}}$, metrized by $\langle\ ,\ \rangle_A$, for $A = \begin{pmatrix} \alpha & \gamma \\ \bar{\gamma} & \beta \end{pmatrix}$, by $(\mathfrak{a}_{\mathfrak{f}} \oplus \mathfrak{b}_{\mathfrak{f}}, A)$. Given any two matrices $A = \begin{pmatrix} \alpha & \gamma \\ \bar{\gamma} & \beta \end{pmatrix}$ and $A' = \begin{pmatrix} \alpha' & \gamma' \\ \bar{\gamma}' & \beta' \end{pmatrix}$ we write

$$A \sim A',$$

if $[(\mathfrak{a}_{\mathfrak{f}} \oplus \mathfrak{b}_{\mathfrak{f}}), A] = [(\mathfrak{a}_{\mathfrak{f}} \oplus \mathfrak{b}_{\mathfrak{f}}), A']$ in $K_0(\bar{\mathcal{O}})$. We now consider the canonical exact sequence

$$0 \longrightarrow \mathfrak{a}_{\mathfrak{f}} \longrightarrow \mathfrak{a}_{\mathfrak{f}} \oplus \mathfrak{b}_{\mathfrak{f}} \longrightarrow \mathfrak{b}_{\mathfrak{f}} \longrightarrow 0.$$

Once we equip $\mathfrak{a}_{\mathfrak{f}}\oplus\mathfrak{b}_{\mathfrak{f}}$ with the metric $\langle\ ,\ \rangle_A$ which is given by $A=\begin{pmatrix}\alpha & \gamma\\ \bar\gamma & \beta\end{pmatrix}$, we obtain the following exact sequence of metrized $\mathcal{O}$-modules:

$$(*)\qquad 0\longrightarrow(\mathfrak{a}_{\mathfrak{f}},\alpha)\longrightarrow(\mathfrak{a}_{\mathfrak{f}}\oplus\mathfrak{b}_{\mathfrak{f}},A)\longrightarrow\left(\mathfrak{b}_{\mathfrak{f}},\beta-\frac{\gamma\bar\gamma}{\alpha}\right)\longrightarrow 0.$$

Indeed, in the exact sequence

$$0\longrightarrow K_{\mathbb{C}}\longrightarrow K_{\mathbb{C}}\oplus K_{\mathbb{C}}\longrightarrow K_{\mathbb{C}}\longrightarrow 0,$$

the restriction of $\langle\ ,\ \rangle_A$ to $K_{\mathbb{C}}\oplus\{0\}$ yields the metric $\alpha x\bar y$ on $K_{\mathbb{C}}$, and the orthogonal complement V of $K_{\mathbb{C}}\oplus\{0\}$ consists of all elements $a+b\in K_{\mathbb{C}}\oplus K_{\mathbb{C}}$ such that

$$\langle x\oplus 0,a\oplus b\rangle=\alpha x\bar a+\gamma x\bar b=0,$$

for all $x\in K_{\mathbb{C}}$, so that

$$V=\left\{(-\bar\gamma/\alpha)b\oplus b\,\middle|\,b\in K_{\mathbb{C}}\right\}.$$

The isomorphism $V\xrightarrow{\pi}K_{\mathbb{C}}$, $(-\bar\gamma/\alpha)b\oplus b\longmapsto b$, transfers the metric $\langle\ ,\ \rangle_A$ on V into the metric $\delta x\bar y$, where δ is determined by the rule

$$\begin{aligned}\delta&=\left\langle\pi^{-1}(1),\pi^{-1}(1)\right\rangle_A=\left\langle(-\bar\gamma/\alpha)1\oplus 1,(-\bar\gamma/\alpha)1\oplus 1\right\rangle_A\\&=\alpha\frac{\bar\gamma\gamma}{\alpha^2}-\gamma\frac{\bar\gamma}{\alpha}-\bar\gamma\frac{\gamma}{\alpha}+\beta=\beta-\frac{\bar\gamma\gamma}{\alpha}.\end{aligned}$$

This shows that $(*)$ is a short exact sequence of metrized $\mathcal{O}$-modules, i.e.,

$$\begin{pmatrix}\alpha & \gamma\\ \bar\gamma & \beta\end{pmatrix}\sim\begin{pmatrix}\alpha & 0\\ 0 & \beta-\frac{\gamma\bar\gamma}{\alpha}\end{pmatrix}.$$

Replacing β by $\beta+\frac{\gamma\bar\gamma}{\alpha}$, we get

$$\begin{pmatrix}\alpha & \gamma\\ \bar\gamma & \beta+\frac{\gamma\bar\gamma}{\alpha}\end{pmatrix}\sim\begin{pmatrix}\alpha & 0\\ 0 & \beta\end{pmatrix}.$$

Applying the same procedure to the exact sequence $0\to\mathfrak{b}_{\mathfrak{f}}\to\mathfrak{a}_{\mathfrak{f}}\oplus\mathfrak{b}_{\mathfrak{f}}\to\mathfrak{a}_{\mathfrak{f}}\to 0$ and the metric $\begin{pmatrix}\alpha' & \gamma\\ \bar\gamma & \beta'\end{pmatrix}$ on $\mathfrak{a}_{\mathfrak{f}}\oplus\mathfrak{b}_{\mathfrak{f}}$, we obtain

$$\begin{pmatrix}\alpha'+\frac{\gamma\bar\gamma}{\beta'} & \gamma\\ \bar\gamma & \beta'\end{pmatrix}\sim\begin{pmatrix}\alpha' & 0\\ 0 & \beta'\end{pmatrix}.$$

Choosing

$$\beta'=\beta+\frac{\gamma\bar\gamma}{\alpha},\quad\alpha'=\frac{\alpha\beta}{\beta+\frac{\gamma\bar\gamma}{\alpha}},$$

makes the matrices on the left equal, and yields

$$\begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} \sim \begin{pmatrix} \frac{\alpha\beta}{\beta+\frac{\gamma\bar{\gamma}}{\alpha}} & 0 \\ 0 & \beta+\frac{\gamma\bar{\gamma}}{\alpha} \end{pmatrix},$$

or, if we put $\delta = \beta + \frac{\gamma\bar{\gamma}}{\alpha}$,

$$(**) \qquad \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} \sim \begin{pmatrix} \frac{\alpha\beta}{\delta} & 0 \\ 0 & \delta \end{pmatrix},$$

which is valid for any F-invariant function $\delta : X(\mathbb{C}) \to \mathbb{R}$ such that $\delta \geq \beta$. This implies furthermore

$$(***) \qquad \begin{pmatrix} \frac{\alpha\beta}{\delta} & 0 \\ 0 & \delta \end{pmatrix} \sim \begin{pmatrix} \frac{\alpha\beta}{\varepsilon} & 0 \\ 0 & \varepsilon \end{pmatrix}$$

for any two F-invariant functions $\delta, \varepsilon : X(\mathbb{C}) \to \mathbb{R}_+^*$. For if $\kappa : X(\mathbb{C}) \to \mathbb{R}$ is an F-invariant function such that $\kappa \geq \delta, \kappa \geq \varepsilon$, then $(**)$ gives

$$\begin{pmatrix} \frac{\alpha\beta}{\delta} & 0 \\ 0 & \delta \end{pmatrix} \sim \begin{pmatrix} \frac{\alpha\beta}{\kappa} & 0 \\ 0 & \kappa \end{pmatrix} \sim \begin{pmatrix} \frac{\alpha\beta}{\varepsilon} & 0 \\ 0 & \varepsilon \end{pmatrix}.$$

Now putting $\delta = \beta$ and $\varepsilon = 1$ in $(***)$, we find

$$[(\mathfrak{a}_f, \alpha)] + [(\mathfrak{b}_f, \beta)] = [(\mathfrak{a}_f, \alpha\beta)] + [\mathfrak{b}_f].$$

For the replete ideals $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$, $\mathfrak{b} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$, this means

$$(1) \qquad [\mathfrak{a}] + [\mathfrak{b}] = [\mathfrak{a}\mathfrak{b}_\infty] + [\mathfrak{b}_f],$$

for if we put $\alpha(\sigma) = e^{2\nu_{\mathfrak{p}\sigma}}$, $\beta(\sigma) = e^{2\mu_{\mathfrak{p}\sigma}}$, then we have

$$(\mathfrak{a}_f, \alpha) = L(\mathfrak{a}), \quad (\mathfrak{b}_f, \beta) = L(\mathfrak{b}), \quad (\mathfrak{a}_f, \alpha\beta) = L(\mathfrak{a}\mathfrak{b}_\infty).$$

On the other hand, we obtain the formula

$$(2) \qquad [\mathfrak{a}] + [\mathfrak{b}_f] = [\mathfrak{a}\mathfrak{b}_f] + 1$$

in the following manner. We have two exact sequences of *coherent* metrized $\mathcal{O}$-modules:

$$0 \longrightarrow (\mathfrak{a}_f\mathfrak{b}_f, \alpha) \longrightarrow (\mathfrak{a}_f, \alpha) \longrightarrow \mathfrak{a}_f/\mathfrak{a}_f\mathfrak{b}_f \longrightarrow 0,$$

$$0 \longrightarrow (\mathfrak{b}_f, 1) \longrightarrow (\mathcal{O}, 1) \longrightarrow \mathcal{O}/\mathfrak{b}_f \longrightarrow 0.$$

As $\mathfrak{a}_f$ and $\mathfrak{b}_f$ are relatively prime, i.e., $\mathfrak{a}_f + \mathfrak{b}_f = \mathcal{O}$, it follows that

$$\mathfrak{a}_f/\mathfrak{a}_f\mathfrak{b}_f \longrightarrow \mathcal{O}/\mathfrak{b}_f$$

is an isomorphism, so that in the group $K^0(\bar{\mathcal{O}})$ one has the identity $[\mathfrak{a}_f/\mathfrak{a}_f\mathfrak{b}_f] = [\mathcal{O}/\mathfrak{b}_f]$, and therefore

$$[(\mathfrak{a}_f, \alpha)] - [(\mathfrak{a}_f\mathfrak{b}_f, \alpha)] = [(\mathcal{O}, 1)] - [(\mathfrak{b}_f, 1)],$$

and so

$$[\mathfrak{a}] - [\mathfrak{a}\mathfrak{b}_f] = 1 - [\mathfrak{b}_f].$$

From (1) and (2) it now follows that

$$[\mathfrak{a}] + [\mathfrak{b}] = [\mathfrak{a}\mathfrak{b}_\infty] + [\mathfrak{b}_f] = [\mathfrak{a}\mathfrak{b}_\infty\mathfrak{b}_f] + 1 = [\mathfrak{a}\mathfrak{b}] + 1.$$

In view of the isomorphism $K_0(\bar{\mathcal{O}}) \cong K^0(\bar{\mathcal{O}})$, this is indeed an identity in $K_0(\bar{\mathcal{O}})$. □

§ 6. The Chern Character

The Grothendieck ring $K_0(\bar{\mathcal{O}})$ is equipped with a canonical surjective homomorphism

$$\mathrm{rk} : K_0(\bar{\mathcal{O}}) \longrightarrow \mathbb{Z}.$$

Indeed, the rule which associates to every isometry class $\{M\}$ of projective metrized $\mathcal{O}$-modules the rank

$$\mathrm{rk}\{M\} = \dim_K(M \otimes_{\mathcal{O}} K)$$

extends by linearity to a ring homomorphism $F_0(\bar{\mathcal{O}}) \to \mathbb{Z}$. For a short exact sequence $0 \to M' \to M \to M'' \to 0$ of metrized $\mathcal{O}$-modules one has $\mathrm{rk}(M) = \mathrm{rk}(M') + \mathrm{rk}(M'')$, and so $\mathrm{rk}(\{M'\} - \{M\} + \{M''\}) = 0$. Thus rk is zero on the ideal $R_0(\bar{\mathcal{O}})$ and induces therefore a homomorphism $K_0(\bar{\mathcal{O}}) \to \mathbb{Z}$. It is called the **augmentation** of $K_0(\bar{\mathcal{O}})$ and its kernel $I = \ker(\mathrm{rk})$ is called the **augmentation ideal**.

(6.1) Proposition. *The ideal I, resp. I^2, is generated as an additive group by the elements $[\mathfrak{a}] - 1$, resp. $([\mathfrak{a}] - 1)([\mathfrak{b}] - 1)$, where $\mathfrak{a}, \mathfrak{b}$ vary over the replete ideals of K.*

Proof: By (5.6), every element $\xi \in K_0(\bar{\mathcal{O}})$ is of the form

$$\xi = \sum_{i=1}^{r} n_i[\mathfrak{a}_i].$$

If $\xi \in I$, then $\mathrm{rk}(\xi) = \sum_{i=1}^{r} n_i = 0$, and thus

$$\xi = \sum_i n_i[\mathfrak{a}_i] - \sum_i n_i = \sum_i n_i([\mathfrak{a}_i] - 1).$$

The ideal I^2 is therefore generated by the elements $([\mathfrak{a}] - 1)([\mathfrak{b}] - 1)$. As

$$[\mathfrak{c}]([\mathfrak{a}] - 1)([\mathfrak{b}] - 1) = \big(([\mathfrak{c}\mathfrak{a}] - 1) - ([\mathfrak{c}] - 1)\big)([\mathfrak{b}] - 1),$$

these elements already form a system of generators of the abelian group I^2. □

By (5.7), this gives us the

(6.2) Corollary. $I^2 = 0$.

We now define

$$\mathrm{gr}\, K_0(\bar{\mathcal{O}}) = \mathbb{Z} \oplus I$$

and turn this additive group into a ring by putting $xy = 0$ for $x, y \in I$.

(6.3) Definition. *The additive homomorphism*

$$c_1 : K_0(\bar{\mathcal{O}}) \longrightarrow I, \quad c_1(\xi) = \xi - \mathrm{rk}(\xi)$$

is called the **first Chern class**. *The mapping*

$$\mathrm{ch} : K_0(\bar{\mathcal{O}}) \to \mathrm{gr}\, K_0(\bar{\mathcal{O}}), \quad \mathrm{ch}(\xi) = \mathrm{rk}(\xi) + c_1(\xi),$$

is called the **Chern character** *of $K_0(\bar{\mathcal{O}})$.*

(6.4) Proposition. *The Chern character*

$$\mathrm{ch} : K_0(\bar{\mathcal{O}}) \longrightarrow \mathrm{gr}\, K_0(\bar{\mathcal{O}})$$

is an isomorphism of rings.

Proof: The mappings rk and c_1 are homomorphisms of additive groups, and both are also multiplicative. For rk this is clear, and for c_1 it is enough to check it on the generators $x = [\mathfrak{a}]$, $y = [\mathfrak{b}]$. This works because

$$c_1(xy) = xy - 1 = (x - 1) + (y - 1) + (x - 1)(y - 1) = c_1(x) + c_1(y),$$

because $(x - 1)(y - 1) = 0$ by (5.7). Therefore ch is a ring homomorphism. The mapping

$$\mathbb{Z} \oplus I \longrightarrow K_0(\bar{\mathcal{O}}), \quad n \oplus \xi \longmapsto \xi + n,$$

is obviously an inverse mapping, so that ch is even an isomorphism. □

We obtain a complete and explicit description of the Chern character by taking into account another homomorphism, as well as the homomorphism $\mathrm{rk} : K_0(\bar{\mathcal{O}}) \to \mathbb{Z}$, namely

$$\det : K_0(\bar{\mathcal{O}}) \longrightarrow Pic(\bar{\mathcal{O}})$$

which is induced by taking determinants $\det M$ of projective $\mathcal{O}$-modules M as follows (see §4). $\det M$ is an invertible metrized $\mathcal{O}$-module, and therefore of the form $L(\mathfrak{a})$ for some replete ideal $\mathfrak{a}$, which is well determined up to isomorphism. Denoting by $[\det M]$ the class of $\mathfrak{a}$ in $Pic(\bar{\mathcal{O}})$, the linear extension of the map $\{M\} \mapsto [\det M]$ gives a homomorphism

$$\det : F_0(\bar{\mathcal{O}}) \longrightarrow Pic(\bar{\mathcal{O}}).$$

It maps the subgroup $R_0(\bar{\mathcal{O}})$ to 1, because it is generated by the elements $\{M'\} - \{M\} + \{M''\}$ which arise from short exact sequences

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

of projective metrized $\mathcal{O}$-modules and which, by (4.7), satisfy

$$\begin{aligned} \det\{M\} &= [\det M] = [\det M' \otimes \det M''] \\ &= [\det M'][\det M''] = \det\{M'\}\det\{M''\}. \end{aligned}$$

Thus we get an induced homomorphism $\det : K_0(\bar{\mathcal{O}}) \to Pic(\bar{\mathcal{O}})$. It satisfies the following proposition.

(6.5) Proposition. (i) *The canonical homomorphism*

$$Pic(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{O}})^*$$

is injective.

(ii) *The restriction of* det *to* I,

$$\det : I \longrightarrow Pic(\bar{\mathcal{O}}),$$

is an isomorphism.

Proof: (i) The composite of both mappings

$$Pic(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{O}})^* \xrightarrow{\det} Pic(\bar{\mathcal{O}})$$

is the identity, since for an invertible metrized $\mathcal{O}$-module M, one clearly has $\det M = M$. This gives (i).

(ii) Next, viewing the elements of $Pic(\bar{\mathcal{O}})$ as elements of $K_0(\bar{\mathcal{O}})$,

$$\delta : Pic(\bar{\mathcal{O}}) \longrightarrow I, \quad \delta(x) = x - 1,$$

gives us an inverse mapping to $\det : I \to Pic(\bar{\mathcal{O}})$. In fact, one has $\det \circ \delta = \mathrm{id}$ since $\det([\mathfrak{a}] - 1) = \det[\mathfrak{a}] = [\mathfrak{a}]$, and $\delta \circ \det = \mathrm{id}$ since $\delta(\det([\mathfrak{a}] - 1)) = \delta(\det[\mathfrak{a}]) = \delta([\mathfrak{a}]) = [\mathfrak{a}] - 1$ and because of the fact that I is generated by elements of the form $[\mathfrak{a}] - 1$ (see (6.1)). $\square$

From the isomorphism $\det : I \xrightarrow{\sim} Pic(\bar{o})$, we now obtain an isomorphism

$$\operatorname{gr} K_0(\bar{o}) \xrightarrow{\sim} \mathbb{Z} \oplus Pic(\bar{o})$$

and the composite

$$K_0(\bar{o}) \xrightarrow{\mathrm{ch}} \operatorname{gr} K_0(\bar{o}) \xrightarrow{\mathrm{id} \oplus \det} \mathbb{Z} \oplus Pic(\bar{o})$$

will again be called the Chern character of $K_0(\bar{o})$. Observing that $\det(c_1(\xi)) = \det(\xi - \operatorname{rk}(\xi) \cdot 1) = \det(\xi)$, this yields the explicit description of the Grothendieck group $K_0(\bar{o})$:

(6.6) Theorem. *The Chern character gives an isomorphism*

$$\mathrm{ch} : K_0(\bar{o}) \xrightarrow{\sim} \mathbb{Z} \oplus Pic(\bar{o}), \quad \mathrm{ch}(\xi) = \operatorname{rk}(\xi) \oplus \det(\xi).$$

The expert should note that this homomorphism is a realization map from K-theory into Chow-theory. Identifying $Pic(\bar{o})$ with the divisor class group $CH^1(\bar{o})$, we have to view $\mathbb{Z} \oplus Pic(\bar{o})$ as the "replete" *Chow ring* $CH(\bar{o})$.

§ 7. Grothendieck-Riemann-Roch

We now consider a finite extension $L|K$ of algebraic number fields and study the relations between the Grothendieck groups of L and K. Let o, resp. $\mathcal{O}$, be the ring of integers of K, resp. L, and write $X(\mathbb{C}) = \operatorname{Hom}(K, \mathbb{C})$, $Y(\mathbb{C}) = \operatorname{Hom}(L, \mathbb{C})$. The inclusion $i : o \to \mathcal{O}$ and the surjection $Y(\mathbb{C}) \to X(\mathbb{C})$, $\sigma \mapsto \sigma|_K$, give two canonical homomorphisms

$$i^* : K_0(\bar{o}) \longrightarrow K_0(\bar{\mathcal{O}}) \quad \text{and} \quad i_* : K_0(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{o}),$$

defined as follows.

If M is a projective metrized o-module, then $M \otimes_o \mathcal{O}$ is a projective $\mathcal{O}$-module. As

$$(M \otimes_o \mathcal{O})_{\mathbb{C}} = M \otimes_o \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{C} = M_{\mathbb{C}} \otimes_{K_{\mathbb{C}}} L_{\mathbb{C}},$$

the hermitian metric on the $K_{\mathbb{C}}$-module $M_{\mathbb{C}}$ extends canonically to an F-invariant metric of the $L_{\mathbb{C}}$-module $(M \otimes_o \mathcal{O})_{\mathbb{C}}$. Therefore $M \otimes_o \mathcal{O}$ is automatically a metrized $\mathcal{O}$-module, which we denote by i^*M. If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is a short exact sequence of projective metrized $\mathcal{o}$-modules, then

$$0 \longrightarrow M' \otimes_{\mathcal{o}} \mathcal{O} \longrightarrow M \otimes_{\mathcal{o}} \mathcal{O} \longrightarrow M'' \otimes_{\mathcal{o}} \mathcal{O} \longrightarrow 0$$

is a short exact sequence of metrized $\mathcal{O}$-modules, because $\mathcal{O}$ is a projective $\mathcal{o}$-module and the metrics in the sequence

$$0 \longrightarrow M'_{\mathbb{C}} \longrightarrow M_{\mathbb{C}} \longrightarrow M''_{\mathbb{C}} \longrightarrow 0$$

simply extend $L_{\mathbb{C}}$-sesquilinearly to metrics in the sequence of $L_{\mathbb{C}}$-modules

$$0 \longrightarrow M'_{\mathbb{C}} \otimes_{K_{\mathbb{C}}} L_{\mathbb{C}} \longrightarrow M_{\mathbb{C}} \otimes_{K_{\mathbb{C}}} L_{\mathbb{C}} \longrightarrow M''_{\mathbb{C}} \otimes_{K_{\mathbb{C}}} L_{\mathbb{C}} \longrightarrow 0.$$

This is why mapping, in the usual way (i.e., via the representation $K_0(\bar{\mathcal{o}}) = F_0(\bar{\mathcal{o}})/R_0(\bar{\mathcal{o}})$),

$$M \longmapsto [i^*M] = [M \otimes_{\mathcal{o}} \mathcal{O}]$$

gives a well-defined homomorphism

$$i^* : K_0(\bar{\mathcal{o}}) \longrightarrow K_0(\bar{\mathcal{O}}).$$

The reader may verify for himself that this is in fact a ring homomorphism.

On the other hand, if M is a projective metrized $\mathcal{O}$-module, then M is automatically also a projective $\mathcal{o}$-module. For the complexification $M_{\mathbb{C}} = M \otimes_{\mathbb{Z}} \mathbb{C}$ we have the decomposition

$$M_{\mathbb{C}} = \bigoplus_{\tau \in Y(\mathbb{C})} M_\tau = \bigoplus_{\sigma \in X(\mathbb{C})} \bigoplus_{\tau | \sigma} M_\tau = \bigoplus_{\sigma \in X(\mathbb{C})} M_\sigma ,$$

where $M_\tau = M \otimes_{\mathcal{O},\tau} \mathbb{C}$ and

$$M_\sigma = M \otimes_{\mathcal{o},\sigma} \mathbb{C} = \bigoplus_{\tau | \sigma} M_\tau .$$

The $\mathbb{C}$-vector spaces M_τ carry hermitian metrics $\langle\ ,\ \rangle_{M_\tau}$, and we define the metric $\langle\ ,\ \rangle_{M_\sigma}$ on the $\mathbb{C}$-vector space M_σ to be the orthogonal sum

$$\langle x, y \rangle_{M_\sigma} = \sum_{\tau | \sigma} \langle x_\tau, y_\tau \rangle_{M_\tau} .$$

This gives a hermitian metric on the $K_{\mathbb{C}}$-module $M_{\mathbb{C}}$, whose F-invariance is clearly guaranteed by the F-invariance of the original metric $\langle\ ,\ \rangle_M$. We denote the metrized $\mathcal{o}$-module M thus constructed by i_*M.

If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of projective metrized $\mathcal{O}$-modules, then

$$0 \longrightarrow i_*M' \longrightarrow i_*M \longrightarrow i_*M'' \longrightarrow 0$$

is clearly an exact sequence of projective metrized $\mathcal{o}$-modules. As before, this is why the correspondence

$$M \longmapsto [i_*M]$$

gives us a well-defined (additive) homomorphism

$$i_* : K_0(\bar{\mathcal{O}}) \longrightarrow K_0(\bar{\mathcal{o}}).$$

(7.1) Proposition (Projection Formula). *The diagram*

$$\begin{array}{ccccc} K_0(\bar{\mathcal{O}}) & \times & K_0(\bar{\mathcal{O}}) & \longrightarrow & K_0(\bar{\mathcal{O}}) \\ {\scriptstyle i_*}\big\downarrow & & \big\uparrow{\scriptstyle i^*} & & \big\downarrow{\scriptstyle i_*} \\ K_0(\bar{o}) & \times & K_0(\bar{o}) & \longrightarrow & K_0(\bar{o}) \end{array}$$

is commutative, where the horizontal arrows are multiplication.

Proof: If M, resp. N, is a projective metrized $\mathcal{O}$-module, resp. o-module, there is an isometry

$$i_*(M \otimes_{\mathcal{O}} i^*N) \cong i_*M \otimes_o N$$

of projective metrized o-modules. Indeed, we have an isomorphism of the underlying o-modules

$$M \otimes_{\mathcal{O}} (N \otimes_o \mathcal{O}) \cong M \otimes_o N\,, \quad a \otimes (b \otimes c) \mapsto ca \otimes b\,.$$

Tensoring with $\mathbb{C}$, it induces an isomorphism

$$M_{\mathbb{C}} \otimes_{L_{\mathbb{C}}} (N_{\mathbb{C}} \otimes_{K_{\mathbb{C}}} L_{\mathbb{C}}) \cong M_{\mathbb{C}} \otimes_{K_{\mathbb{C}}} N_{\mathbb{C}}\,.$$

That this is an isometry of metrized $K_{\mathbb{C}}$-modules results from the distributivity

$$\sum_{\tau|\sigma} \langle\ ,\ \rangle_{M_\tau} \langle\ ,\ \rangle_{N_\sigma} = \Big(\sum_{\tau|\sigma} \langle\ ,\ \rangle_{M_\tau}\Big) \langle\ ,\ \rangle_{N_\sigma}$$

by applying mathematical grammar. □

The Riemann-Roch problem in Grothendieck's perspective is the task of computing the Chern character $\mathrm{ch}(i_*M)$ for a projective metrized $\mathcal{O}$-module M in terms of $\mathrm{ch}(M)$. By (6.6), this amounts to computing $\det(i_*M)$ in terms of $\det M$. But $\det M$ is an invertible metrized $\mathcal{O}$-module and is therefore isometric by (4.5) to the metrized $\mathcal{O}$-module $L(\mathfrak{A})$ of a replete ideal $\mathfrak{A}$ of L. $N_{L|K}(\mathfrak{A})$ is then a replete ideal of K, and we put

$$N_{L|K}(\det M) := L\big(N_{L|K}(\mathfrak{A})\big)\,.$$

This is an invertible metrized o-module which is well determined by M up to isometry. With this notation we first establish the following theorem.

(7.2) Theorem. *For any projective metrized $\mathcal{O}$-module M one has:*

$$\mathrm{rk}(i_*M) = \mathrm{rk}(M)\,\mathrm{rk}(\mathcal{O})\,,$$

$$\det(i_*M) \cong N_{L|K}(\det M) \otimes_o (\det i_*\mathcal{O})^{\mathrm{rk}(M)}\,.$$

Here we have $\mathrm{rk}(\mathcal{O}) = [L : K]$.

Proof: One has $M_K := M \otimes_{\mathcal{o}} K = M \otimes_{\mathcal{O}} \mathcal{O} \otimes_{\mathcal{o}} K = M \otimes_{\mathcal{O}} L =: M_L$ and therefore

$$\mathrm{rk}(i_* M) = \dim_K(M_K) = \dim_K(M_L) = \dim_L(M_L)[L:K] = \mathrm{rk}(M)\,\mathrm{rk}(\mathcal{O}).$$

In order to prove the second equation, we first reduce to a special case. Let

$$\lambda(M) = \det(i_* M) \quad \text{and} \quad \rho(M) = N_{L|K}(\det M) \otimes_{\mathcal{o}} (\det i_* \mathcal{O})^{\mathrm{rk}(M)}.$$

If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of projective metrized $\mathcal{O}$-modules, one has

$$(*) \qquad \lambda(M) \cong \lambda(M') \otimes_{\mathcal{o}} \lambda(M'') \quad \text{and} \quad \rho(M) \cong \rho(M') \otimes_{\mathcal{o}} \rho(M'').$$

The isomorphism on the left follows from the exact sequence $0 \to i_* M' \to i_* M \to i_* M'' \to 0$ by (4.7), and the one on the right from (4.7) also, from the multiplicativity of the norm $N_{L|K}$ and the additivity of the rank rk. As in the proof of (5.6), we now make use of the fact that every projective metrized $\mathcal{O}$-module M projects via an admissible epimorphism onto a suitable $\mathcal{O}$-module of the form $L(\mathfrak{A})$ for some replete ideal $\mathfrak{A}$. Thus $(*)$ allows us to reduce by induction on $\mathrm{rk}(M)$ to the case $M = L(\mathfrak{A})$. Here $\mathrm{rk}(M) = 1$, so we have to establish the isomorphism

$$\det\big(i_* L(\mathfrak{A})\big) = L\big(N_{L|K}(\mathfrak{A})\big) \otimes_{\mathcal{o}} \det_{\mathcal{o}} \mathcal{O}.$$

For the underlying $\mathcal{o}$-modules this amounts to the identity

$$(**) \qquad \det_{\mathcal{o}} \mathfrak{A}_{\mathrm{f}} = N_{L|K}(\mathfrak{A}_{\mathrm{f}}) \det_{\mathcal{o}} \mathcal{O},$$

which has to be viewed as inside $\det_K L$ and which is proved as follows. If $\mathcal{O}$ and $\mathcal{o}$ were principal ideal domains, it would be obvious. In fact, in that case we could choose a generator α of $\mathfrak{A}_{\mathrm{f}}$ and an integral basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}$ over $\mathcal{o}$. Since $N_{L|K}(\alpha)$ is by definition the determinant $\det(T_\alpha)$ of the transformation $T_\alpha : L \to L$, $x \mapsto \alpha x$, we would get the equation

$$\alpha\omega_1 \wedge \ldots \wedge \alpha\omega_n = N_{L|K}(\alpha)(\omega_1 \wedge \ldots \wedge \omega_n),$$

the left-hand side, resp. right-hand side, of which would, by (1.6), generate the left-hand side, resp. right-hand side, of $(**)$. But we may always produce a principal ideal domain as desired by passing from $\mathcal{O}|\mathcal{o}$ to the localization $\mathcal{O}_{\mathfrak{p}}|\mathcal{o}_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of $\mathcal{o}$ (see chap. I, § 11 and § 3, exercise 4). The preceding argument then shows that

$$(\det_{\mathcal{o}} \mathfrak{A}_{\mathrm{f}})_{\mathfrak{p}} = \det_{\mathcal{o}_{\mathfrak{p}}} \mathfrak{A}_{\mathrm{f}_{\mathfrak{p}}} = N_{L|K}(\mathfrak{A}_{\mathrm{f}_{\mathfrak{p}}}) \det_{\mathcal{o}_{\mathfrak{p}}} \mathcal{O}_{\mathfrak{p}} = (N_{L|K}(\mathfrak{A}_{\mathrm{f}}) \det_{\mathcal{o}} \mathcal{O})_{\mathfrak{p}},$$

and since this identity is valid for all prime ideals $\mathfrak{p}$ of $\mathcal{o}$, we deduce the equality $(**)$.

In order to prove that the metrics agree on both sides of $(**)$, we put $M = L(\mathfrak{A})$, $N = L(\mathcal{O})$, $\mathfrak{a} = N_{L|K}(\mathfrak{A})$ and we view $M, N, \mathfrak{a}$ as metrized $\mathcal{o}$-modules. One has $M_{\mathbb{C}} = N_{\mathbb{C}} = L_{\mathbb{C}}$ and $\mathfrak{a}_{\mathbb{C}} = K_{\mathbb{C}}$, and we consider the metrics on the components

$$M_\sigma = \bigoplus_{\tau|\sigma} \mathbb{C}, \quad N_\sigma = \bigoplus_{\tau|\sigma} \mathbb{C}, \quad \mathfrak{a}_\sigma = \mathbb{C},$$

where $\sigma \in \mathrm{Hom}(K, \mathbb{C})$ and $\tau \in \mathrm{Hom}(L, \mathbb{C})$ is such that $\tau|\sigma$. We have to show that, for $\xi, \eta \in \det_{\mathbb{C}} M_\sigma$ and $a, b \in \mathbb{C}$, one has the identity

$$\langle a\xi, b\eta\rangle_{\det M_\sigma} = \langle a, b\rangle_{\mathfrak{a}_\sigma} \langle \xi, \eta\rangle_{\det N_\sigma}.$$

For this, let $\mathfrak{A}_\infty = \prod_{\mathfrak{P}|\infty} \mathfrak{P}^{\nu_{\mathfrak{P}}}$, so that one gets

$$\mathfrak{a}_\infty = N_{L|K}(\mathfrak{A}_\infty) = \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_{\mathfrak{p}}}$$

with $\nu_{\mathfrak{p}} = \sum_{\mathfrak{P}|\mathfrak{p}} f_{\mathfrak{P}|\mathfrak{p}} \nu_{\mathfrak{P}}$. Then

$$\langle x, y\rangle_{N_\sigma} = \sum_{\tau|\sigma} x_\tau \bar{y}_\tau, \qquad \langle x, y\rangle_{M_\sigma} = \sum_{\tau|\sigma} e^{2\nu_{\mathfrak{P}_\tau}} x_\tau \bar{y}_\tau,$$

$$\langle a, b\rangle_{\mathfrak{a}_\sigma} = e^{2\nu_{\mathfrak{p}_\sigma}} a\bar{b}, \qquad \nu_{\mathfrak{p}_\sigma} = \sum_{\mathfrak{P}|\mathfrak{p}_\sigma} f_{\mathfrak{P}|\mathfrak{p}_\sigma} \nu_{\mathfrak{P}} = \sum_{\tau|\sigma} \nu_{\mathfrak{P}_\tau}.$$

Let $\xi = x_1 \wedge \ldots \wedge x_n$, $\eta = y_1 \wedge \ldots \wedge y_n$. We number the embeddings $\tau|\sigma$, $\tau_1, \ldots, \tau_n$, put $\nu_k = \nu_{\mathfrak{P}_{\tau_k}}$ and form the matrices

$$A = (x_{i\tau_k}), \quad B = (\bar{y}_{i\tau_k}), \quad D = \begin{pmatrix} e^{\nu_1} & & 0 \\ & \ddots & \\ 0 & & e^{\nu_n} \end{pmatrix}.$$

Then, observing that

$$\det(D) = \prod_\tau e^{\nu_{\mathfrak{P}_\tau}} = \prod_{\mathfrak{P}|\mathfrak{p}_\sigma} e^{f_{\mathfrak{P}|\mathfrak{p}_\sigma} \nu_{\mathfrak{P}}} = e^{\nu_{\mathfrak{p}_\sigma}},$$

we do indeed get

$$\begin{aligned}\langle a\xi, b\eta\rangle_{\det M_\sigma} &= a\bar{b}\, \langle \xi, \eta\rangle_{\det M_\sigma} \\ &= a\bar{b} \det\big((AD)(BD)^t\big) = a\bar{b}\, (\det D)^2 \det(AB^t) \\ &= e^{2\nu_{\mathfrak{p}_\sigma}} a\bar{b}\, \langle \xi, \eta\rangle_{\det N_\sigma} = \langle a, b\rangle_{\mathfrak{a}_\sigma} \langle \xi, \eta\rangle_{\det N_\sigma}.\end{aligned}$$

This proves our theorem. □

Extending the formulas of (7.2) to the free abelian group

$$F_0(\bar{\mathcal{O}}) = \bigoplus_{\{M\}} \mathbb{Z}\{M\}$$

by linearity, and passing to the quotient group $K_0(\bar{\mathcal{O}}) = F_0(\bar{\mathcal{O}})/R_0(\bar{\mathcal{O}})$ yields the following corollary.

(7.3) Corollary. *For every class $\xi \in K_0(\overline{\mathcal{O}})$, one has the formulas*

$$\begin{aligned} \operatorname{rk}(i_*\xi) &= [L:K]\operatorname{rk}(\xi),\\ \det(i_*\xi) &= [\det i_*\mathcal{O}]^{\operatorname{rk}(\xi)} N_{L|K}(\det \xi). \end{aligned}$$

The square of the metrized o-module $\det i_*\mathcal{O}$ appearing in the second formula can be computed to be the *discriminant* $\mathfrak{d}_{L|K}$ of the extension $L|K$, which we view as a metrized o-module with the *trivial* metric.

(7.4) Proposition. *There is a canonical isomorphism*

$$(\det i_*\mathcal{O})^{\otimes 2} \cong \mathfrak{d}_{L|K}$$

of metrized o-modules.

Proof: Consider on $\mathcal{O}$ the bilinear trace map

$$T : \mathcal{O} \times \mathcal{O} \longrightarrow o, \quad (x,y) \longmapsto Tr_{L|K}(xy).$$

It induces an o-module homomorphism

$$T : \det \mathcal{O} \otimes \det \mathcal{O} \longrightarrow o,$$

given by

$$T\big((\alpha_1 \wedge \ldots \wedge \alpha_n) \otimes (\beta_1 \wedge \ldots \wedge \beta_n)\big) = \det\big(Tr_{L|K}(\alpha_i \beta_j)\big).$$

The image of T is the discriminant ideal $\mathfrak{d}_{L|K}$, which, by definition, is generated by the discriminants

$$d(\omega_1, \ldots, \omega_n) = \det\big(Tr_{L|K}(\omega_i \omega_j)\big)$$

of all bases of $L|K$ which are contained in $\mathcal{O}$. This is clear if $\mathcal{O}$ admits an integral basis over o, since the α_i and β_i can be written in terms of such a basis with coefficients in o. If there is no such integral basis, it will exist after localizing $\mathcal{O}_\mathfrak{p}|o_\mathfrak{p}$ at every prime ideal $\mathfrak{p}$ (see chap. I, (2.10)). The image of

$$T_\mathfrak{p} : (\det \mathcal{O}_\mathfrak{p}) \otimes (\det \mathcal{O}_\mathfrak{p}) \longrightarrow o_\mathfrak{p}$$

is therefore the discriminant ideal of $\mathcal{O}_\mathfrak{p}|o_\mathfrak{p}$ and at the same time the localization of the image of T. Since two ideals are equal when their localizations are, we find $\operatorname{image}(T) = \mathfrak{d}_{L|K}$. Furthermore, T has to be injective since $(\det \mathcal{O})^{\otimes 2}$ is an invertible o-module. Therefore T is an o-module isomorphism.

We now check that

$$T_{\mathbb{C}} : \big(\det(\mathcal{O})^{\otimes 2}\big)_{\mathbb{C}} \longrightarrow (\mathfrak{d}_{L|K})_{\mathbb{C}}$$

is indeed an isometry. For $\mathcal{O}_{\mathbb{C}} = \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{C}$, we obtain the $K_{\mathbb{C}}$-module decomposition

$$\mathcal{O}_{\mathbb{C}} = \bigoplus_{\sigma} \mathcal{O}_{\sigma},$$

where σ varies over the set $\mathrm{Hom}(K, \mathbb{C})$, and the direct sum

$$\mathcal{O}_{\sigma} = \bigoplus_{\tau|\sigma} (\mathcal{O} \otimes_{\mathcal{O},\tau} \mathbb{C}) = \bigoplus_{\tau|\sigma} \mathbb{C}$$

is taken over all $\tau \in \mathrm{Hom}(L, \mathbb{C})$ such that $\tau|_K = \sigma$. The mapping $\mathcal{O}_{\mathbb{C}} \to K_{\mathbb{C}}$ induced by $Tr_{L|K} : \mathcal{O} \to \mathcal{o}$ is given, for $x = \bigoplus_{\sigma} x_{\sigma}$, $x_{\sigma} \in \mathcal{O}_{\sigma}$, by

$$Tr_{L|K}(x) = \sum_{\sigma} Tr_{\sigma}(x_{\sigma}),$$

where $Tr_{\sigma}(x_{\sigma}) = \sum_{\tau|\sigma} x_{\sigma,\tau}$, the $x_{\sigma,\tau} \in \mathbb{C}$ being the components of x_{σ}. The metric on $(i_*\mathcal{O})_{\mathbb{C}} = \mathcal{O}_{\mathbb{C}}$ is the orthogonal sum of the standard metrics

$$\langle x, y\rangle_{\sigma} = \sum_{\tau|\sigma} x_{\tau}\bar{y}_{\tau} = Tr_{\sigma}(x\bar{y})$$

on the $\mathbb{C}$-vector spaces $(i_*\mathcal{O})_{\sigma} = \mathcal{O}_{\sigma} = \bigoplus_{\tau|\sigma} \mathbb{C}$. Now let x_i, $y_i \in \mathcal{O}_{\sigma}$, $i = 1, \dots, n$, and write $x = x_1 \wedge \dots \wedge x_n$, $y = y_1 \wedge \dots \wedge y_n \in \det(\mathcal{O}_{\sigma})$. The map $T_{\mathbb{C}}$ splits into the direct sum $T_{\mathbb{C}} = \bigoplus_{\sigma} T_{\sigma}$ of the maps

$$T_{\sigma} : \det(\mathcal{O}_{\sigma}) \otimes_{\mathbb{C}} \det(\mathcal{O}_{\sigma}) \to (\mathfrak{d}_{L|K})_{\sigma} = \mathbb{C}$$

which are given by

$$T_{\sigma}(x \otimes y) = \det\big(Tr_{\sigma}(x_i y_j)\big).$$

For any two n-tuples x_i', $y_i' \in \mathcal{O}_{\sigma}$ we form the matrices

$$A = \big(Tr_{\sigma}(x_i y_j)\big),\ A' = \big(Tr_{\sigma}(\bar{x}_i'\bar{y}_j')\big),\ B = \big(Tr_{\sigma}(x_i\bar{x}_j')\big),\ B' = \big(Tr_{\sigma}(y_i\bar{y}_j')\big).$$

Then one has $AA' = BB'$, and we obtain

$$\begin{aligned}
\big\langle T_{\sigma}(x \otimes y), T_{\sigma}(x' \otimes y')\big\rangle_{(\mathfrak{d}_{L|K})_{\sigma}} &= T_{\sigma}(x \otimes y)\,\overline{T_{\sigma}(x' \otimes y')} \\
&= \det\big(Tr_{\sigma}(x_i y_j)\big)\det\big(Tr_{\sigma}(\bar{x}_i'\bar{y}_j')\big) = \det(AA') = \det(BB') \\
&= \det\big(Tr_{\sigma}(x_i\bar{x}_j')\big)\det\big(Tr_{\sigma}(y_i\bar{y}_j')\big) = \det\big(\langle x_i, x_j'\rangle_{\sigma}\big)\det\big(\langle y_i, y_j'\rangle_{\sigma}\big) \\
&= \langle x, x'\rangle_{\det\mathcal{O}_{\sigma}}\langle y, y'\rangle_{\det\mathcal{O}_{\sigma}} = \langle x \otimes y, x' \otimes y'\rangle_{(\det\mathcal{O}_{\sigma})^{\otimes 2}}.
\end{aligned}$$

This shows that $T_{\mathbb{C}}$ is an isometry. $\square$

We now set out to rewrite the results obtained in (7.2) and (7.4) in the language of GROTHENDIECK's general formalism. For the homomorphism i_* there is the commutative diagram

$$\begin{array}{ccc} K_0(\bar{\mathcal{O}}) & \xrightarrow{\text{rk}} & \mathbb{Z} \\ {\scriptstyle i_*}\downarrow & & \downarrow{\scriptstyle [L:K]} \\ K_0(\bar{o}) & \xrightarrow{\text{rk}} & \mathbb{Z}, \end{array}$$

because $[L:K]$ times the rank of an $\mathcal{O}$-module M is its rank as o-module. Therefore i_* induces a homomorphism

$$i_* : I(\bar{\mathcal{O}}) \longrightarrow I(\bar{o})$$

between the kernels of both rank homomorphisms, so that there is a homomorphism

$$i_* : \operatorname{gr} K_0(\bar{\mathcal{O}}) \longrightarrow \operatorname{gr} K_0(\bar{o}).$$

It is called the **Gysin map**. (7.3) immediately gives the following explicit description of it.

(7.5) Corollary. *The diagram*

$$\begin{array}{ccc} \operatorname{gr} K_0(\bar{\mathcal{O}}) & \xrightarrow[\sim]{\text{id}\oplus\det} & \mathbb{Z} \oplus Pic(\bar{\mathcal{O}}) \\ {\scriptstyle i_*}\downarrow & & \downarrow{\scriptstyle [L:K]\oplus N_{L|K}} \\ \operatorname{gr} K_0(\bar{o}) & \xrightarrow[\sim]{\text{id}\oplus\det} & \mathbb{Z} \oplus Pic(\bar{o}) \end{array}$$

is commutative.

We now consider the following diagram

$$\begin{array}{ccc} K_0(\bar{\mathcal{O}}) & \xrightarrow{\text{ch}} & \operatorname{gr} K_0(\bar{\mathcal{O}}) \\ {\scriptstyle i_*}\downarrow & & \downarrow{\scriptstyle i_*} \\ K_0(\bar{o}) & \xrightarrow{\text{ch}} & \operatorname{gr} K_0(\bar{o}) \end{array}$$

where the Gysin map i_* on the right is explicitly given by (7.5), whereas the determination of the composite $\text{ch} \circ i_*$ is precisely the Riemann-Roch problem. The difficulty that confronts us here lies in the fact that the diagram is *not commutative*. In order to make it commute, we need a correction, which will be provided via the module of differentials (with trivial metric), by the *Todd class*, which is defined as follows.

The module $\Omega^1_{\mathcal{O}|\mathfrak{o}}$ of differentials is only a coherent, and not a projective $\mathcal{O}$-module. But its class $[\Omega^1_{\mathcal{O}|\mathfrak{o}}]$ is viewed as an element of $K_0(\overline{\mathcal{O}})$ via the Poincaré isomorphism

$$K_0(\overline{\mathcal{O}}) \xrightarrow{\sim} K^0(\overline{\mathcal{O}}),$$

and since $\operatorname{rk}_{\mathcal{O}}(\Omega^1_{\mathcal{O}|\mathfrak{o}}) = 0$, it lies in $I(\overline{\mathcal{O}})$.

(7.6) Definition. *The* **Todd class** *of $\mathcal{O}|\mathfrak{o}$ is defined to be the element*

$$\operatorname{Td}(\mathcal{O}|\mathfrak{o}) = 1 - \frac{1}{2}c_1\big([\Omega^1_{\mathcal{O}|\mathfrak{o}}]\big) = 1 - \frac{1}{2}[\Omega^1_{\mathcal{O}|\mathfrak{o}}] \in \operatorname{gr} K_0(\overline{\mathcal{O}}) \otimes \mathbb{Z}[\tfrac{1}{2}].$$

Because of the factor $\frac{1}{2}$, the Todd class does not belong to the ring $\operatorname{gr} K_0(\overline{\mathcal{O}})$ itself, but is only an element of $\operatorname{gr} K_0(\overline{\mathcal{O}}) \otimes \mathbb{Z}[\frac{1}{2}]$. The module of differentials $\Omega^1_{\mathcal{O}|\mathfrak{o}}$ is connected with the **different** $\mathfrak{D}_{L|K}$ of the extension $L|K$ by the exact sequence

$$0 \longrightarrow \mathfrak{D}_{L|K} \longrightarrow \mathcal{O} \longrightarrow \Omega^1_{\mathcal{O}|\mathfrak{o}} \longrightarrow 0$$

of $\mathcal{O}$-modules (with trivial metrics) (see §2, exercise 3). This implies that $[\Omega^1_{\mathcal{O}|\mathfrak{o}}] = 1 - [\mathfrak{D}_{L|K}]$. We may therefore describe the Todd class also by the different:

$$\operatorname{Td}(\mathcal{O}|\mathfrak{o}) = 1 + \frac{1}{2}([\mathfrak{D}_{L|K}] - 1) = \frac{1}{2}\big(1 + [\mathfrak{D}_{L|K}]\big).$$

The main result now follows from (7.3) using the Todd class.

(7.7) Theorem (Grothendieck-Riemann-Roch). *The diagram*

$$\begin{array}{ccc}
K_0(\overline{\mathcal{O}}) & \xrightarrow{\operatorname{Td}(\mathcal{O}|\mathfrak{o})\operatorname{ch}} & \operatorname{gr} K_0(\overline{\mathcal{O}}) \\
{\scriptstyle i_*}\downarrow & & \downarrow{\scriptstyle i_*} \\
K_0(\overline{\mathfrak{o}}) & \xrightarrow{\operatorname{ch}} & \operatorname{gr} K_0(\overline{\mathfrak{o}})
\end{array}$$

is commutative.

Proof: For $\xi \in K_0(\overline{\mathcal{O}})$, we have to show the identity

$$\operatorname{ch}(i_*\xi) = i_*\big(\operatorname{Td}(\mathcal{O}|\mathfrak{o})\operatorname{ch}(\xi)\big).$$

Decomposing $\mathrm{ch}(i_*\xi) = \mathrm{rk}(i_*\xi) \oplus c_1(i_*\xi)$ and $\mathrm{ch}(\xi) = \mathrm{rk}(\xi) \oplus c_1(\xi)$ and observing that

$$\begin{aligned}\mathrm{Td}(\mathcal{O}|\mathcal{o})\,\mathrm{ch}(\xi) &= \big(1 + \tfrac{1}{2}\big([\mathfrak{D}_{L|K}] - 1\big)\big)\,(\mathrm{rk}(\xi) + c_1(\xi))\\ &= \mathrm{rk}(\xi) + \big[c_1(\xi) + \tfrac{1}{2}\,\mathrm{rk}(\xi)([\mathfrak{D}_{L|K}] - 1)\big],\end{aligned}$$

it suffices to check the equations

(a) $$\mathrm{rk}(i_*\xi) = \mathrm{rk}(\xi)\,\mathrm{rk}(i_*[\mathcal{O}]),$$

(b) $$c_1(i_*\xi) = i_*(c_1(\xi)) + \mathrm{rk}(\xi)c_1(i_*[\mathcal{O}])$$

and

(c) $$\mathrm{rk}\big(i_*[\mathcal{O}]\big) = i_*(1),$$

(d) $$2c_1\big(i_*[\mathcal{O}]\big) = i_*\big([\mathfrak{D}_{L|K}] - 1\big)$$

in $\mathrm{gr}\,K_0(\bar{\mathcal{o}})$. The equations (a) and (c) are clear because of $\mathrm{rk}(i_*[\mathcal{O}]) = \mathrm{rk}(i_*\mathcal{O}) = [L : K]$. To show (b) and (d), we apply det to both sides and are reduced by the commutative diagram (7.5) to the equations

(e) $$\det(i_*\xi) = N_{L|K}\big((\det\xi)\big)\,[\det i_*\mathcal{O}]^{\mathrm{rk}(\xi)},$$

(f) $$(\det i_*\mathcal{O})^{\otimes 2} = N_{L|K}(\det\mathfrak{D}_{L|K}).$$

But (e) is the second identity of (7.3), and (f) follows from (7.4) and (2.9). □

With this final theorem, the theory of algebraic integers can be integrated completely into a general programme of algebraic geometry as a special case. What is needed is the use of the geometric language for the objects considered. Thus the ring $\mathcal{o}$ is interpreted as the scheme $X = \mathrm{Spec}(\mathcal{o})$, the projective metrized $\mathcal{o}$-modules as metrized *vector bundles*, the invertible $\mathcal{o}$-modules as *line bundles*, the inclusion $i : \mathcal{o} \to \mathcal{O}$ as morphism $f : Y = \mathrm{Spec}(\mathcal{O}) \to X$ of schemes, the class $\Omega^1_{\mathcal{O}|\mathcal{o}}$ as the *cotangent element*, etc. In this way one realizes in the present context the old idea of viewing number theory as part of geometry.

§ 8. The Euler-Minkowski Characteristic

Considering the theorem of Grothendieck-Riemann-Roch in the special case of an extension $K|\mathbb{Q}$, amounts to revisiting the Riemann-Roch theory of § 3 from our new point of view. At the center of that theory was the Euler-Minkowski characteristic

$$\chi(\mathfrak{a}) = -\log \operatorname{vol}(\mathfrak{a})$$

of replete ideals $\mathfrak{a}$ of K. Here, $\operatorname{vol}(\mathfrak{a})$ was the *canonical measure* of a fundamental mesh of the lattice in Minkowski space $K_{\mathbb{R}} = \mathfrak{a} \otimes_{\mathbb{Z}} \mathbb{R}$ defined by $\mathfrak{a}$. This definition is properly explained in the theory of metrized modules of higher rank. More precisely, instead of considering $\mathfrak{a}$ as a metrized $\mathcal{O}$-module of rank 1, it should be viewed as a metrized $\mathbb{Z}$-module of rank $[K : \mathbb{Q}]$. This point of view leads us necessarily to the following definition of the Euler-Minkowski characteristic.

(8.1) Proposition. *The degree map*

$$\deg_K : Pic(\bar{\mathcal{O}}) \longrightarrow \mathbb{R}, \quad \deg_K([\mathfrak{a}]) = -\log \mathfrak{N}(\mathfrak{a}),$$

extends uniquely to a homomorphism

$$\chi_K : K_0(\bar{\mathcal{O}}) \longrightarrow \mathbb{R}$$

on $K_0(\bar{\mathcal{O}})$, and thereby on $K^0(\bar{\mathcal{O}})$. It is given by

$$\chi_K = \deg \circ \det$$

and called the **Euler-Minkowski characteristic** *over K.*

Proof: Since, by (5.6), $K_0(\bar{\mathcal{O}})$ is generated as an additive group by the elements $[\mathfrak{a}] \in Pic(\bar{\mathcal{O}})$, the map $\deg_K$ on $Pic(\bar{\mathcal{O}})$ determines a unique homomorphism $K_0(\bar{\mathcal{O}}) \to \mathbb{R}$ which extends $\deg_K$. But such a homomorphism is given by the composite of the homomorphisms

$$K_0(\bar{\mathcal{O}}) \xrightarrow{\det} Pic(\bar{\mathcal{O}}) \xrightarrow{\deg} \mathbb{R},$$

as the composite $Pic(\bar{\mathcal{O}}) \hookrightarrow K_0(\bar{\mathcal{O}}) \xrightarrow{\det} Pic(\bar{\mathcal{O}})$ is the identity. □

Via the Poincaré isomorphism $K_0(\bar{\mathcal{O}}) \xrightarrow{\sim} K^0(\bar{\mathcal{O}})$, we transfer the maps det and χ_K to the Grothendieck group $K^0(\bar{\mathcal{O}})$ of coherent metrized $\mathcal{O}$-modules. Then proposition (8.1) is equally valid for $K^0(\bar{\mathcal{O}})$ as for $K_0(\bar{\mathcal{O}})$. We define in what follows $\chi_K(M) = \chi_K([M])$ for a metrized $\mathcal{O}$-module M. If $L|K$ is an extension of algebraic number fields and $i : \mathcal{O} \to \mathcal{O}$ the inclusion of the maximal orders of K, resp. L, then applying $\deg_K$ to the formula (7.2) and using

$$\deg_L(\mathfrak{A}) = -\log \mathfrak{N}(\mathfrak{A}) = -\log \mathfrak{N}\big(N_{L|K}(\mathfrak{A})\big) = \deg_K\big(N_{L|K}(\mathfrak{A})\big)$$

(see (1.6), (iii)) gives the

(8.2) Theorem. *For every coherent $\mathcal{O}$-module M, the Riemann-Roch formula*

$$\chi_K(i_*M) = \deg_L(\det M) + \mathrm{rk}(M)\,\chi_K(i_*\mathcal{O})$$

is valid, and in particular, for an invertible metrized $\mathcal{O}$-module M, we have

$$\chi_K(i_*M) = \deg_L(M) + \chi_K(i_*\mathcal{O}).$$

We now specialize to the case of the base field $K = \mathbb{Q}$, that is, we consider metrized $\mathbb{Z}$-modules. Such a module is simply a finitely generated abelian group M together with a euclidean metric on the real vector space

$$M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}.$$

Indeed, since $\mathbb{Q}$ has only a single embedding into $\mathbb{C}$, i.e., $\mathbb{Q}_{\mathbb{C}} = \mathbb{C}$, a metric on M is simply given by a hermitian scalar product on the $\mathbb{C}$-vector space $M_{\mathbb{C}} = M_{\mathbb{R}} \otimes \mathbb{C}$. Restricting this to $M_{\mathbb{R}}$ gives a euclidean metric the sesquilinear extension of which reproduces the original metric.

If M is a *projective* metrized $\mathbb{Z}$-module, then the underlying $\mathbb{Z}$-module is a finitely generated free abelian group. The canonical map $M \to M \otimes \mathbb{R}$, $a \mapsto a \otimes 1$, identifies M with a complete lattice in $M_{\mathbb{R}}$. If $\alpha_1, \dots, \alpha_n$ is a $\mathbb{Z}$-basis of M, then the set

$$\Phi = \big\{ x_1\alpha_1 + \cdots + x_n\alpha_n \bigm| x_i \in \mathbb{R},\ 0 \le x_i < 1 \big\}$$

is a *fundamental mesh* of the lattice M. The euclidean metric $\langle\ ,\ \rangle_M$ defines a Haar measure on $M_{\mathbb{R}}$. Once we choose an orthonormal basis $e_1, \dots, e_n$ of $M_{\mathbb{R}}$, this Haar measure can be expressed, via the isomorphism $M_{\mathbb{R}} \xrightarrow{\sim} \mathbb{R}^n$, $x_1e_1 + \cdots + x_ne_n \longmapsto (x_1, \dots, x_n)$, by the Lebesgue measure on $\mathbb{R}^n$. With respect to this measure, the volume of the fundamental mesh Φ is given by

$$\mathrm{vol}(\Phi) = \big|\det(\langle\alpha_i, \alpha_j\rangle)\big|^{1/2}.$$

It will be denoted by $\mathrm{vol}(M)$ for short. It does not depend on the choice of $\mathbb{Z}$-basis $\alpha_1, \dots, \alpha_n$ because a different choice is linked to the original one by a matrix with integer coefficients which also has an inverse with integer coefficients, hence has determinant of absolute value 1.

A more elegant definition of $\mathrm{vol}(M)$ can be given in terms of the invertible metrized $\mathbb{Z}$-module $\det M$. $\det M_{\mathbb{R}}$ is a one-dimensional $\mathbb{R}$-vector space with metric $\langle\ ,\ \rangle_{\det M}$, and with the lattice $\det M$ isomorphic to $\mathbb{Z}$. If $x \in \det M$ is a generator (for instance, $x = \alpha_1 \wedge \ldots \wedge \alpha_n$), then

$$\mathrm{vol}(M) = \|x\|_{\det M} = \sqrt{\langle x, x\rangle_{\det M}}\,.$$

In the present case, where the base field is $\mathbb{Q}$, the degree map

$$\deg : Pic(\bar{\mathbb{Z}}) \longrightarrow \mathbb{R}$$

is an isomorphism (see § 1, exercise 3), and we call the unique homomorphism arising from this,

$$\chi = \deg \circ \det : K^0(\bar{\mathbb{Z}}) \longrightarrow \mathbb{R},$$

the Euler-Minkowski characteristic. It is computed explicitly as follows.

(8.3) Proposition. *For a coherent metrized $\mathbb{Z}$-module M, one has*

$$\chi(M) = \log \#M_{\text{tor}} - \log \operatorname{vol}(M/M_{\text{tor}})\,.$$

In this formula M_{tor} denotes the torsion subgroup of M and M/M_{tor} the projective metrized $\mathbb{Z}$-module which receives its metric from M via $M \otimes \mathbb{R} = M/M_{\text{tor}} \otimes \mathbb{R}$.

Proof of (8.3): If M is a finite $\mathbb{Z}$-module, then the determinant of the class $[M] \in K^0(\bar{\mathbb{Z}})$ is computed from a free resolution

$$0 \longrightarrow E \longrightarrow F \xrightarrow{\alpha} M \longrightarrow 0,$$

where $F = \mathbb{Z}^n$ and $E = \ker(\alpha) \cong \mathbb{Z}^n$. If we equip $F \otimes \mathbb{R} = E \otimes \mathbb{R} = \mathbb{R}^n$ with the standard metric, the sequence becomes a short exact sequence of metrized $\mathbb{Z}$-modules, because $M \otimes \mathbb{R} = 0$. We therefore have in $K^0(\bar{\mathbb{Z}})$:

$$[M] = [F] - [E]\,.$$

Let A be the matrix corresponding to the change of basis from the standard basis $e_1, \dots, e_n$ of F to a $\mathbb{Z}$-basis $e_1', \dots, e_n'$ of E. Then $x = e_1 \wedge \dots \wedge e_n$, resp. $x' = e_1' \wedge \dots \wedge e_n'$, is a generator of $\det F$, resp. $\det E$, and

$$x' = \det A \cdot x = (F : E) \cdot x = \#M \cdot x\,.$$

The metric $\|\ \|$ on $\det E$ is the same as that on $\det F$, so that

$$\chi(E) = \deg(\det E) = -\log \|x'\| = -\log\bigl(\#M \|x\|\bigr) = -\log \#M + \chi(F)\,,$$

and then

$$\chi(M) = \chi\bigl([F] - [E]\bigr) = \chi(F) - \chi(E) = \log \#M\,.$$

For an arbitrary coherent metrized $\mathbb{Z}$-module M we have the direct sum decomposition

$$M = M_{\text{tor}} \bigoplus M/M_{\text{tor}}$$

into metrized $\mathbb{Z}$-modules. If $\alpha_1, \dots, \alpha_n$ is a basis of the lattice M/M_{tor}, then $x = \alpha_1 \wedge \dots \wedge \alpha_n$ is a generator of $\det M/M_{\text{tor}}$; then $\chi(M/M_{\text{tor}}) = \deg(\det M/M_{\text{tor}}) = -\log \|x\| = -\log \operatorname{vol}(M/M_{\text{tor}})$. We therefore conclude that

$$\chi(M) = \chi(M_{\text{tor}}) + \chi(M/M_{\text{tor}}) = \log \# M_{\text{tor}} - \log \operatorname{vol}(M/M_{\text{tor}}). \qquad \square$$

The Euler-Minkowski characteristic of a replete ideal $\mathfrak{a}$,

$$\chi(\mathfrak{a}) = -\log \operatorname{vol}(\mathfrak{a}),$$

which we defined *ad hoc* in § 3 via the Minkowski measure $\operatorname{vol}(\mathfrak{a})$ now appears as a simple special case of the Euler-Minkowski characteristic for metrized $\mathbb{Z}$-modules to which the detailed development of the theory has led us. Indeed, viewing the metrized $\mathcal{O}$-module $L(\mathfrak{a})$ of rank 1 associated to $\mathfrak{a}$ as the metrized $\mathbb{Z}$-module $i_*L(\mathfrak{a})$ of rank $[K : \mathbb{Q}]$, we get the

(8.4) Proposition. $\chi(\mathfrak{a}) = \chi\big(i_*L(\mathfrak{a})\big)$.

Proof: Let $\mathfrak{a} = \mathfrak{a}_f \mathfrak{a}_\infty = \mathfrak{a}_f \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_\mathfrak{p}}$. The metric $\langle\ ,\ \rangle_{i_*L(\mathfrak{a})}$ on the $\mathbb{C}$-vector space $K_\mathbb{C} = \prod_{\tau \in X(\mathbb{C})} \mathbb{C}$ is then given by

$$\langle x, y \rangle_{i_*L(\mathfrak{a})} = \sum_\tau e^{2\nu_{\mathfrak{p}_\tau}} x_\tau \bar{y}_\tau,$$

where $\mathfrak{p}_\tau$ is the infinite place of K corresponding to the embedding $\tau : K \to \mathbb{C}$. It results from the standard metric $\langle\ ,\ \rangle$ via the F-invariant transformation

$$T : K_\mathbb{C} \longrightarrow K_\mathbb{C}, \quad (x_\tau)_{\tau \in X(\mathbb{C})} \longmapsto (e^{\nu_{\mathfrak{p}_\tau}} x_\tau)_{\tau \in X(\mathbb{C})}.$$

Equivalently,

$$\langle x, y \rangle_{i_*L(\mathfrak{a})} = \langle Tx, Ty \rangle.$$

The volume $\operatorname{vol}(i_*L(\mathfrak{a}))$ of a fundamental mesh of the lattice $\mathfrak{a}_f$ in $K_\mathbb{R}$ with respect to the Haar measure defined by the euclidean metric on $K_\mathbb{R}$ is then the volume of a fundamental mesh of the lattice $T\mathfrak{a}_f$ with respect to the canonical measure defined by $\langle\ ,\ \rangle$. Thus

$$\operatorname{vol}\big(i_*L(\mathfrak{a})\big) = \operatorname{vol}(T\mathfrak{a}_f).$$

In the representation $K_\mathbb{R} = \prod_{\mathfrak{p}|\infty} K_\mathfrak{p}$, the canonical embedding

$$K_\mathbb{R} = K \otimes_\mathbb{Q} \mathbb{R} \longrightarrow K_\mathbb{C} = K \otimes_\mathbb{Q} \mathbb{C}$$

maps an element $(x_\mathfrak{p})_{\mathfrak{p}|\infty}$ to the element $(x_\tau)_{\tau \in X(\mathbb{C})}$ with $x_\tau = \tau x_{\mathfrak{p}_\tau}$. Here we extend τ to $K_{\mathfrak{p}_\tau}$. The restriction of the transformation $T : (x_\tau) \mapsto (e^{\nu_{\mathfrak{p}_\tau}} x_\tau)$ to $K_\mathbb{R} = \prod_{\mathfrak{p}|\infty} K_\mathfrak{p}$ is therefore given by $(x_\mathfrak{p}) \mapsto (e^{\nu_\mathfrak{p}} x_\mathfrak{p})$. The lattice $T\mathfrak{a}_f$ is then the same lattice which was denoted $\mathfrak{a}$ in §3. So we obtain

$$\operatorname{vol}\big(i_*L(\mathfrak{a})\big) = \operatorname{vol}(\mathfrak{a}),$$

i.e., $\chi(i_*L(\mathfrak{a})) = \chi(\mathfrak{a})$. □

Given this identification, the Riemann-Roch theorem (3.4) proven in §3 for replete ideals $\mathfrak{a}$,

$$\chi(\mathfrak{a}) = \deg(\mathfrak{a}) + \chi(\mathcal{O}),$$

now appears as a special case of theorem (8.2), which says that

$$\chi\big(i_*L(\mathfrak{a})\big) = \deg\big(L(\mathfrak{a})\big) + \chi(i_*\mathcal{O}).$$